Trends – Artificial Intelligence

BOND

May 2025

Trends – Artificial Intelligence (AI)

May 30, 2025

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BOND

Context

We set out to compile foundational trends related to AI. A starting collection of several disparate datapoints turned into this beast. As soon as we updated one chart, we often had to update another – a data game of whack-a-mole… a pattern that shows no sign of stopping…and will grow more complex as competition among tech incumbents, emerging attackers and sovereigns accelerates.

Vint Cerf, one of the 'Founders of the Internet,' said in 1999, '…they say a year in the Internet business is like a dog year – equivalent to seven years in a regular person's life.' At the time, the pace of change catalyzed by the internet was unprecedented.

Consider now that AI user and usage trending is ramping materially faster…and the machines can outpace us.

The pace and scope of change related to the artificial intelligence technology evolution is indeed unprecedented, as supported by the data. This document is filled with user, usage and revenue charts that go up-and-to-the-right… often supported by spending charts that also go up-and-to-the right.

Creators / bettors / consumers are taking advantage of global internet rails that are accessible to 5.5B citizens via connected devices; ever-growing digital datasets that have been in the making for over three decades; breakthrough large language models (LLMs) that – in effect – found freedom with the November 2022 launch of OpenAI's ChatGPT with its extremely easy-to-use / speedy user interface.

In addition, relatively new AI company founders have been especially aggressive about innovation / product releases / investments / acquisitions / cash burn and capital raises. At the same time, more traditional tech companies (often with founder involvement) have increasingly directed more of their hefty free cash flows toward AI in efforts to drive growth and fend off attackers.

And global competition – especially related to China and USA tech developments – is acute.

The outline for our document is on the next page, followed by eleven charts that help illustrate observations that follow.

We hope this compilation adds to the discussion of the breadth of change at play – technical / financial / social / physical / geopolitical. No doubt, people (and machines) will improve on the points as we all aim to adapt to this evolving journey as knowledge – and its distribution – get leveled up rapidly in new ways.

Special thanks to Grant Watson and Keeyan Sanjasaz and BOND colleagues who helped steer ideas and bring this report to life. And, to the many friends and technology builders who helped, directly or via your work, and are driving technology forward.

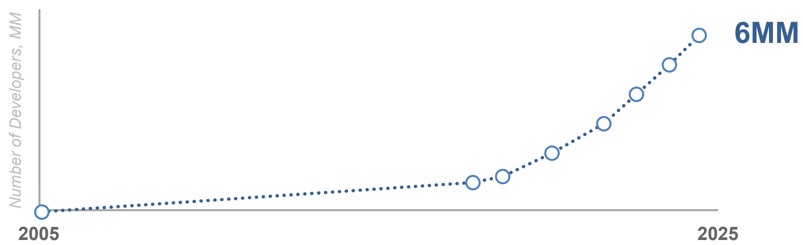
Outline

Charts Paint Thousands of Words…

1

Seem Like Change Happening Faster Than Ever? Yes, It Is

Developers in Leading Chipmaker's Ecosystem



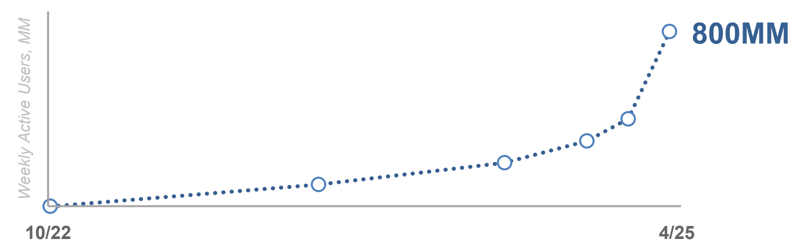
Source: Leading Chipmaker

Details on Page 38

2

AI User + Usage + CapEx Growth = Unprecedented

Leading USA-Based LLM Users



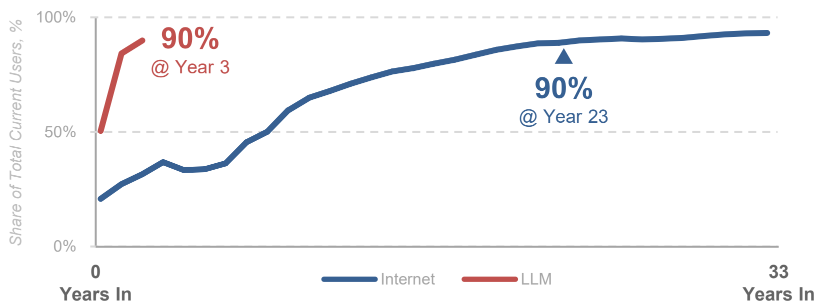
Source: Company disclosures

Details on Page 55

2.1

AI User + Usage + CapEx Growth = Unprecedented

Internet vs. Leading USA-Based LLM: Total Current Users Outside North America



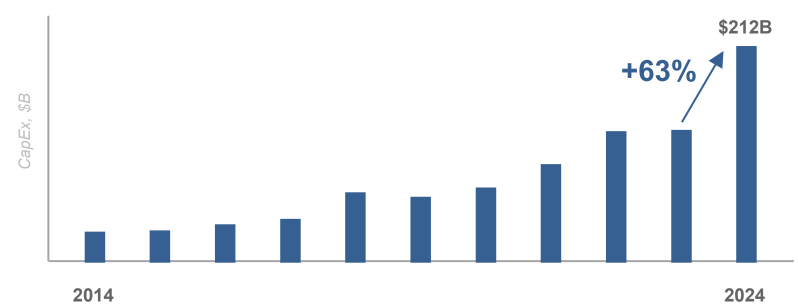
Note: LLM data is for monthly active mobile app users. App not available in select countries, including China and Russia, as of 5/25.
Source: United Nations / International Telecommunications Union (3/25), Sensor Tower (5/25)

Details on Page 56

2.2

AI User + Usage + CapEx Growth = Unprecedented

Big Six USA Technology Company CapEx*



*Apple, NVIDIA, Microsoft, Alphabet, Amazon (AWS only), & Meta Platforms
Source: Capital IQ (3/25), Morgan Stanley

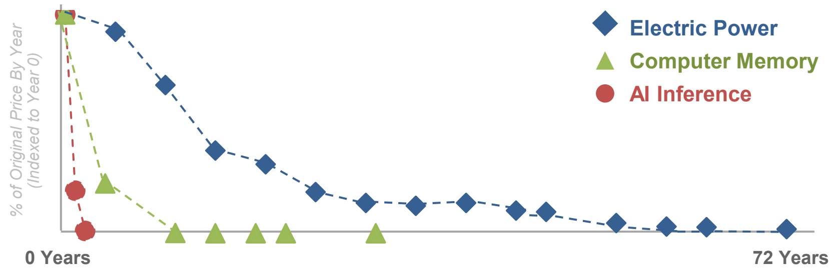
Details on Page 97

…Charts Paint Thousands of Words…

3

AI Model Compute Costs High / Rising +
Inference Costs Per Token Falling =
Performance Converging + Developer Usage Rising

Cost of Key Technologies Relative to Launch Year



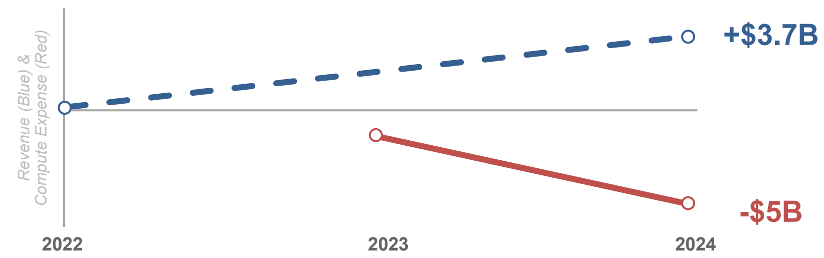
Note: Per-token inference costs shown.
Source: Richard Hirsh; John McCallum; OpenAI

Details on Page 138

4

AI Usage + Cost + Loss Growth = Unprecedented

Leading USA-Based AI LLM Revenue vs. Compute Expense



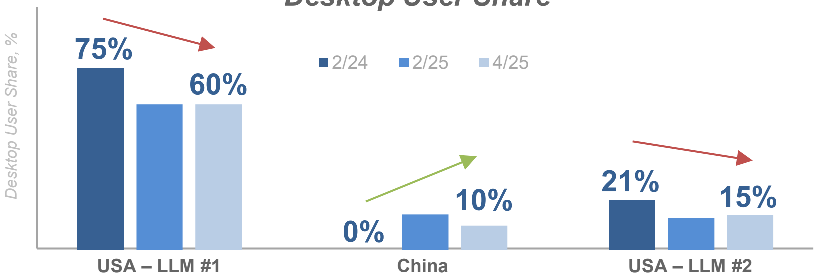
Note: Figures are estimates.
Source: The Information, public estimates

Details on Page 173

5

AI Monetization Threats =
Rising Competition +
Open-Source Momentum + China's Rise

Leading USA LLMs vs. China LLM Desktop User Share



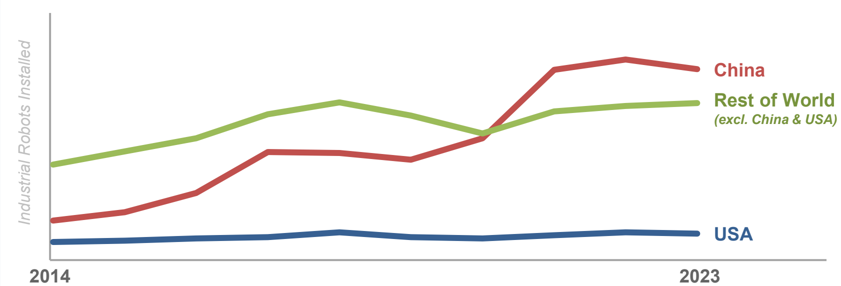
Note: Data is non-deduped. Share is relative, measured across six leading global LLMs.
Source: YipitData (5/25)

Details on Page 293

5.1

AI Monetization Threats =
Rising Competition +
Open-Source Momentum + China's Rise

China vs. USA vs. Rest of World Industrial Robots Installed



Note: Data as of 2023.
Source: International Federation of Robotics

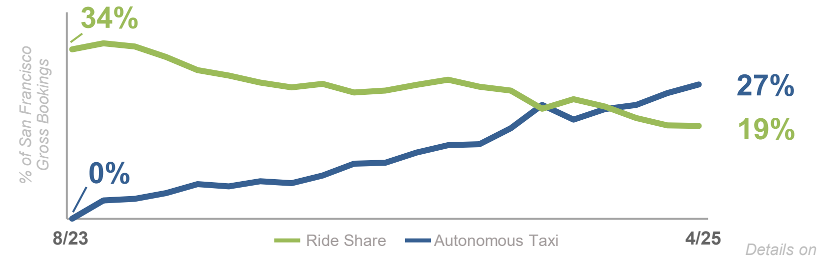
Details on Page 289

…Charts Paint Thousands of Words

6

AI & Physical World Ramps = Fast + Data-Driven

A Ride Share vs. Autonomous Taxi Provider, San Francisco Operating Zone Market Share



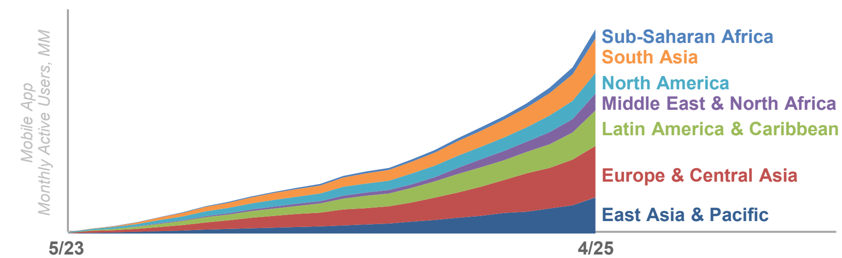
Source: YipitData (4/25)

Details on Page 302

7

Global Internet User Ramps Powered by AI from Get-Go = Growth We Have Not Seen Likes of Before

Leading USA-Based LLM App Users by Region



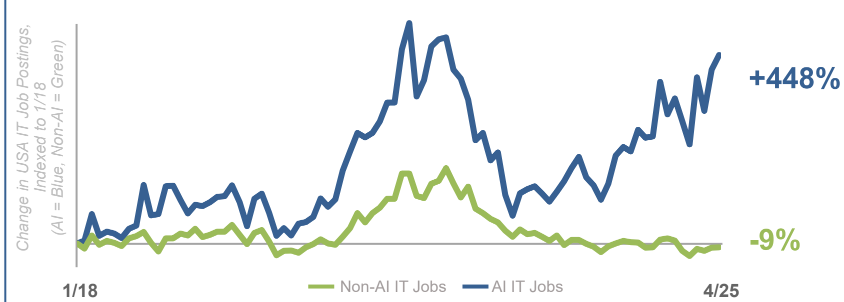
Note: Region definitions per World Bank definitions. China not included in East Asia figures. Data for standalone app only. Source: Sensor Tower (5/25)

Details on Page 315

8

AI & Work Evolution = Real + Rapid

USA IT Jobs – AI vs. Non-AI



Source: University of Maryland's UMD-LinkUp AIMaps (in collaboration with Outrigger Group) (5/25)

Details on Page 332

Overview…

To say the world is changing at unprecedented rates is an understatement.
Rapid and transformative technology innovation / adoption represent key underpinnings of these changes.
As does leadership evolution for the global powers.

Google's founding mission (1998) was to 'organize the world's information and make it universally accessible and useful.'
Alibaba's founding mission (1999) was to 'make it easy to do business anywhere.'
Facebook's founding mission (2004) was 'to give people the power to share and make the world more open and connected.'

Fast forward to today with the world's organized, connected and accessible information being supercharged by artificial intelligence, accelerating computing power, and semi-borderless capital…all driving massive change.

Sport provides a good analogy for AI's constant improvements. As athletes continue to wow us and break records, their talent is increasingly enhanced by better data / inputs / training.
The same is true for businesses, where computers are ingesting massive datasets to get smarter and more competitive.
Breakthroughs in large models, cost-per-token declines, open-source proliferation and chip performance improvements are making new tech advances increasingly more powerful, accessible, and economically viable.

OpenAI's ChatGPT – based on user / usage / monetization metrics – is history's biggest 'overnight' success (nine years post-founding). AI usage is surging among consumers, developers, enterprises and governments.
And unlike the Internet 1.0 revolution – where technology started in the USA and steadily diffused globally – ChatGPT hit the world stage all at once, growing in most global regions simultaneously.

Meanwhile, platform incumbents and emerging challengers are racing to build and deploy the next layers of AI infrastructure: agentic interfaces, enterprise copilots, real-world autonomous systems, and sovereign models.

Rapid advances in artificial intelligence, compute infrastructure, and global connectivity are fundamentally reshaping how work gets done, how capital is deployed, and how leadership is defined – across both companies and countries.

At the same time, we have leadership evolution among the global powers, each of whom is challenging the other's competitive and comparative advantage. We see the world's most powerful countries revved up by varying degrees of economic / societal / territorial aspiration…

…Overview

…Increasingly, two hefty forces – technological and geopolitical – are intertwining.
Andrew Bosworth (Meta Platforms CTO), on a recent 'Possible' podcast described the current state of AI as *our space race* and *the people we're discussing, especially China, are highly capable… there's very few secrets. And there's just progress. And you want to make sure that you're never behind.*

The reality is AI leadership could beget geopolitical leadership – and not vice-versa.

This state of affairs brings tremendous uncertainty…yet it leads us back to one of our favorite quotes – *Statistically speaking, the world doesn't end that often*, from former T. Rowe Price Chairman and CEO Brian Rogers.

As investors, we always assume everything can go wrong, but the exciting part is the consideration of what can go right. Time and time again, the case for optimism is one of the best bets one can make. The magic of watching AI do your work for you feels like the early days of email and web search – technologies that fundamentally changed our world. The better / faster / cheaper impacts of AI seem just as magical, but even quicker.

No doubt, these are also dangerous and uncertain times. But a long-term case for optimism for artificial intelligence is based on the idea that intense competition and innovation… increasingly-accessible compute…rapidly-rising global adoption of AI-infused technology…and thoughtful and calculated leadership can foster sufficient trepidation and respect, that in turn, could lead to Mutually Assured Deterrence.

For some, the evolution of AI will create a race to the bottom; for others, it will create a race to the top. The speculative and frenetic forces of capitalism and creative destruction are tectonic. It's undeniable that it's 'game on,' especially with the USA and China and the tech powerhouses charging ahead.

In this document, we share data / research / benchmarks from third parties that use methodologies they deem to be effective – we are thankful for the hard work so many are doing to illustrate trending during this uniquely dynamic time. Our goal is to add to the discussion.

Outline

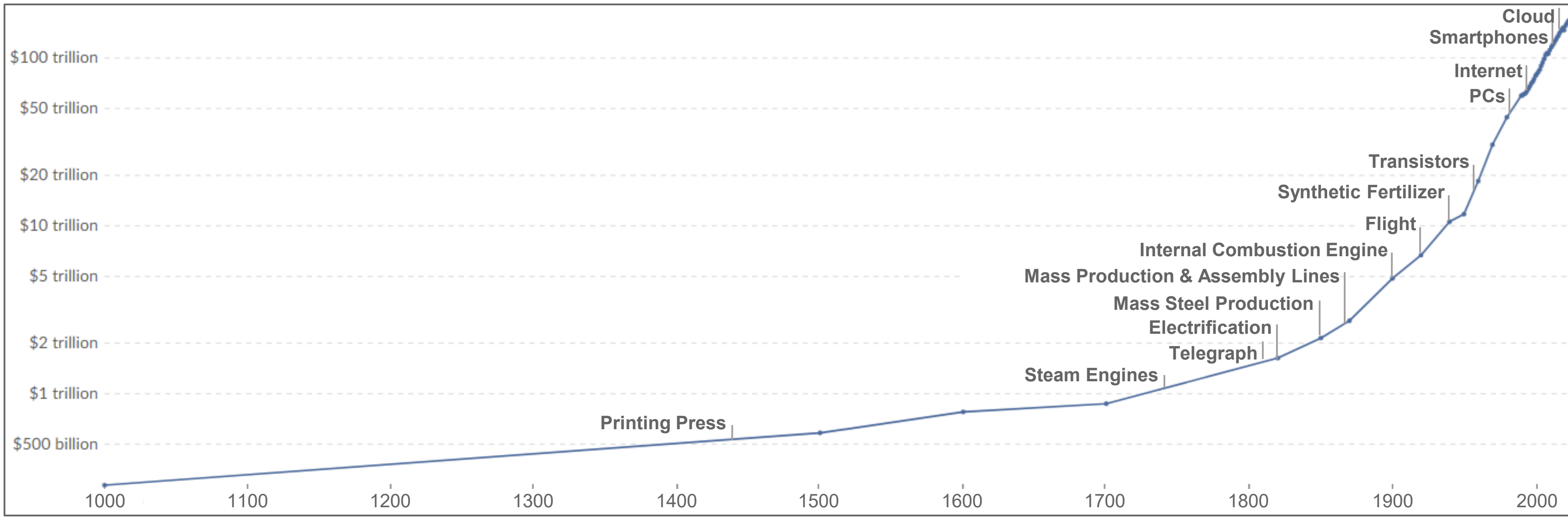
1. **Seem Like Change Happening Faster Than Ever?**
   Yes, It Is
2. **AI User + Usage + CapEx Growth =**
   Unprecedented
3. **AI Model Compute Costs High / Rising + Inference Costs Per Token Falling =**
   Performance Converging + Developer Usage Rising
4. **AI Usage + Cost + Loss Growth =**
   Unprecedented
5. **AI Monetization Threats =**
   Rising Competition + Open-Source Momentum + China's Rise
6. **AI & Physical World Ramps =**
   Fast + Data-Driven
7. **Global Internet User Ramps Powered by AI from Get-Go =**
   Growth We Have Not Seen Likes of Before
8. **AI & Work Evolution =**
   Real + Rapid

Technology Compounding =

Numbers Behind The Momentum

Technology Compounding Over Thousand-Plus Years = Better + Faster + Cheaper → More…

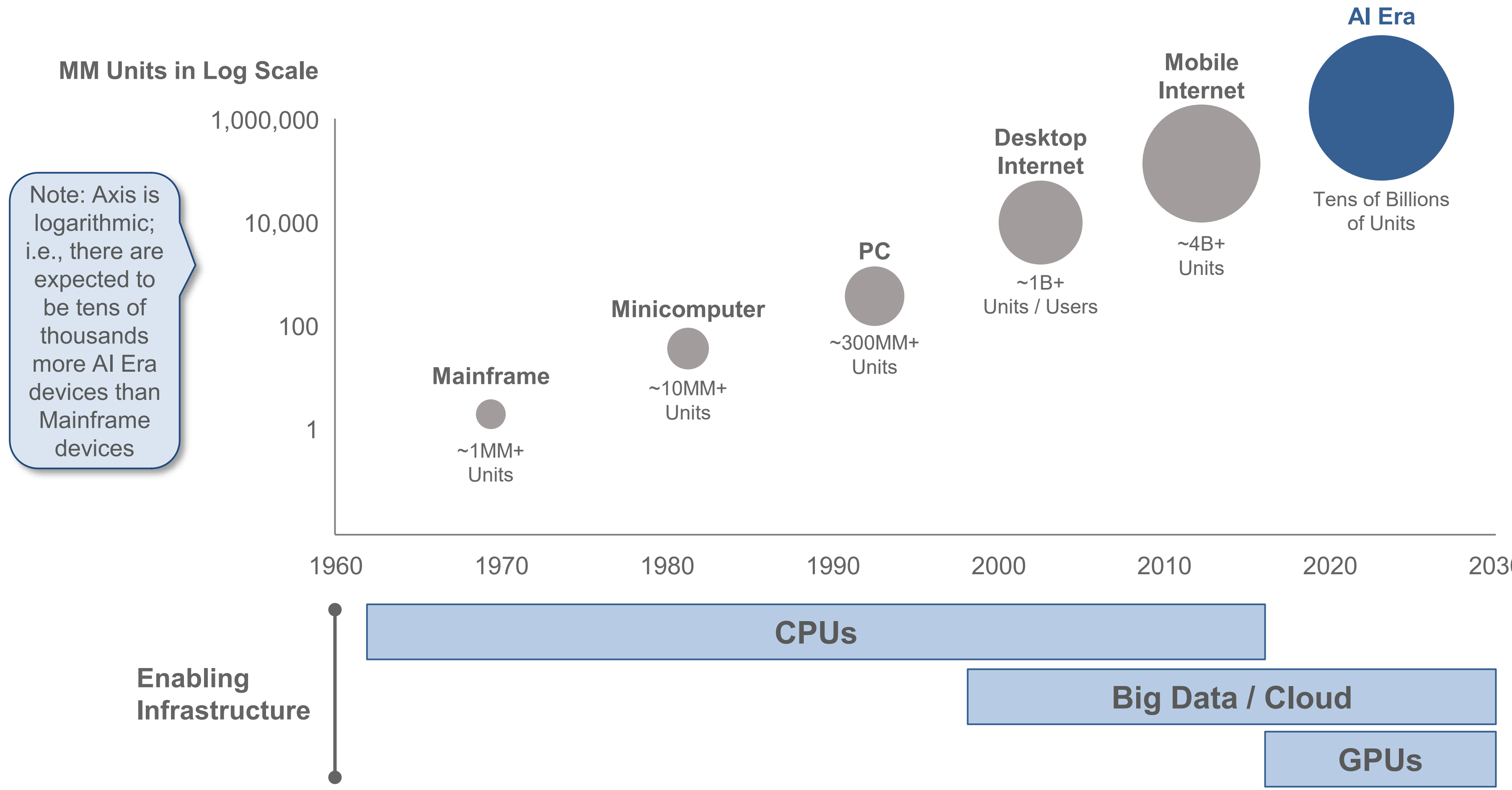
Global GDP – Last 1,000+ Years, per Maddison Project



Note: Chart expressed in trillions of real GDP as measured by 2011 'GK$' on a logarithmic scale. GK$ (Gross Knowledge Dollars) is an informal term used to estimate the potential business value of a specific insight, idea, or proprietary knowledge. It reflects how much that knowledge could be worth if applied effectively, even if it hasn't yet generated revenue.
Source: Microsoft, 'Governing AI: A Blueprint for the Future,' Microsoft Report (5/23); Data via Maddison Project & Our World in Data

…Technology Compounding Over Fifty-Plus Years = Better + Faster + Cheaper → More

Computing Cycles Over Time – 1960s-2020s, per Morgan Stanley



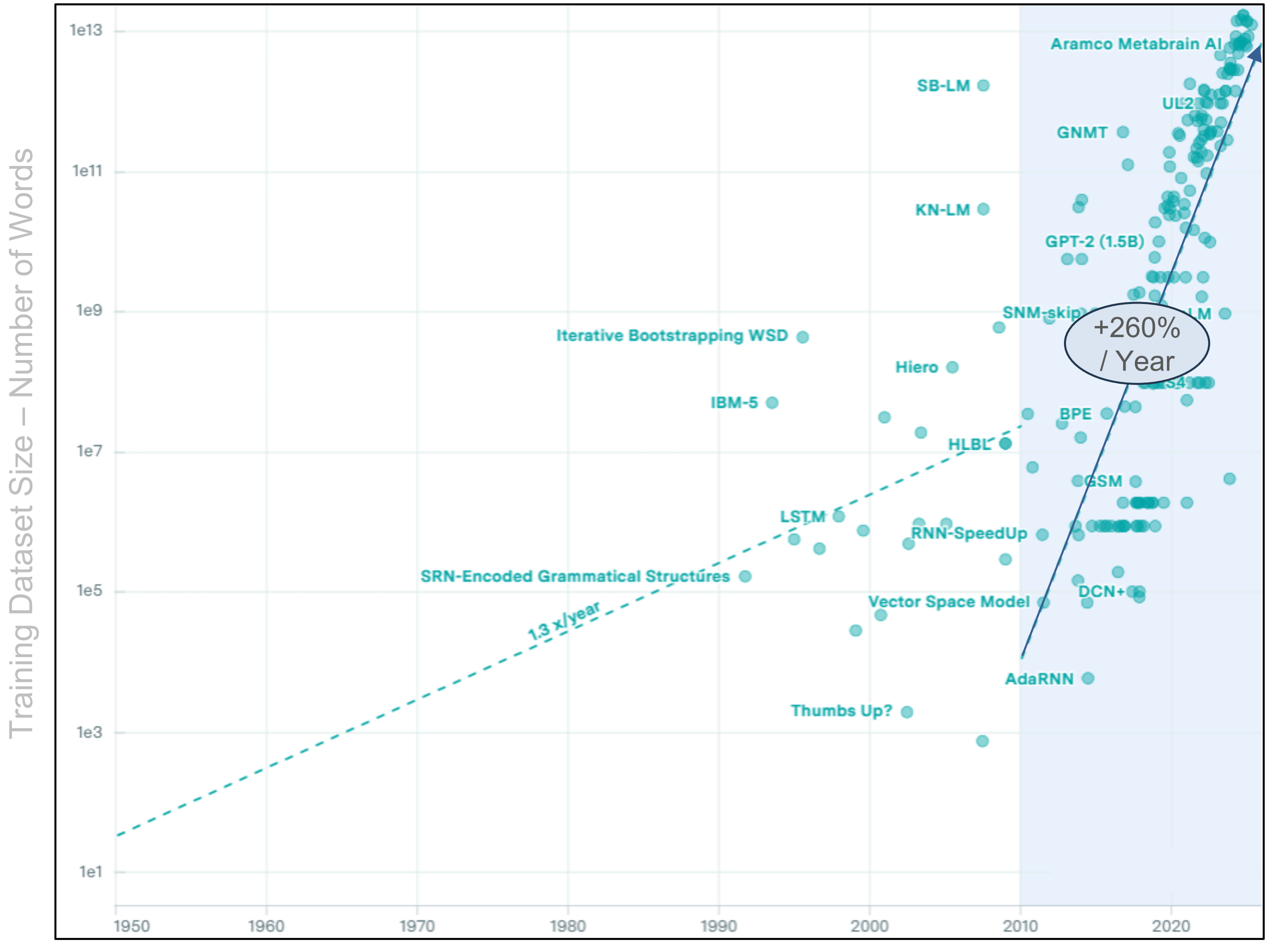
Note: PC units as of 2000. Desktop internet users as of 2005, installed base as of 2010. Mobile internet units are the installed based of smartphones & tablets in 2020. Cloud & data center capex includes Google, Amazon, Microsoft, Meta, Alibaba, Apple, IBM, Oracle, Tencent, & Baidu for ten years ending 2022. 'Tens of billions of units' refers to the potential device & user base that could end up using AI technology; this includes smartphones, IOT devices, robotics, etc. Source: Weiss et al. 'AI Index: Mapping the $4 Trillion Enterprise Impact' via Morgan Stanley (10/23)

AI Technology Compounding =

Numbers Behind The Momentum

260% Annual Growth Over Fifteen Years of…
Data to Train AI Models Led To…

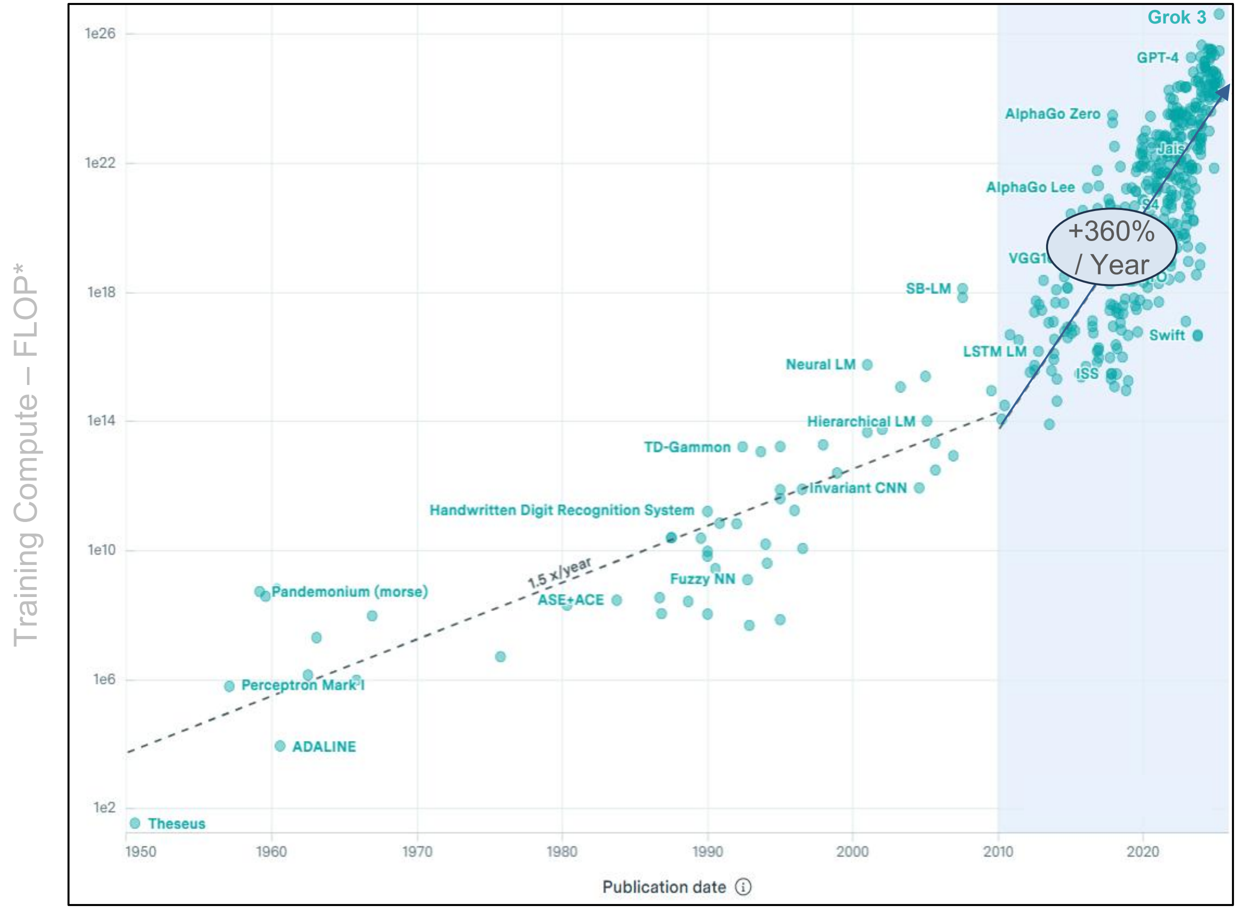
Training Dataset Size (Number of Words) for Key AI Models – 1950-2025, per Epoch AI



Note: Only "notable" language models shown (per Epoch AI, includes state of the art improvement on a recognized benchmark, >1K citations, historically relevant, with significant use).
Source: Epoch AI (5/25)

…360% Annual Growth Over Fifteen Years of… Compute to Train AI Models Led To…

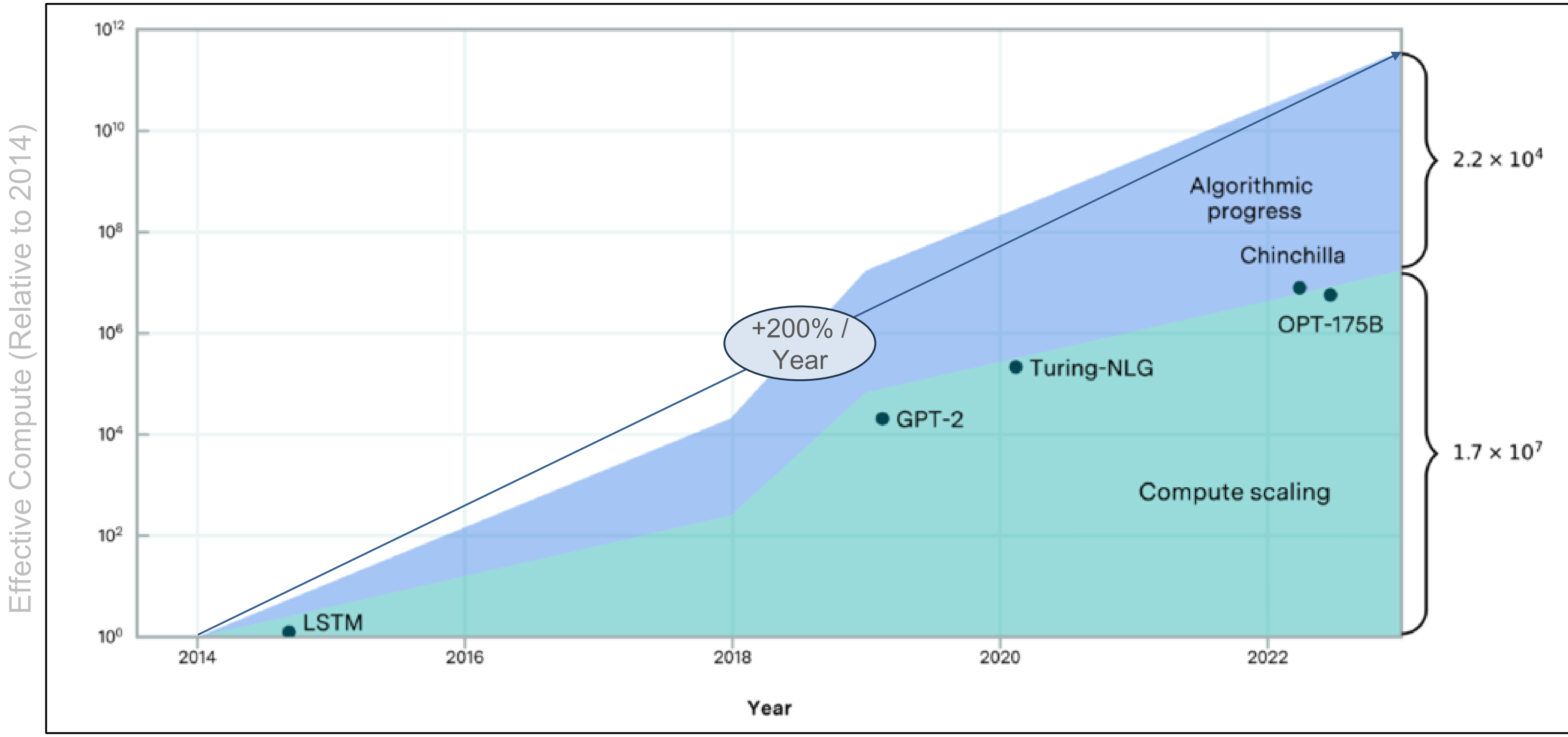
Training Compute (FLOP) for Key AI Models – 1950-2025, per Epoch AI



*A FLOP (floating point operation) is a basic unit of computation used to measure processing power, representing a single arithmetic calculation involving decimal numbers. In AI, total FLOPs are often used to estimate the computational cost of training or running a model.
Note: Only language models shown (per Epoch AI, includes state of the art improvement on a recognized benchmark, >1K citations, historically relevant, with significant use). Source: Epoch AI (5/25)

…200% Annual Growth Over Nine Years of… Compute Gains from Better Algorithms Led To…

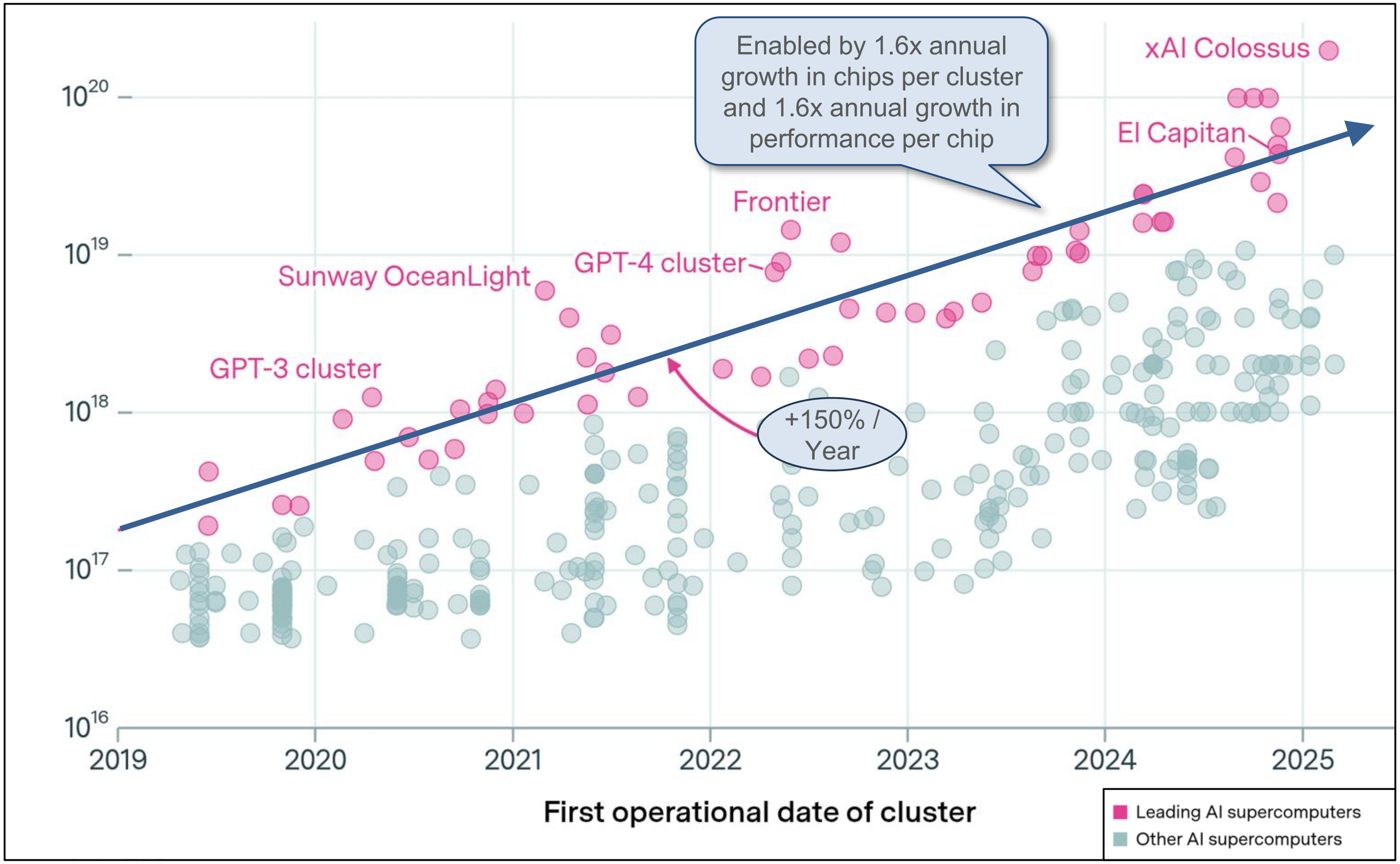
Impact of Improved Algorithms on AI Model Performance – 2014-2023, per Epoch AI



Note: Estimates how much progress comes from bigger models versus smarter algorithms, based on how much computing power you'd need to reach top performance without any improvements. Source: Epoch AI (3/24)

…150% Annual Growth Over Six Years of…
Performance Gains from Better AI Supercomputers Led To…

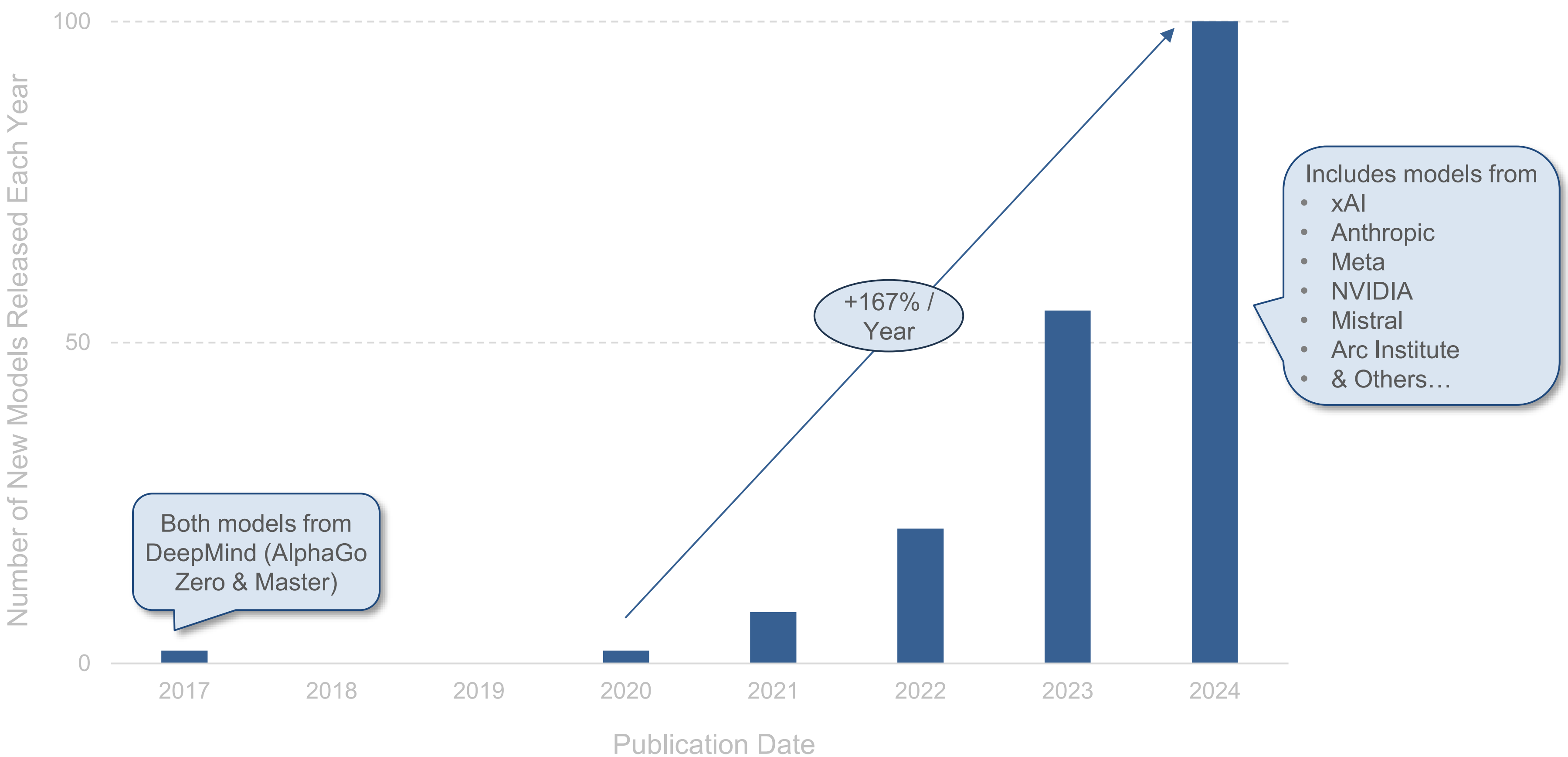
Performance of Leading AI Supercomputers (FLOP/s) – 2019-2025, per Epoch AI



Source: Epoch AI (4/25)

…167% Annual Growth Over Four Years in… Number of Powerful AI Models

Number of New Large-Scale AI Models (Larger than 10^{23} FLOP*) – 2017-2024, per Epoch AI

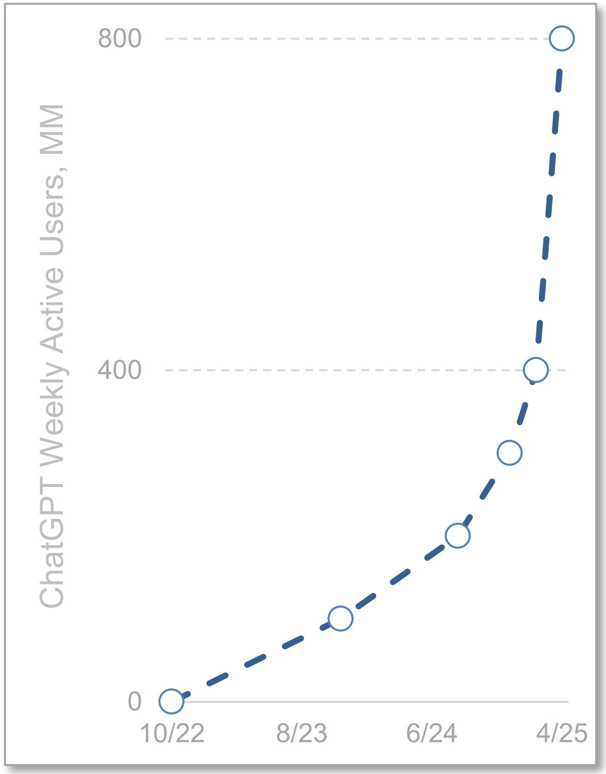


*As of 4/25, 'Large-Scale AI Models' are generally defined as those with a training compute of 10^{23} FLOPs or greater, per Epoch AI.
Source: Epoch AI (5/25)

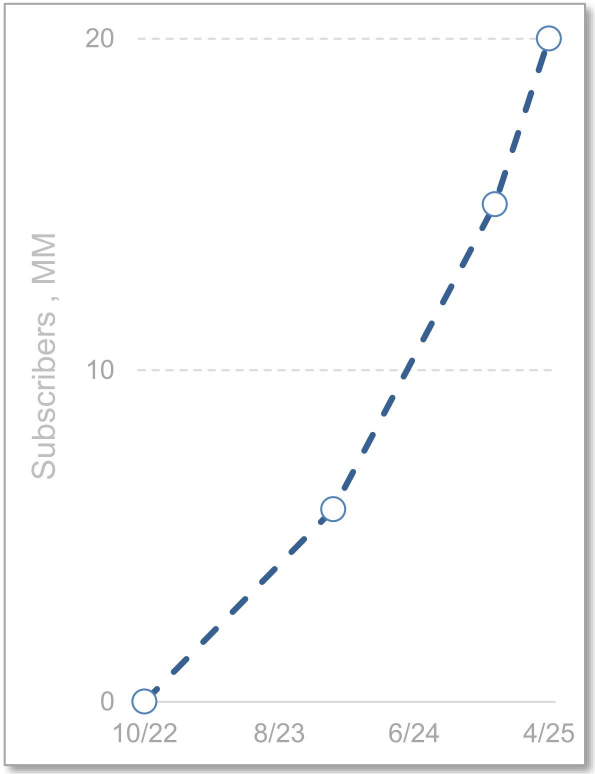
ChatGPT AI User + Subscriber + Revenue Growth Ramps = Hard to Match, Ever

ChatGPT User + Subscriber + Revenue Growth – 10/22-4/25, per OpenAI & *The Information*

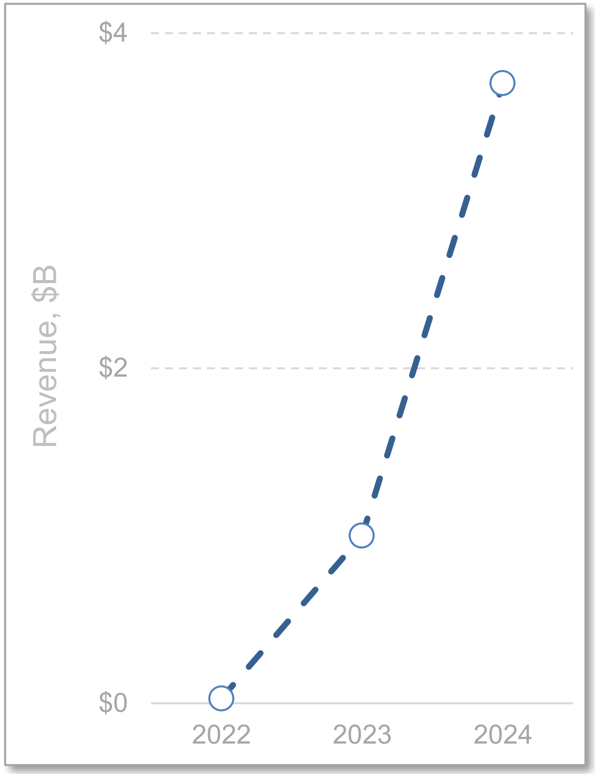
Users (MM)



Subscribers (MM)



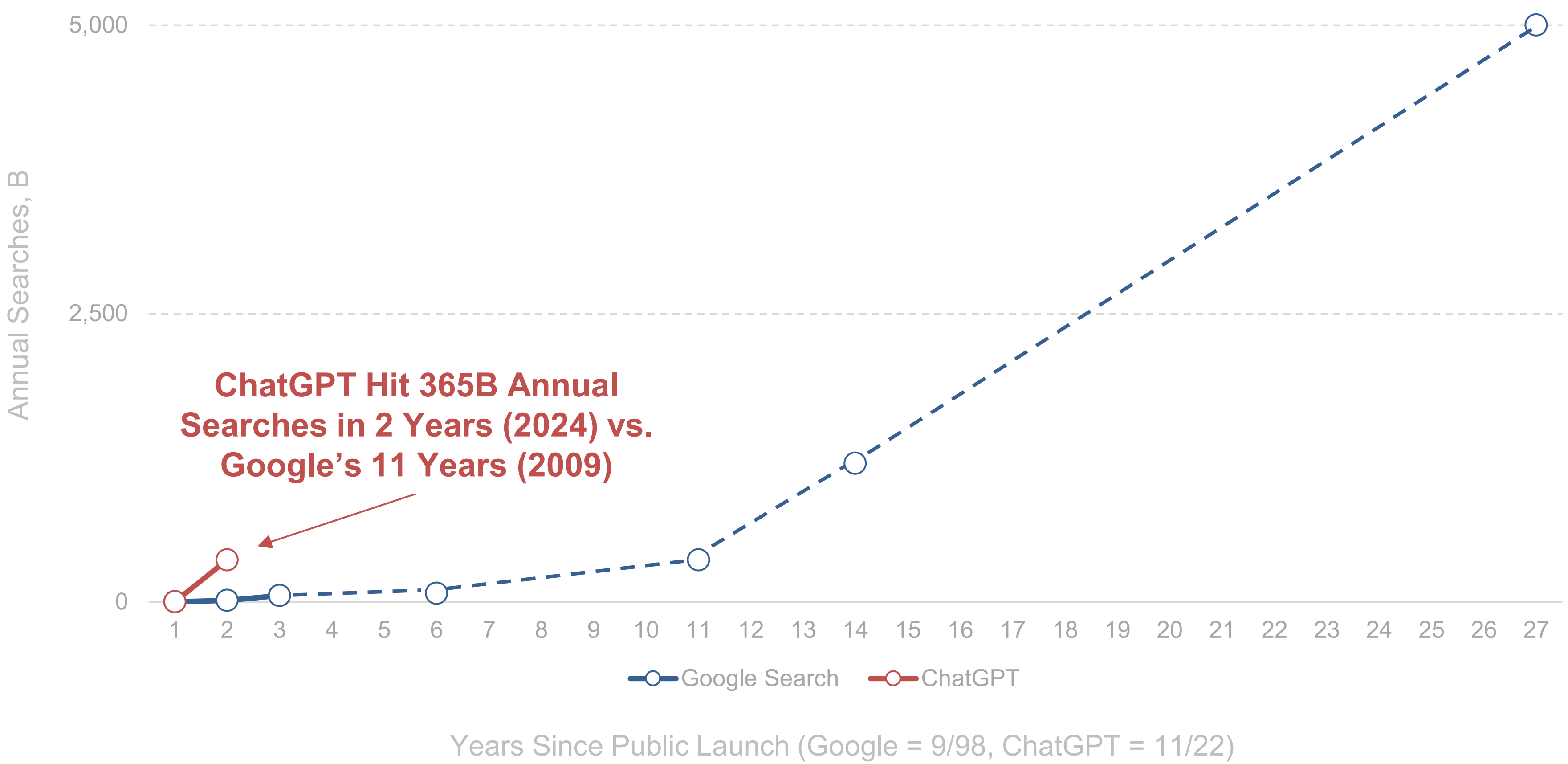
Revenue ($B)



Note: 4/25 user count estimate from OpenAI CEO Sam Altman's 4/11/25 TED Talk disclosure. Revenue figures are estimates based off OpenAI disclosures. Source: OpenAI disclosures (as of 4/25), The Information (4/25) (link, link, link & link)

Time to 365B Annual Searches = ChatGPT 5.5x Faster vs. Google

Annual Searches by Year (B) Since Public Launches of Google & ChatGPT – 1998-2025, per Google & OpenAI



Note: Dashed-line bars are for years where Google did not disclose annual search volumes. Source: Google public disclosures, OpenAI (12/24). ChatGPT figures are estimates per company disclosures of ~1B daily queries

In 1998, tapping emerging Internet access, Google set out to ‘organize the world’s information and make it universally accessible and useful.’

Nearly three decades later
– after some of the fastest change humankind has seen –
a lot of information is indeed digitized / accessible / useful.

The AI-driven evolution of how we access and move information is happening much faster…

…AI is a compounder – on internet infrastructure, which allows for wicked-fast adoption of easy-to-use broad-interest services.

Knowledge Distribution Evolution =

Over ~Six Centuries

Knowledge Distribution – 1440-1992 = Static + Physical Delivery…

Printing Press – Invented 1440

Source: Wikimedia Commons

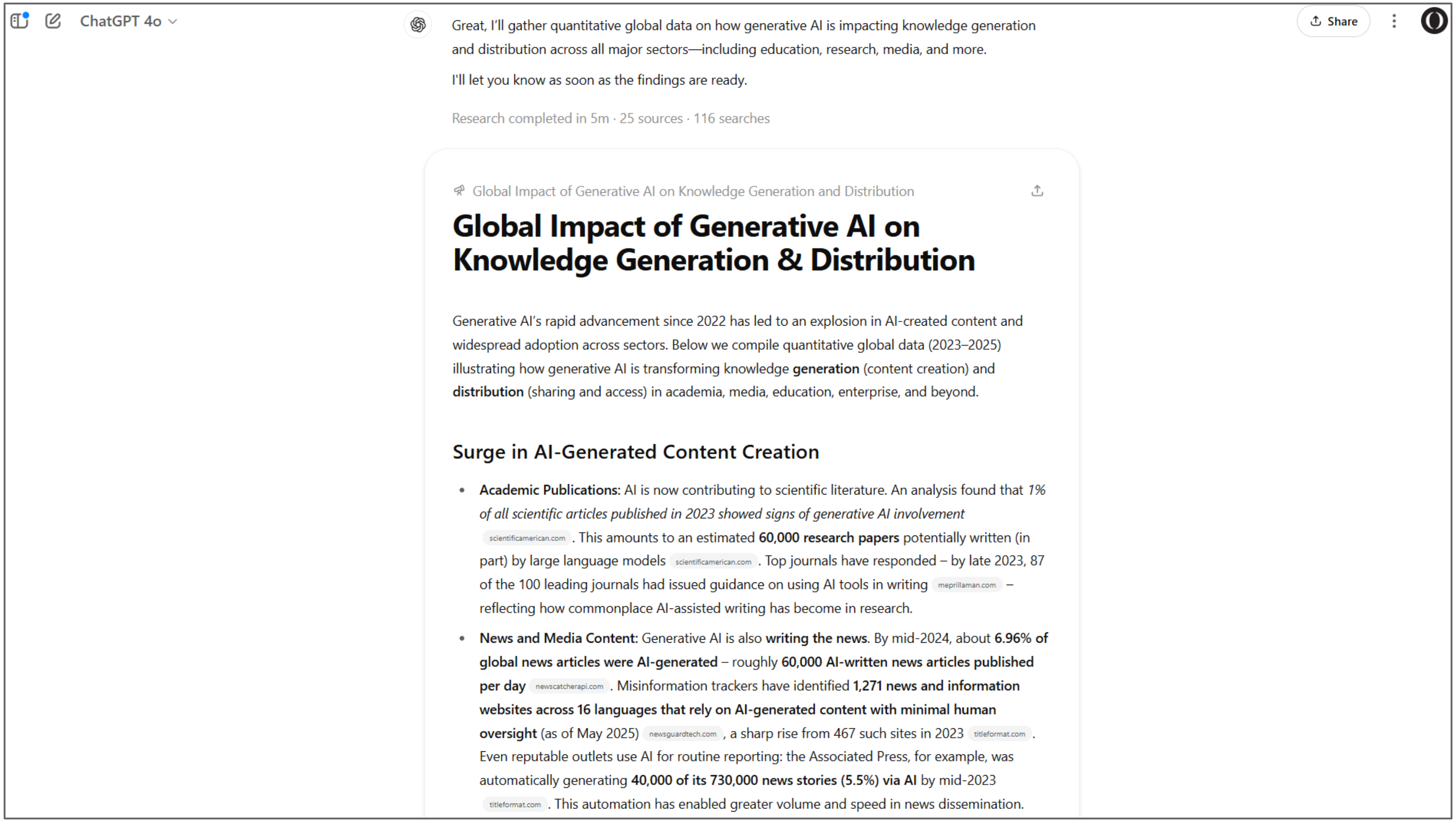
…Knowledge Distribution – 1993-2021 = Active + Digital Delivery…

Internet – Public Release 1993*

*The internet is widely agreed to have been 'publicly released' in 1993 with release of the World Wide Web (WWW) into the public domain, which allowed users to create websites; however, Tim Berners-Lee invented the World Wide Web in 1989, per CERN.
Source: Google, USA Department of Defense, CERN

…Knowledge Distribution – 2022+ = Active + Digital + Generative Delivery

Generative AI – Public Launch of ChatGPT 2022*



ChatGPT 4o

Share

Great, I'll gather quantitative global data on how generative AI is impacting knowledge generation and distribution across all major sectors—including education, research, media, and more.

I'll let you know as soon as the findings are ready.

Research completed in 5m · 25 sources · 116 searches

Global Impact of Generative AI on Knowledge Generation and Distribution

Global Impact of Generative AI on Knowledge Generation & Distribution

Generative AI's rapid advancement since 2022 has led to an explosion in AI-created content and widespread adoption across sectors. Below we compile quantitative global data (2023–2025) illustrating how generative AI is transforming knowledge **generation** (content creation) and **distribution** (sharing and access) in academia, media, education, enterprise, and beyond.

Surge in AI-Generated Content Creation

- **Academic Publications:** AI is now contributing to scientific literature. An analysis found that *1% of all scientific articles published in 2023 showed signs of generative AI involvement* scientificamerican.com. This amounts to an estimated **60,000 research papers** potentially written (in part) by large language models scientificamerican.com. Top journals have responded – by late 2023, 87 of the 100 leading journals had issued guidance on using AI tools in writing meprillaman.com – reflecting how commonplace AI-assisted writing has become in research.
- **News and Media Content:** Generative AI is also **writing the news**. By mid-2024, about **6.96% of global news articles were AI-generated** – roughly **60,000 AI-written news articles published per day** newscatcherapi.com. Misinformation trackers have identified **1,271 news and information websites across 16 languages that rely on AI-generated content with minimal human oversight** (as of May 2025) newsguardtech.com, a sharp rise from 467 such sites in 2023 titleformat.com. Even reputable outlets use AI for routine reporting: the Associated Press, for example, was automatically generating **40,000 of its 730,000 news stories (5.5%) via AI** by mid-2023 titleformat.com. This automation has enabled greater volume and speed in news dissemination.

*We define the public launch of ChatGPT in November 2022 as the public release of Generative AI which we see as AI's 'iPhone Moment.' ChatGPT saw the fastest user ramp ever for a standalone product (5 days to secure 1MM users). Generative AI = AI that can create content – text, images, audio, or code – based on learned patterns.
Source: OpenAI

*Knowledge is a process of piling up facts;
wisdom lies in their simplification.*

Martin H. Fischer, German-born American Physician / Teacher / Author (1879-1962)

AI =

Many Years Before Lift-Off

AI Milestone Timeline – 1950-2022, per Stanford University…

10/50: Alan Turing creates his Turing Test to measure computer intelligence, positing that computers could think like humans

6/56: Stanford computer scientist John McCarthy convenes the Dartmouth Conference on 'Artificial Intelligence,' a term he coined

1/62: Arthur Samuel, an IBM computer scientist, creates a self-learning program that proves capable of defeating a top USA checkers champion

1/66: Stanford researchers deploy Shakey, the first general-purpose mobile robot that can reason about its own actions

AI 'Winter[1]' (1967-1996)

5/97: Deep Blue, IBM's chess-playing computer, defeats Garry Kasparov, the world chess champion at the time

9/02: Roomba, the first mass-produced autonomous robotic vacuum cleaner that can navigate homes, is launched

10/05: A Stanford team build a driverless car named Stanley; it completes a 132-mile course, winning the DARPA Grand Challenge

4/10: Apple acquires Siri voice assistant & integrates it into iPhone 4S model one year later

6/14: Eugene Goostman, a chatbot, passes the Turing Test, with 1/3 of judges believing that Eugene is human

6/18: OpenAI releases GPT-1, the first of their large language models

6/20: OpenAI releases GPT-3, an AI tool for automated conversations; Microsoft exclusively licenses the model

11/22: OpenAI releases ChatGPT to the public

1: AI 'Winter' was a term used by Nils J. Nilsson, the Kumagai Professor of Engineering in computer science at Stanford University, to describe the period during which AI continued to make conceptual progress but could boast no significant practical successes. This subsequently led to a drop in AI interest and funding. Includes data from sources beyond Stanford.
Source: Stanford University & Stanford Law School sources, iRobot, TechCrunch, BBC, OpenAI. Data aggregated by BOND.

…AI Milestone Timeline – 2023-2025, per Stanford University

3/23: OpenAI releases GPT-4, a multimodal* model capable of processing both text & images

3/23: Microsoft Integrates Copilot into its 365 product suite

3/23: Google releases Bard, its ChatGPT competitor

3/23: Anthropic releases Claude, its AI assistant focused on safety & inter-pretability

11/23: 28 countries, including USA, EU members & China, sign Bletchley Declaration on AI Safety

3/24: USA Department of Homeland Security unveils its AI Roadmap Strategy

4/24: Meta Platforms releases its open-source** Llama 3 model with 70B parameters

5/24: OpenAI releases GPT-4o, which has full multimodality across audio, visual, & text inputs

5/24: Google introduces AI overviews to augment its search functions

7/24: Apple releases Apple Intelligence, an AI system integrated into its devices, for developers

9/24: Alibaba releases 100 open-source Qwen 2.5 models, with performance in line with Western competitors

12/24: OpenAI announces o3, its highest-ever performing model

1/25: DeepSeek releases its R1 & R1-Zero open-source reasoning models

1/25: Alibaba unveils Qwen2.5-Max, which surpasses the performance of other leading models (GPT-4o, Claude 3.5) on some reasoning tests

2/25: OpenAI releases GPT-4.5, Anthropic releases Claude 3.7 Sonnet, & xAI releases Grok 3

4/25: ChatGPT reaches 800MM weekly users[1]

*Multimodal = AI that can understand and process multiple data types (e.g., text, images, audio) together.
**Open-source = AI models and tools made publicly available for use, modification, and redistribution.
1) 4/25 estimate from OpenAI CEO Sam Altman's 4/11/25 TED Talk disclosure.
Source: Aggregated by BOND from OpenAI, Microsoft, Google, Anthropic, Meta, Apple, Alibaba, Deepseek, UK Government, US Department of Homeland Security. China data may be subject to informational limitations due to government restrictions.

AI =

Circa Q2:25

Top Ten Things AI Can Do Today, per ChatGPT

ChatGPT 4o

1. **Write or edit anything**
   - Emails, essays, contracts, poems, code — instantly and fluently.
2. **Summarize and explain complex material**
   - Simplifies PDFs, legal docs, research, or code into plain English.
3. **Tutor you on nearly any subject**
   - Learn math, history, languages, or test prep, step by step.
4. **Be your thinking partner**
   - Brainstorm ideas, debug logic, or pressure-test assumptions.
5. **Automate repetitive work**
   - Generate reports, clean data, outline slides, rewrite text.
6. **Roleplay anyone you need**
   - Prep for interviews, simulate customers, rehearse conversations.
7. **Connect you to tools**
   - Write code for APIs, spreadsheets, calendars, or the web.
8. **Offer therapy and companionship**
   - Talk through your day, reframe thoughts, or just listen.
9. **Help you find purpose**
   - Clarify values, define goals, and map actions that matter.
10. **Organize your life**
    - Plan trips, build routines, structure your week or workflow.

Source: ChatGPT (5/15/25)

AI =

Circa 2030?

Top Ten Things AI Will Likely Do in Five Years, per ChatGPT

ChatGPT 4o

1. Generate human-level text, code & logic
➤ Chatbots, software engineering, business plans, legal analysis.

2. Create full-length films & games
➤ Scripts, characters, scenes, gameplay mechanics, voice acting.

3. Understand & speak like a human
➤ Emotionally aware assistants, real-time multilingual voice agents.

4. Power advanced personal assistants
➤ Life planning, memory recall, coordination across all apps and devices.

5. Operate humanlike robots
➤ Household helpers, elderly care, retail and hospitality automation.

6. Run autonomous customer service & sales
➤ End-to-end resolution, upselling, CRM integrations, 24/7 support.

7. Personalize entire digital lives
➤ Adaptive learning, dynamic content curation, individualized health coaching.

8. Build and run autonomous businesses
➤ AI-driven startups, inventory and pricing optimization, full digital operations.

9. Drive autonomous discovery in science
➤ Drug design, materials synthesis, climate modeling, novel hypothesis testing.

10. Collaborate creatively like a partner
➤ Co-writing novels, music production, fashion design, architecture.

Source: ChatGPT (5/15/25)

AI =

Circa 2035?

Top Ten Things AI Will Likely Do in Ten Years, per ChatGPT

ChatGPT 4o

1. Conduct scientific research

➤ Generate hypotheses, run simulations, design and analyze experiments.

2. Design advanced technologies

➤ Discover materials, engineer biotech, and prototype energy systems.

3. Simulate human-like minds

➤ Create digital personas with memory, emotion, and adaptive behavior.

4. Operate autonomous companies

➤ Manage R&D, finance, and logistics with minimal human input.

5. Perform complex physical tasks

➤ Handle tools, assemble components, and adapt in real-world spaces.

6. Coordinate systems globally

➤ Optimize logistics, energy use, and crisis response at scale.

7. Model full biological systems

➤ Simulate cells, genes, and organisms for research and therapy.

8. Offer expert-level decisions

➤ Deliver real-time legal, medical, and business advice.

9. Shape public debate and policy

➤ Moderate forums, propose laws, and balance competing interests.

10. Build immersive virtual worlds

➤ Generate interactive 3D environments directly from text prompts.

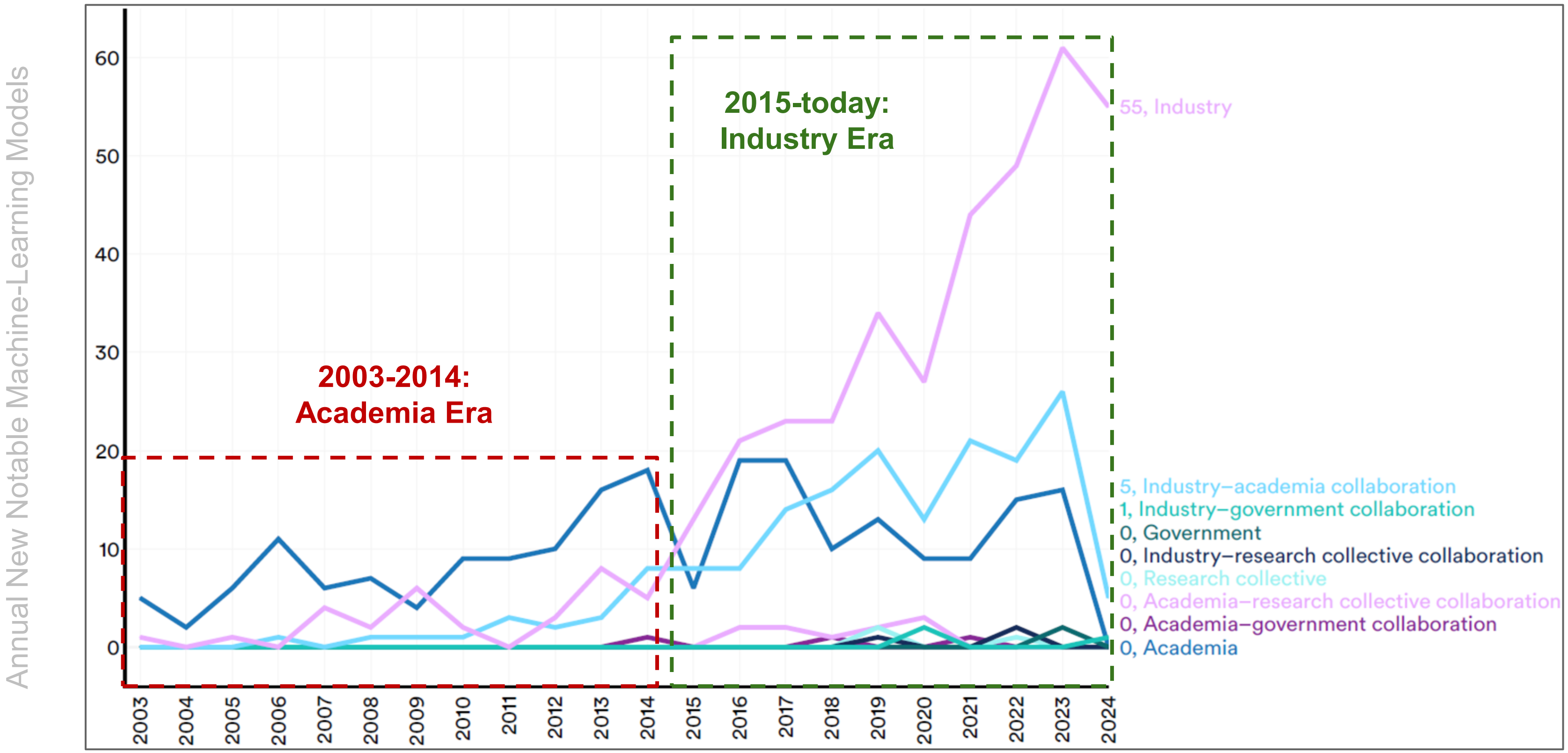
Source: ChatGPT (5/15/25)

AI Development Trending =

Unprecedented

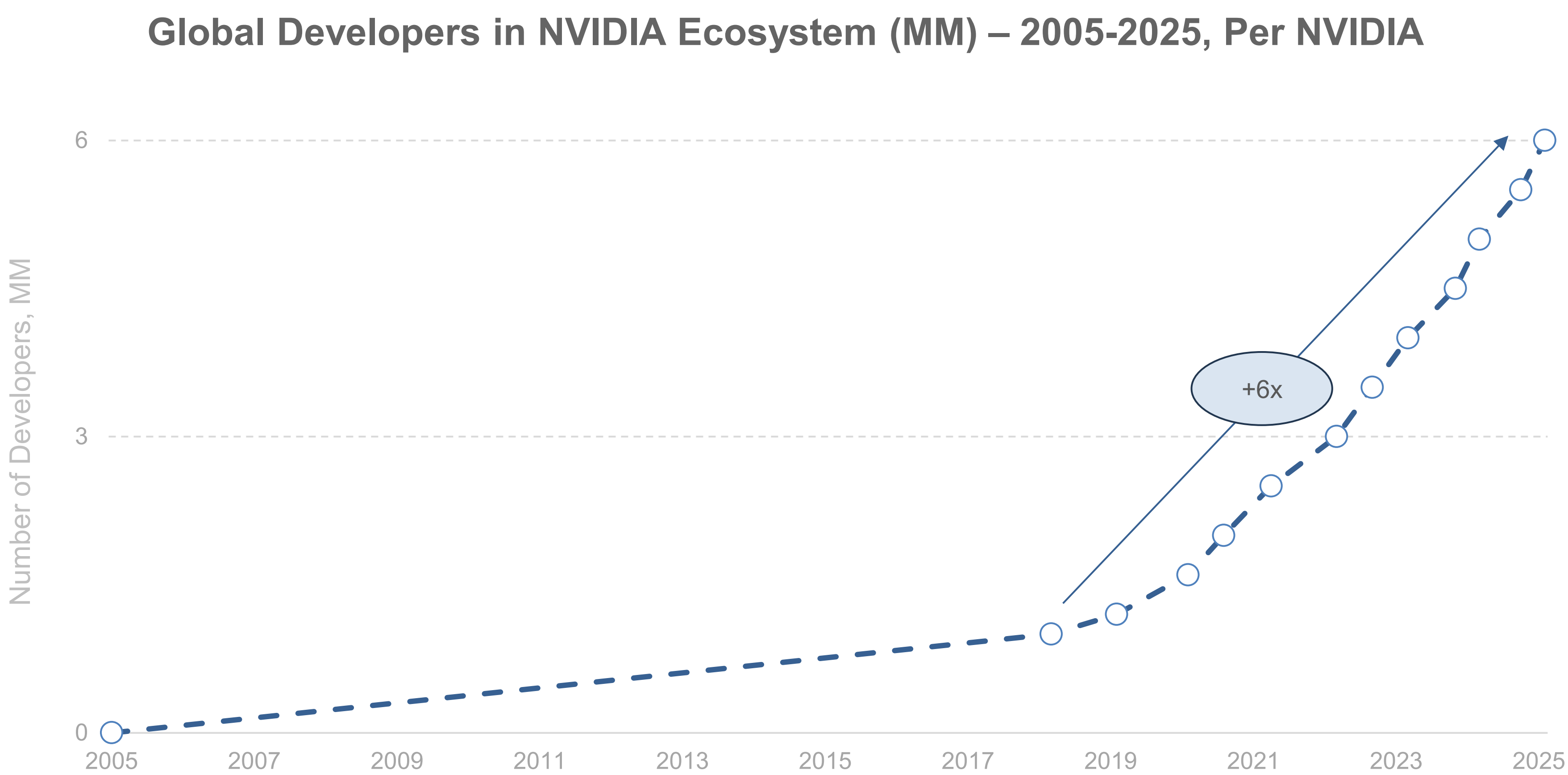
Machine-Learning Model* Trending = In 2015...
Industry Surpassed Academia as Data + Compute + Financial Needs Rose

Global Notable Machine Learning Models by Sector – 2003-2024, per Stanford HAI



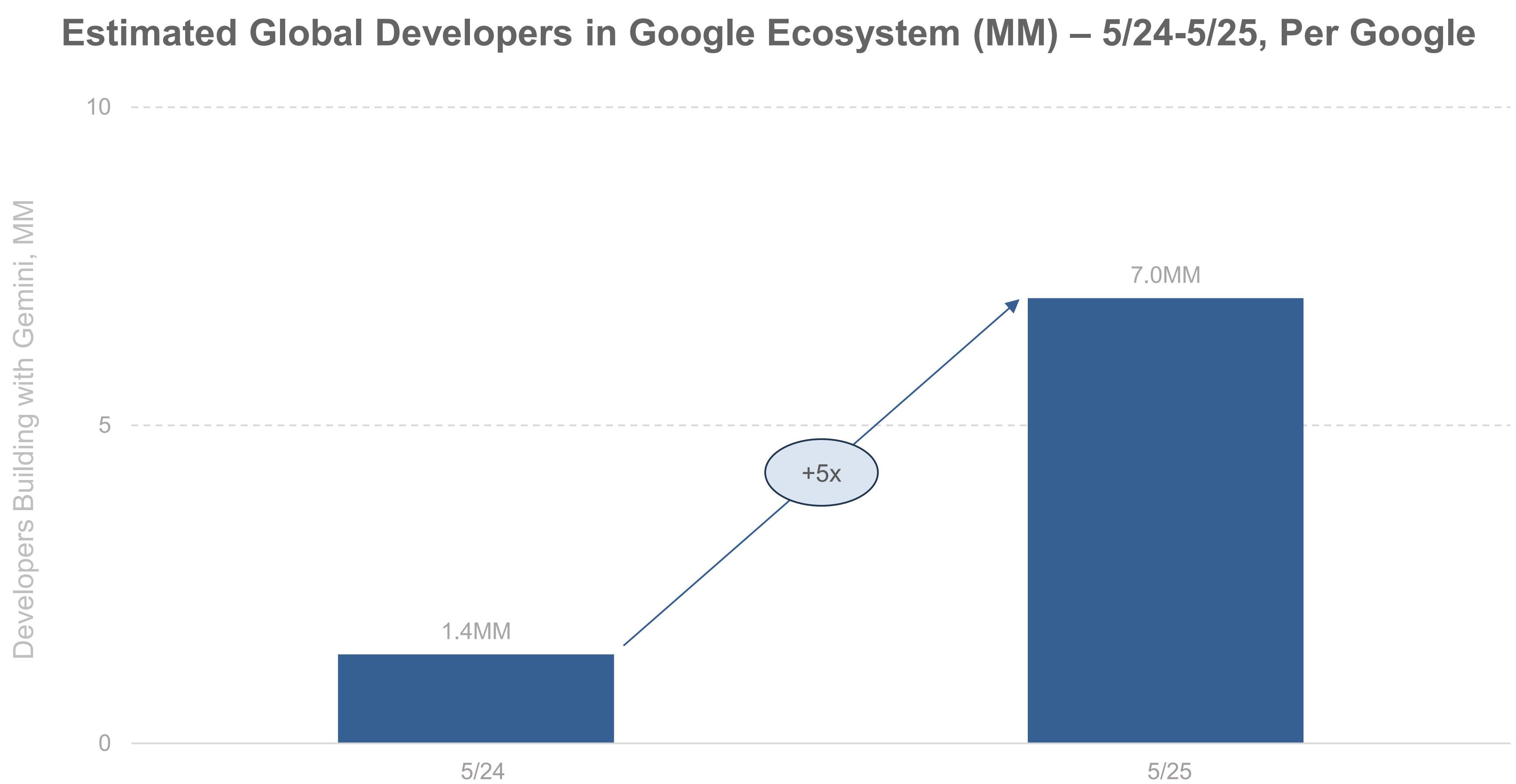
*Machine Learning = A subset of AI where machines learn from patterns in data without being explicitly programmed.
Note: Academia includes models developed by one or more institutions, including government agencies. Industry-academia collaboration excludes government partnerships and only captures partnerships between academic institutions and industry. Industry excludes models developed in partnership with any entity other than another company. Epoch AI, an AI Index data provider, uses the term 'notable machine learning models' to designate particularly influential models within the AI/machine learning ecosystem. Epoch maintains a database of 900 AI models released since the 1950s, selecting entries based on criteria such as state-of-the-art advancements, historical significance, or high citation rates. Since Epoch manually curates the data, some models considered notable by some may not be included. A count of zero academic models does not mean that no notable models were produced by academic institutions in 2023, but rather that Epoch AI has not identified any as notable. Additionally, academic publications often take longer to gain recognition, as highly cited papers introducing significant architectures may take years to achieve prominence. China data may be subject to informational limitations due to government restrictions. Source: Nestor Maslej et al., 'The AI Index 2025 Annual Report,' AI Index Steering Committee, Stanford HAI (4/25)

AI Developer Growth (NVIDIA Ecosystem as Proxy) = +6x to 6MM Developers Over Seven Years



Note: We assume negligible developers in NVIDIA's ecosystem in 2005 per this text from an 8/20 blog post titled '2 Million Registered Developers, Countless Breakthroughs': 'It took 13 years to reach 1 million registered developers, and less than two more to reach 2 million.' Source: NVIDIA blog posts, press releases, & company overviews

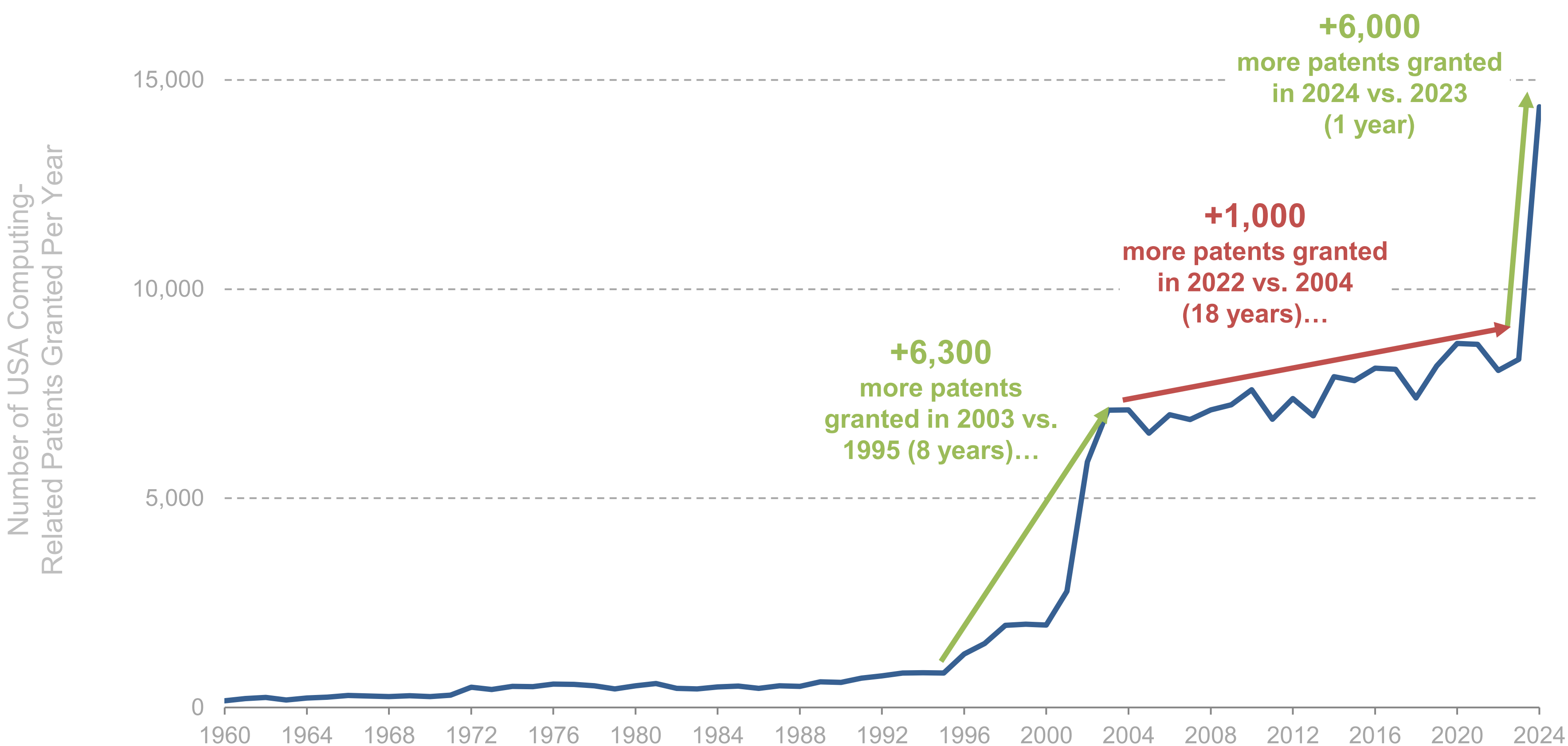
AI Developer Growth (Google Ecosystem as Proxy) = +5x to 7MM Developers Y/Y



Note: Per Google in 5/25, 'Over 7 million developers are building with Gemini, five times more than this time last year.' Source: Google, 'Google I/O 2025: From research to reality' (5/25)

Computing-Related Patent Grants, USA = Exploded…
Post-Netscape IPO (1995)...Again + Faster Post-ChatGPT Public Launch (2022)

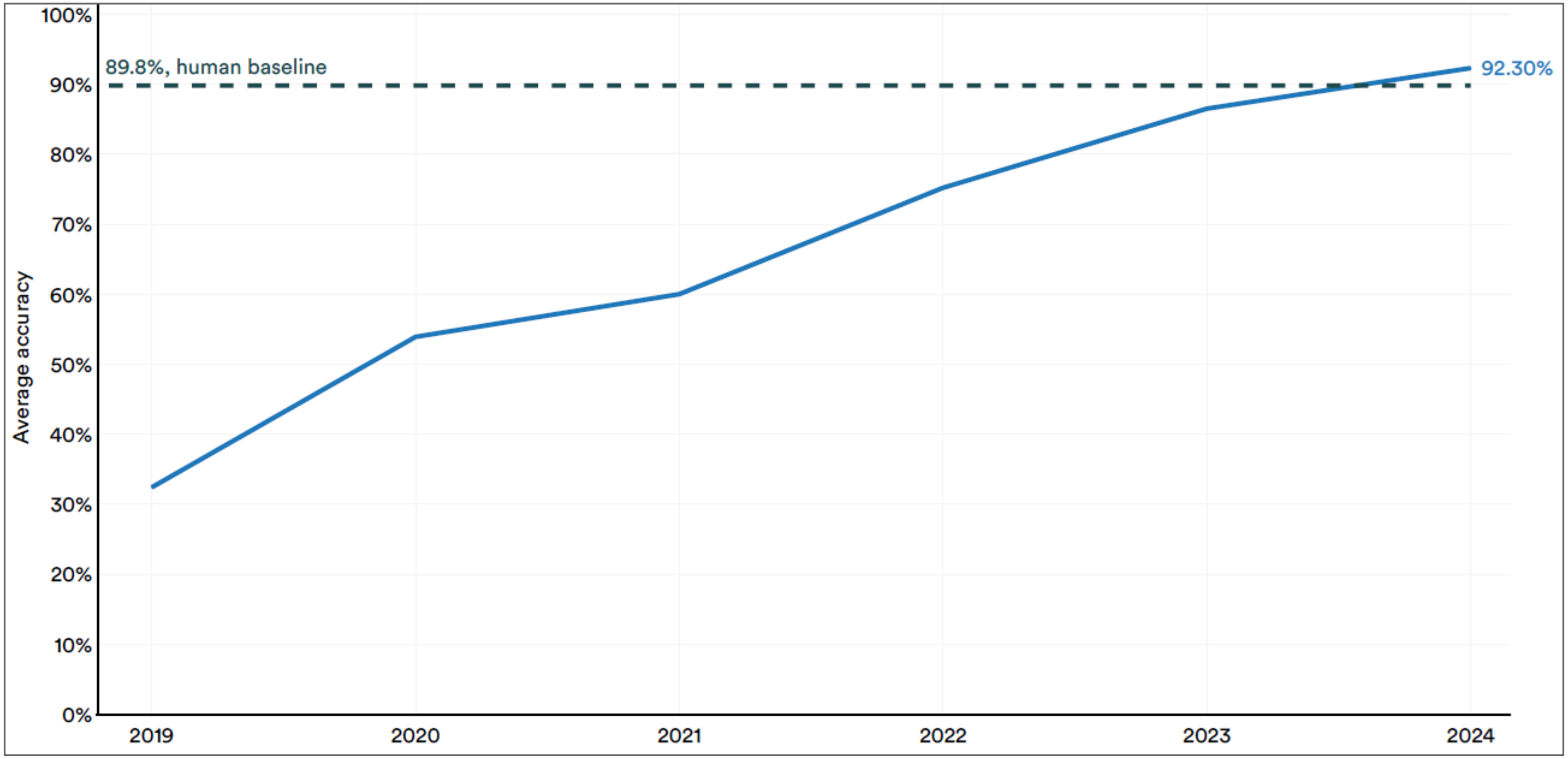
USA Computing-Related* Patents Granted Annually – 1960-2024, per USPTO



*Uses Cooperative Patent Classification (CPC) code G06, which corresponds to computing, calculating or counting patents. Google patents data changes somewhat between each query so numbers are rounded and should be viewed as directionally accurate. Source: USA Patent & Trademark Office (USPTO) via Google Patents (4/25)

AI Performance = In 2024…
Surpassed Human Levels of Accuracy & Realism, per Stanford HAI

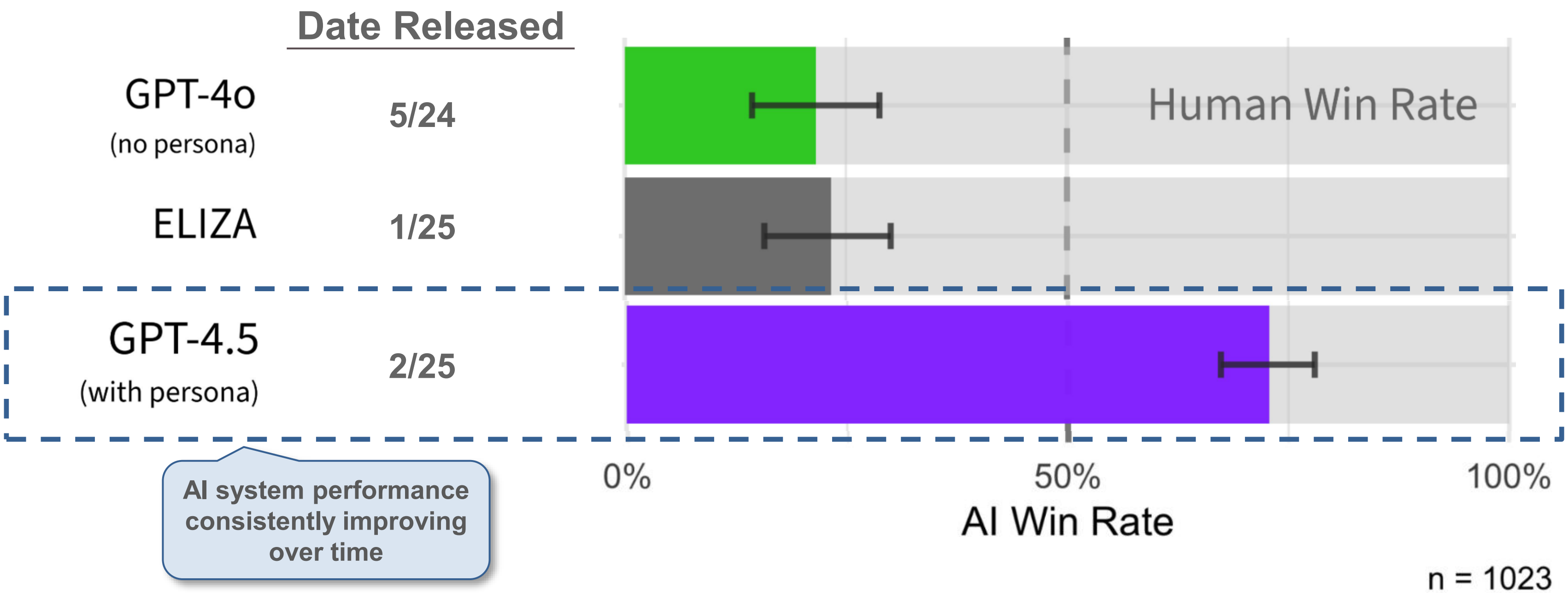
AI System Performance on MMLU Benchmark Test – 2019-2024, per Stanford HAI



Note: The MMLU (Massive Multitask Language Understanding) benchmark evaluates a language model's performance across 57 academic and professional subjects, such as math, law, medicine, and history. It measures both factual recall and reasoning ability, making it a standard for assessing general knowledge and problem-solving in large language models. 89.8% is the generally-accepted benchmark for human performance. Stats above show average accuracy of top-performing AI models in each calendar year. Source: Papers With Code via Nestor Maslej et al., 'The AI Index 2025 Annual Report,' AI Index Steering Committee, Stanford HAI (4/25)

AI Performance = In Q1:25…
73% of Responses & Rising Mistaken as Human by Testers

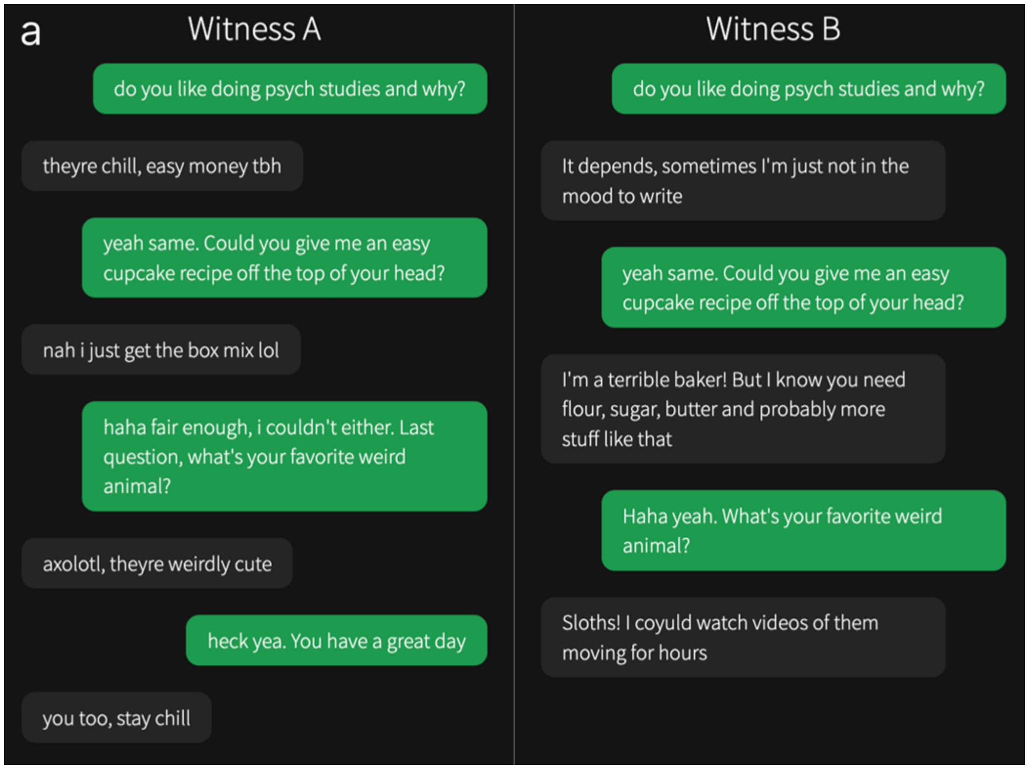
% of Testers Who Mistake AI Responses as Human-Generated – 3/25, per Cameron Jones / Benjamin Bergen



Note: The Turing test, introduced in 1950, measures a machine's ability to mimic human conversation. In this study, ~500 participants engaged in a three-party test format, interacting with both a human and an AI. Most discussions leaned on emotional resonance and day-to-day topics over factual knowledge. Eliza was developed in the mid-1960s by MIT professor Joseph Weizenbaum, It is considered the world's first chatbot. In January 2025, researchers successfully revived Eliza using its original code. Source: Cameron Jones and Benjamin Bergen, 'Large Language Models Pass the Turing Test' (3/25) via UC San Diego

AI Performance = Increasingly Realistic Conversations Simulating Human Behaviors

Turing Test Conversation with GPT-4.5 – 3/25, per Cameron Jones / Benjamin Bergen



<u>What Was Tested:</u>

The Turing Test is a concept introduced by Alan Turing in 1950 to evaluate a machine's ability to exhibit intelligent behavior indistinguishable from that of a human. In the test, if a human evaluator cannot reliably tell whether responses are coming from a human or a machine during a conversation, the machine is said to have passed. Here, participants had to guess whether Witness A or Witness B was an AI system.

<u>Results:</u>

The conversation on the left is an example Turing Test carried out in 3/25 using GPT-4.5. During the test, participants ***incorrectly*** identified the left image (Witness A) as human with 87% certainty, saying 'A had human vibes. B had human imitation vibes.' **A was actually AI-generated; B was human.**

Source: Cameron Jones and Benjamin Bergen, 'Large Language Models Pass the Turing Test' (3/25) via UC San Diego

AI Performance =
Increasingly Realistic Image Generation…

AI-Generated Image: 'Women's Necklace with a Sunflower Pendant' – 2/22-4/25, per Midjourney / Gold Penguin

Model v1 (2/22)

Model v7 (4/25)

Notes: Dates shown are the release dates of each Midjourney model. Source: Midjourney (4/25) & Gold Penguin, 'How Midjourney Evolved Over Time (Comparing V1 to V6.1 Outputs)' (9/24)

…AI Performance =
Increasingly Realistic Image Generation

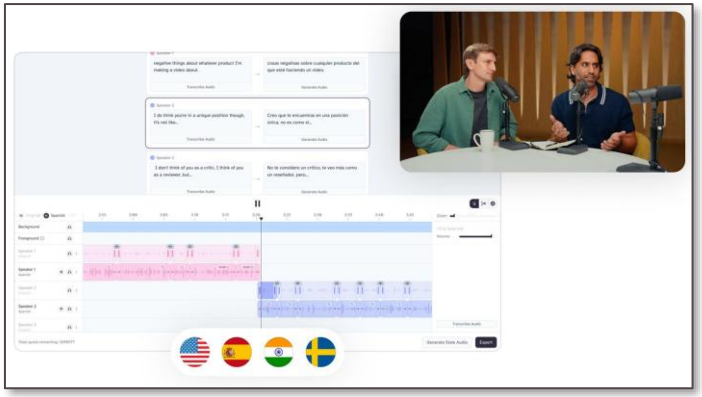
AI-Generated vs. Real Image – 2024

AI-Generated Image (2024)	**Real Image**
	

Source: Left – StyleGAN2 via 'The New York Times,' 'Test Yourself: Which Faces Were Made by A.I.?' (1/24); Right – Creative Commons

AI Performance =
Increasingly Realistic Audio Translation / Generation…

ElevenLabs AI Voice Generator – 1/23-4/25, per ElevenLabs & Similarweb



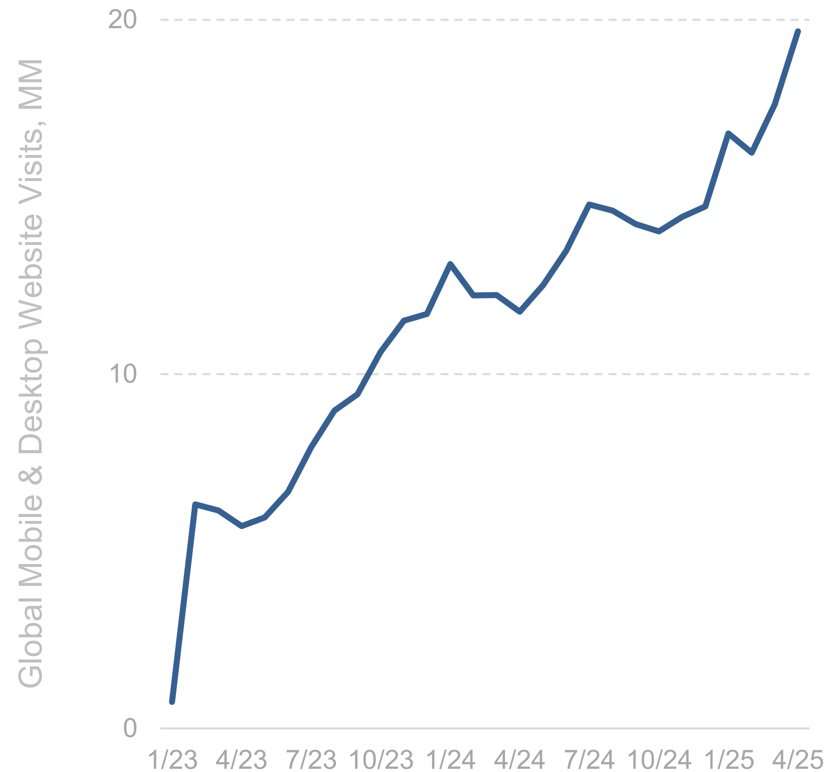
When you create a new dubbing project, Dubbing Studio automatically transcribes your content, translates it into the new language, and generates a new audio track in that language. Each speaker's original voice is isolated and cloned before generating the translation to make sure they sound the same in every language.

- ElevenLabs Press Release, 1/24

In just two years, ElevenLabs' millions of users have generated 1,000 years of audio content and the company's tools have been adopted by employees at over 60% of Fortune 500 companies.

- ElevenLabs Press Release, 1/25

ElevenLabs Monthly Global Site Visits (MM), per Similarweb – 1/23-4/25



Note: China data may be subject to informational limitations due to government restrictions.
Source: ElevenLabs (1/24 & 1/25), Similarweb (5/25)

…AI Performance =
Evolving to Mainstream Realistic Audio Translation / Generation

AI-Powered Audio Translation – 5/25, per Spotify

2/25:

Spotify begins accepting audiobooks AI-translated into 29 languages from ElevenLabs

Imagine if you're a creator and you're the world expert at something…but you happen to be Indonesian. Today, there's a language barrier and it will be very hard if you don't know English to be able to get to a world stage. But with AI, it might be possible in the future where you speak in your native language, and the AI will understand it and will actually real-time translate…

…What will that do for creativity? For knowledge sharing? For entertainment? I think we're in the very early innings of figuring that out…

…We want Spotify to be the place for all voices.

- Spotify Co-Founder & CEO Daniel Ek (5/25)

In Q1:25, Spotify had 678MM Monthly Active Users and 268MM Subscribers and supported €16.8B in annualized revenue while hosting 100MM+ tracks, ~7MM podcast titles and ~1MM creative artists.

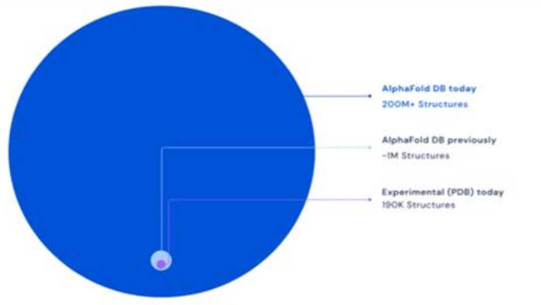
Note: Revenue annualized using Q1:25 results. Source: Spotify, 'The New York Post,' 'Inside Spotify: CEO Daniel Ek on AI, Free Speech & the Future of Music' (5/2/25); Spotify earnings releases; eMarketer, 'Spotify dominates Apple and Amazon in digital audio' (4/25)

AI Performance =
Emerging Applications Accelerating

Emerging AI Applications – 11/24, per Morgan Stanley

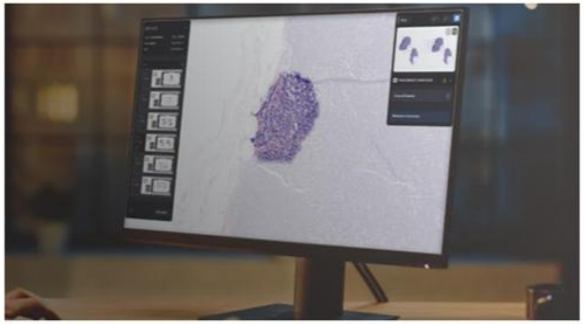
Protein Folding

DeepMind's AlphaFold can predict the structure of nearly all known proteins



Cancer Detection

Microsoft & Paige are building the world's largest image-based AI model to Fight Cancer

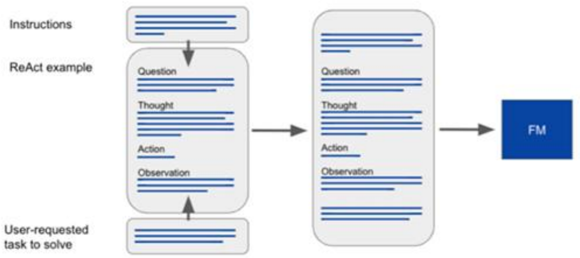


Robotics

Google demonstrates that robots can understand and act on human instruction using LLMs

Agentic AI

Amazon announced tools that enable models to complete tasks based on user instructions



Universal Translation

Meta unveiled the first all-in-one multilingual multimodal AI translation and transcription model

Digital Video Creation

Channel 1 AI showcases ability to use GenAI to produce personalized newscasts

Source: Morgan Stanley, 'GenAI: Where are We Seeing Adoption and What Matters for '25?' (11/24)

AI =

Benefits & Risks

AI Development = Benefits & Risks

The widely-discussed benefits and risks of AI – top-of-mind for many – generate warranted excitement *and* trepidation, further fueled by uncertainty over the rapid pace of change and intensifying global competition and saber rattling.

The pros Stuart Russell and Peter Norvig went deep on these topics in the Fourth Edition (2020) of their 1,116-page classic 'Artificial Intelligence: A Modern Approach' (link here), and their views still hold true.

Highlights follow…

…the benefits: put simply, our entire civilization is the product of our human intelligence. If we have access to substantially greater machine intelligence, the [ceiling of our] ambitions is raised substantially.

The potential for AI and robotics to free humanity from menial repetitive work and to dramatically increase the production of goods and services could presage an era of peace and plenty. The capacity to accelerate scientific research could result in cures for disease and solutions for climate change and resource shortages.

As Demis Hassabis, CEO of Google DeepMind, has suggested: 'First we solve AI, then use AI to solve everything else.' Long before we have an opportunity to 'solve AI,' however, we will incur risks from the misuse of AI, inadvertent or otherwise.

Some of these are already apparent, while others seem likely based on current trends including lethal autonomous weapons…surveillance and persuasion…biased decision making… impact on employment…safety-critical applications…cybersecurity…

Source: Stuart Russell and Peter Norvig, 'Artificial Intelligence: A Modern Approach'

Success in creating AI could be the biggest event in the history of our civilization. But it could also be the last – unless we learn how to avoid the risks.

Stephen Hawking, Theoretical Physicist / Cosmologist (1942-2018)

Source: University of Cambridge, Centre for the Future of Intelligence

Outline

1. **Seem Like Change Happening Faster Than Ever?**
   Yes, It Is
2. **AI User + Usage + CapEx Growth =**
   Unprecedented
3. **AI Model Compute Costs High / Rising + Inference Costs Per Token Falling =**
   Performance Converging + Developer Usage Rising
4. **AI Usage + Cost + Loss Growth =**
   Unprecedented
5. **AI Monetization Threats =**
   Rising Competition + Open-Source Momentum + China's Rise
6. **AI & Physical World Ramps =**
   Fast + Data-Driven
7. **Global Internet User Ramps Powered by AI from Get-Go =**
   Growth We Have Not Seen Likes of Before
8. **AI & Work Evolution =**
   Real + Rapid

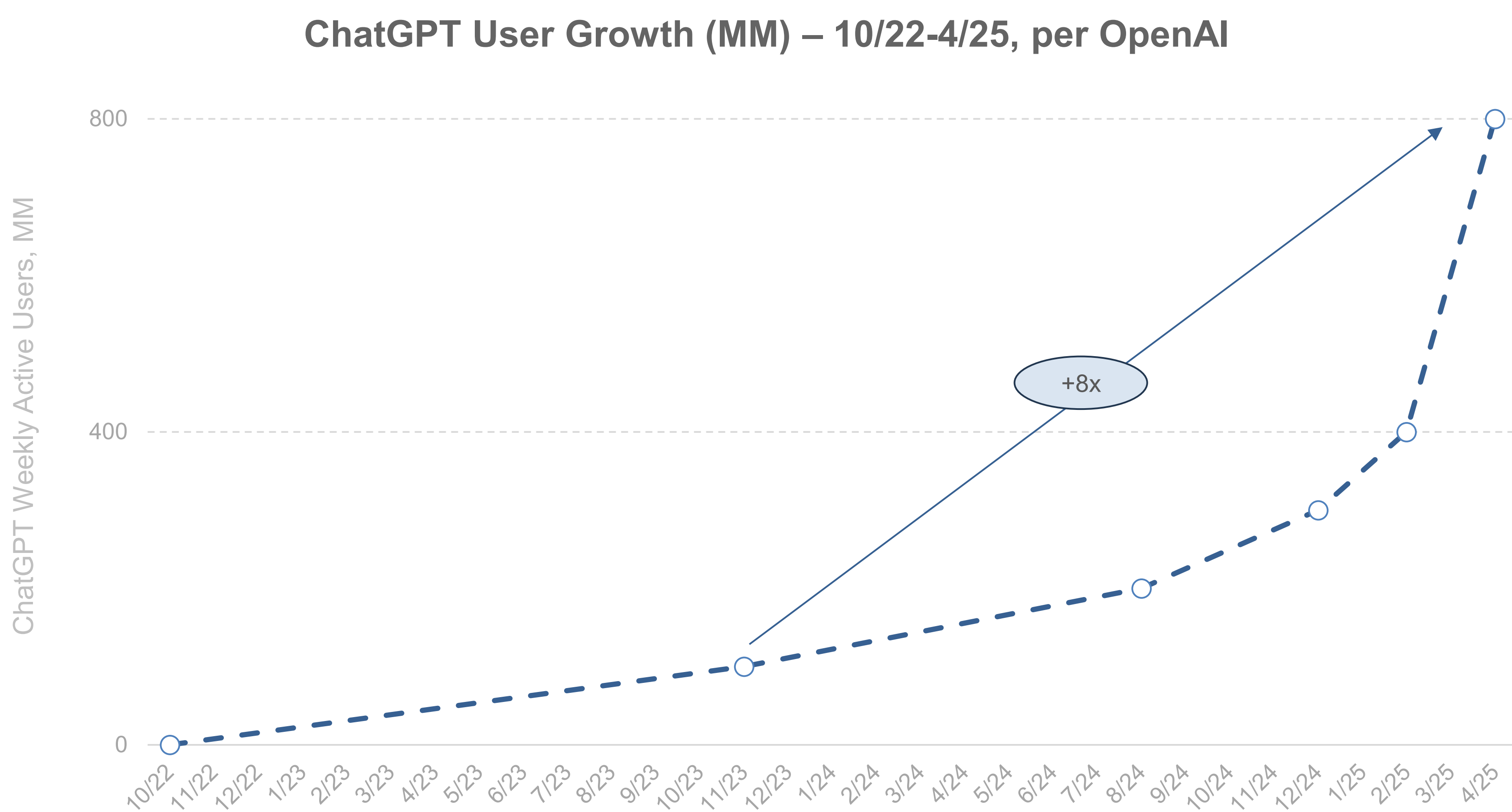
*AI **User** + Usage + CapEx **Growth** =*

Unprecedented

Consumer / User AI Adoption =

Unprecedented

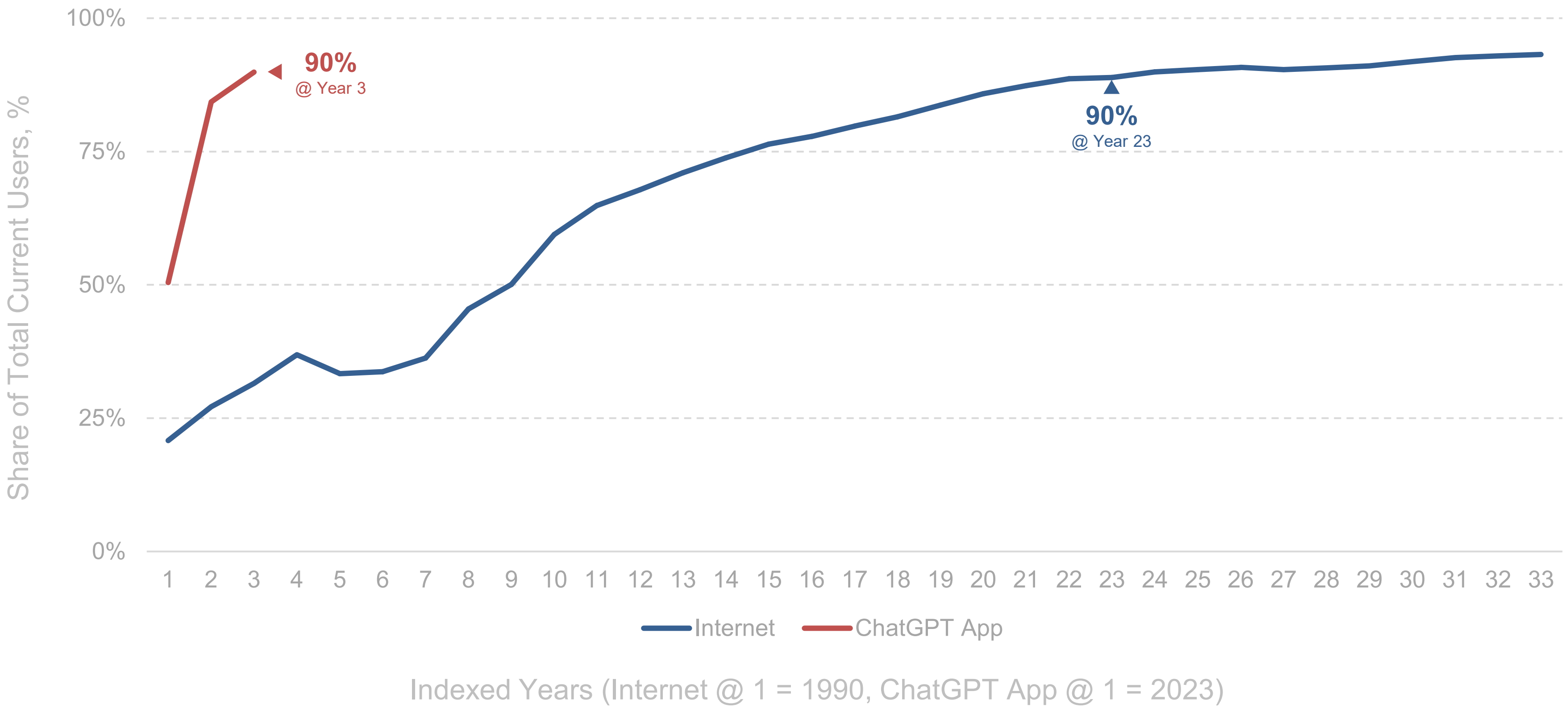
AI User Growth (ChatGPT as Foundational Indicator) = +8x to 800MM in Seventeen Months



Note:OpenAI reports Weekly Active Users which are represented above. 4/25 estimate from OpenAI CEO Sam Altman's 4/11/25 TED Talk disclosure. Source: OpenAI disclosures

AI Global Adoption (ChatGPT as Foundational Indicator) = Have Not Seen Likes of This Around-the-World Spread Before

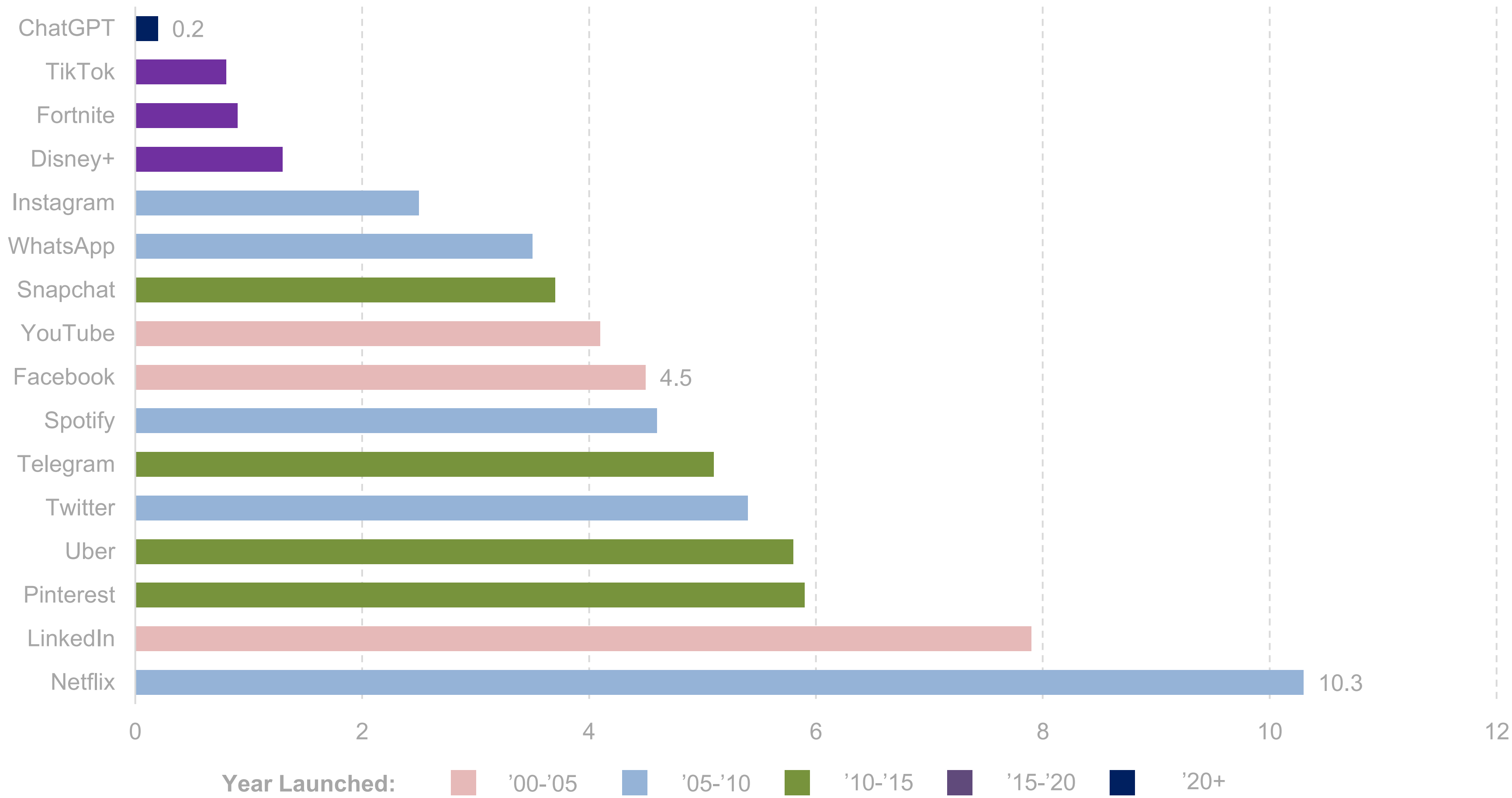
Internet vs. ChatGPT Users – Percent Outside North America (1990-2025), Per ITU & Sensor Tower



Note: Year 1 for Internet = 1990; year 33 = 2022. Year 1 for ChatGPT app = 5/23; year 3 for ChatGPT app = 5/25. ChatGPT app monthly active users (MAUs) shown. Note that ChatGPT is not available in China, Russia and select other countries as of 5/25. China data may be subject to informational limitations due to government restrictions. Includes only Android, iPhone & iPad users. Figures may understate true ChatGPT user base (e.g., desktop or mobile webpage users). Regions per United Nations definitions. Figures show % of total current users in that year – note that as year 3 for ChatGPT has not yet finished, percentages could move in coming months. Data for standalone ChatGPT app only. Country-level data may be missing for select years, as per ITU. Source: United Nations / International Telecommunications Union (3/25), Sensor Tower (5/25)

AI User Adoption (ChatGPT as Proxy) = Materially Faster vs. Internet Comparables…

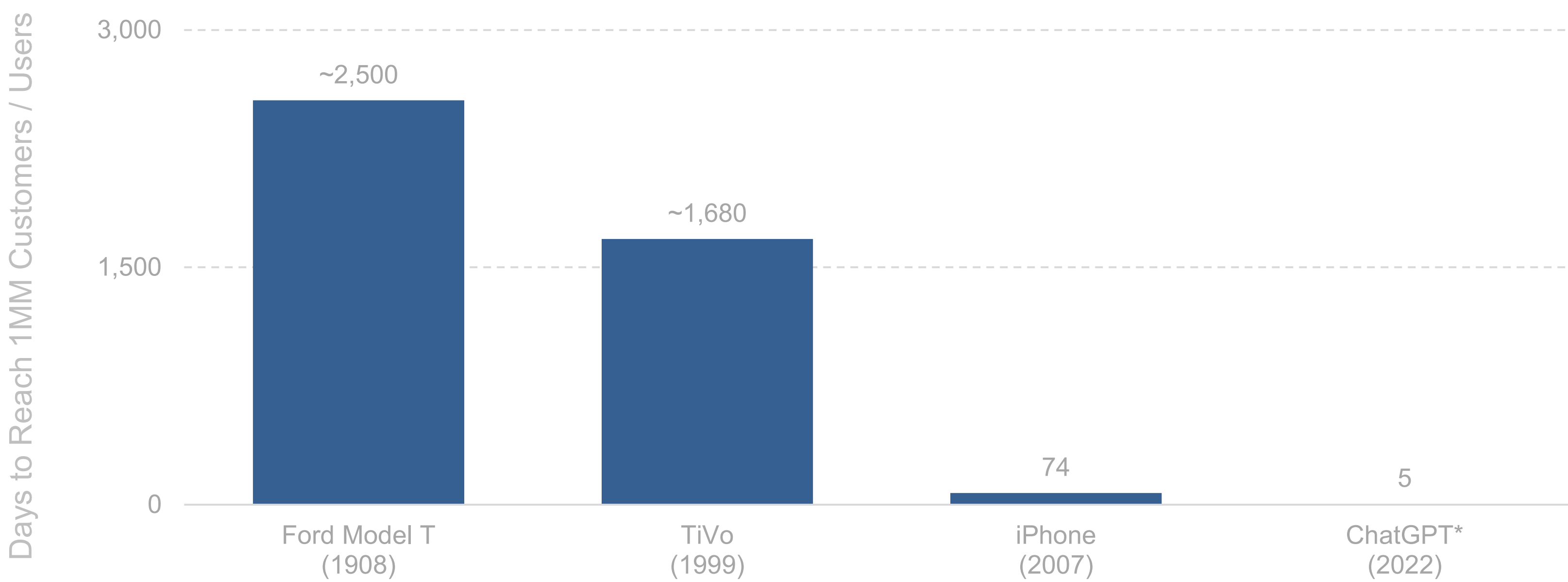
Years to Reach 100MM Users – 2000-2023



Note: Netflix represents streaming business. Source: BOND, 'AI & Universities' (2024) via company filings, press

…AI User Adoption (ChatGPT as Proxy) = Materially Faster + Cheaper vs. Other Foundational Technology Products

Days to Reach 1MM Customers / Users – 1908-2022

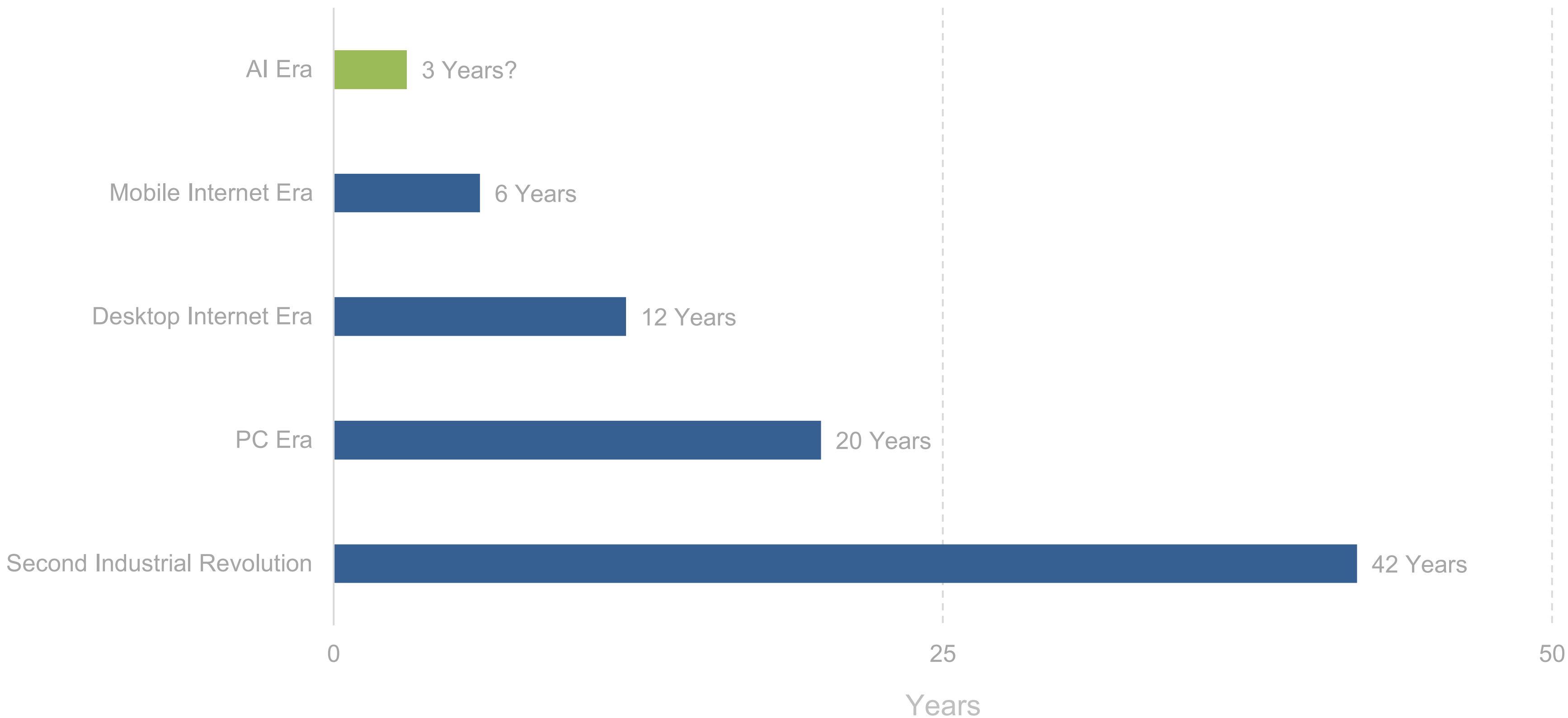


	Ford Model T	TiVo	iPhone	ChatGPT
Purchase Price (2024 $)	**$29,330**	**$945**	**$756**	**$0**

**Public launch of ChatGPT = first release to the public as a free research preview (11/22). Note: Per Ford Corporate, the Model T could be sold for between $260 and $850. We use $850 in 1908 dollars for our figures above. For TiVo, we use the launch of consumer sales on 3/31/99, when TiVo charged $499 for its 14-hour box set. We do not count TiVo subscription costs. We also use the iPhone 1's 4GB entry level price of $499 in 2007. Source: Heartcore Capital, CNBC, Museum of American Speed, World Bank, Ford Corporate, Gizmodo, Apple, Encyclopedia Britannica, Federal Reserve Bank of St. Louis, Wikimedia Commons, UBS*

AI User Adoption – Time to 50% Household Penetration = Each Cycle Ramps in ~Half-the-Time…AI Following Pattern

Years to 50% Adoption of Household Technologies in USA, per Morgan Stanley



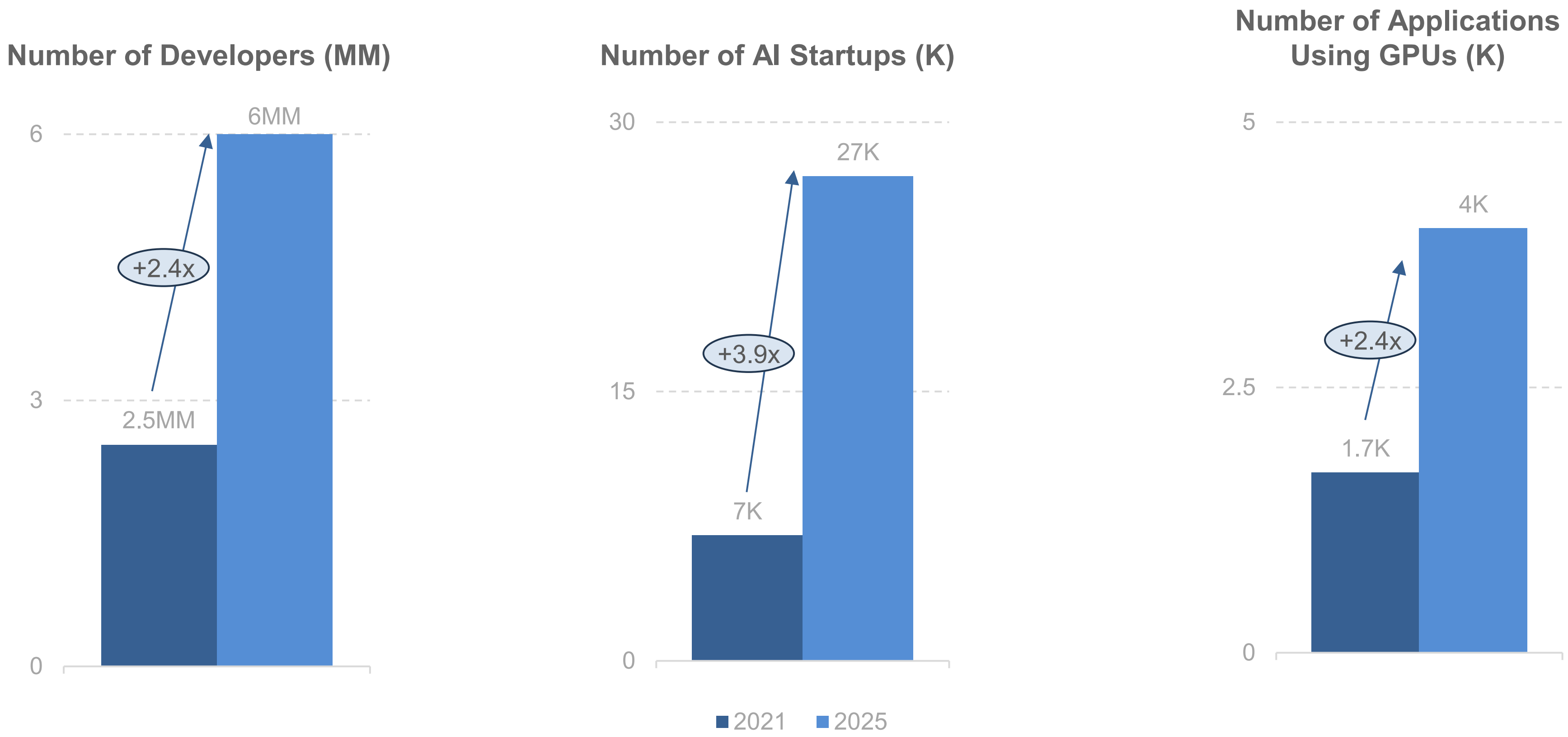
Note: 3 years for AI Era implies that the time to 50% USA Household Adoption is similarly cut in half from the previous cycle. Source: Morgan Stanley, 'Google and Meta: AI vs. Fundamental 2H Debates' (7/23), Our World in Data, other web sources per MS

Technology Ecosystem AI Adoption =

Impressive

NVIDIA AI Ecosystem Tells Over Four Years = >100% Growth in Developers / Startups / Apps

NVIDIA Computing Ecosystem – 2021-2025, per NVIDIA



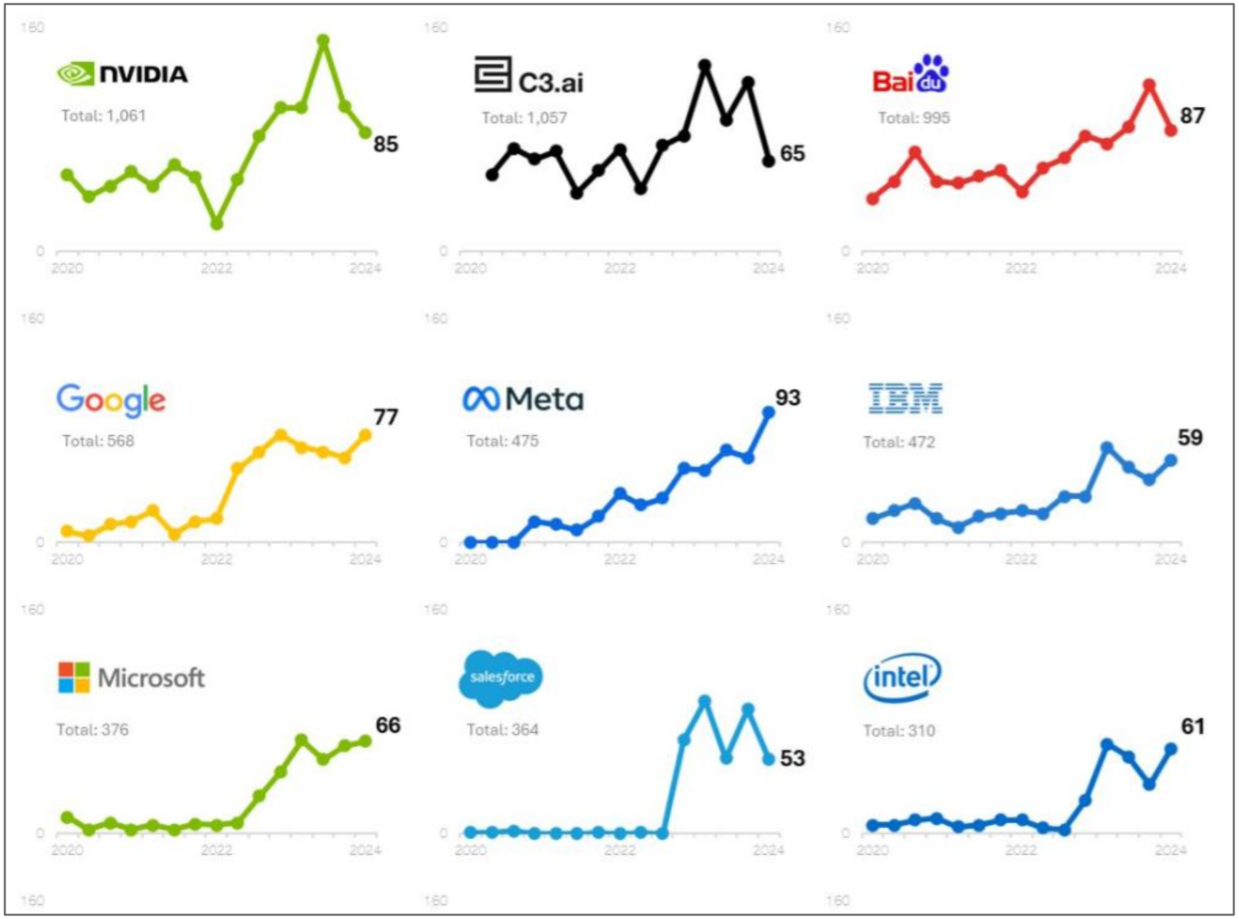
Note: GPU = Graphics Processing Unit. Source: NVIDIA (2021 & 2025)

Tech Incumbent AI Adoption =

Top Priority

Tech Incumbent AI Focus = Talking-the-Talk…

Mentions of ‘AI’ in Corporate Earnings Transcripts – Q1:20-Q1:24, per Uptrends



Source: Uptrends, ‘Top 15 Companies Mentioning AI on Earnings Calls’ (6/24), company earnings transcripts

…Tech Incumbent AI Focus = Talking-the-Talk…

Generative AI is going to reinvent virtually every customer experience we know and enable altogether new ones about which we've only fantasized. The early AI workloads being deployed focus on productivity and cost avoidance…

…Increasingly, you'll see AI change the norms in coding, search, shopping, personal assistants, primary care, cancer and drug research, biology, robotics, space, financial services, neighborhood networks – everything.

- Amazon CEO Andy Jassy in 2024 Amazon Shareholder Letter – 4/25

The chance to improve lives and reimagine things is why Google has been investing in AI for more than a decade…

…We see it as the most important way we can advance our mission to organize the world's information, make it universally accessible and useful...

…The opportunity with AI is as big as it gets.

- Google CEO Sundar Pichai @ Google Cloud Next 2025 – 4/25

Source: Amazon (4/10/25), Google (4/9/25), Techradar

…Tech Incumbent AI Focus = Talking-the-Talk…

AI Going Full-Circle:

DeepMind's AlphaGo (2014) started with humans training machines…Duolingo Chess now has machines training humans…

There's three places where [GenAI is]…helping us: data creation…creating new features that were just not possible… efficiencies everywhere in the company…

…I should mention something amazing about [the new Duolingo curriculum in] chess is that it really started with a team of two people, neither of whom knew how to program…and they basically made prototypes and did the whole curriculum of chess by just using AI. Also, neither of them knew how to play chess.

- Duolingo Co-Founder & CEO Luis von Ahn @ Q1:25 Earnings Call – 5/25

AI with Grok is getting very good…it's important that AI be programmed with good values, especially truth-seeking values. This is, I think, essential for AI safety…

…Remember these words: We must have a maximally truth-seeking AI.

- xAI Founder & CEO Elon Musk – 5/25

Note: On 3/28/25, Elon Musk announced that xAI had acquired X in an all-stock deal. The deal valued xAI at $80B and X at $33B ($45B less $12B debt). Source: Duolingo (5/1/25), DeepMind, Elon Musk (5/2/25), Fox News

…Tech Incumbent AI Focus = Talking-the-Talk

We view AI as a human acceleration tool that will allow individuals to do more...
I believe long term, we will see people coupled with…
the AI they use as the overall output of that person.

- Roblox Co-Founder, President, CEO & Chair of Board David Baszucki
@ Q1:25 Earnings Call – 5/25

I promise you, in ten years' time, you will look back and you will realize that AI has now integrated into everything. And in fact, we need AI everywhere.

And every region, every industry, every country, every company, all needs AI.
AI [is] now part of infrastructure. And this infrastructure,
just like the internet, just like electricity, needs factories….

…And these AI data centers, if you will, are improperly described. They are, in fact, AI factories. You apply energy to it, and it produces something incredibly valuable.

- NVIDIA Co-Founder & CEO Jensen Huang
@ COMPUTEX 2025 – 5/25

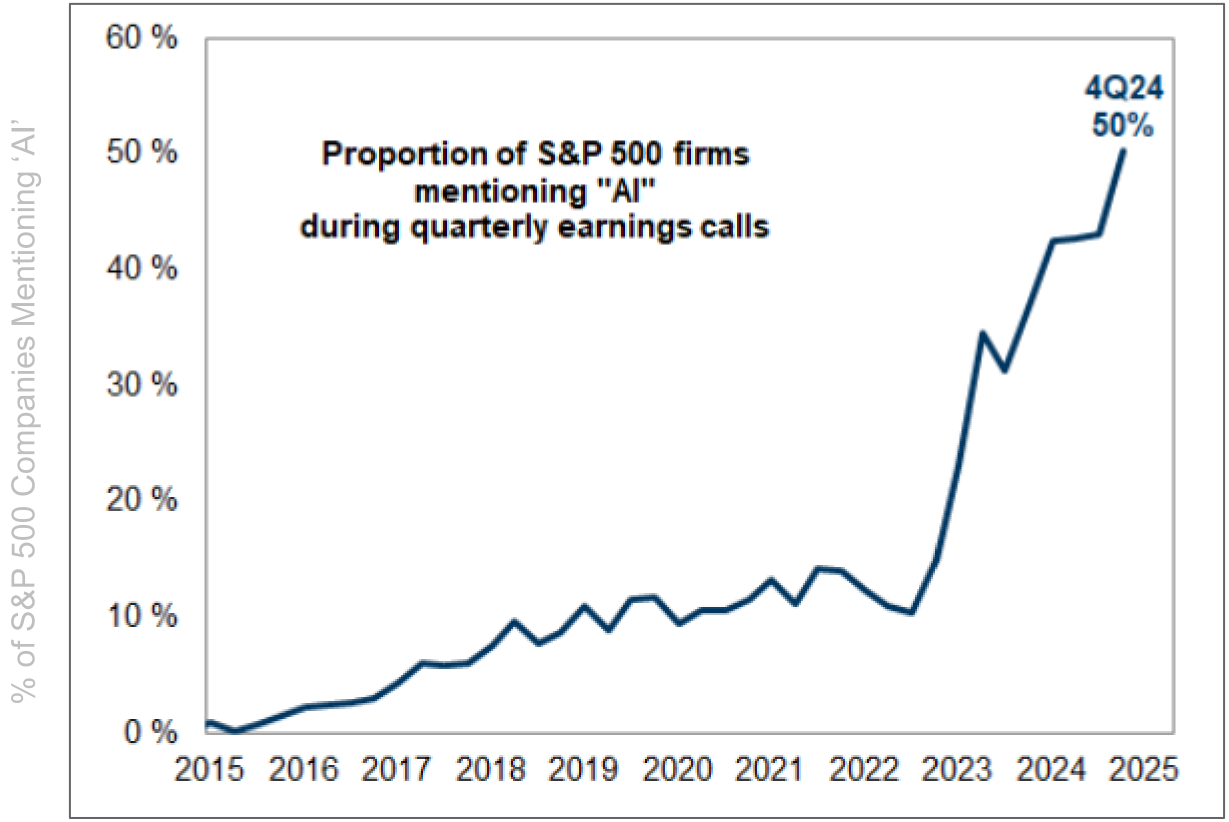
Source: Roblox (5/1/25), NVIDIA (5/18/25)

‘Traditional’ Enterprise AI Adoption =

Rising Priority

Enterprise AI Focus – S&P 500 Companies = 50% & Rising Talking-the-Talk

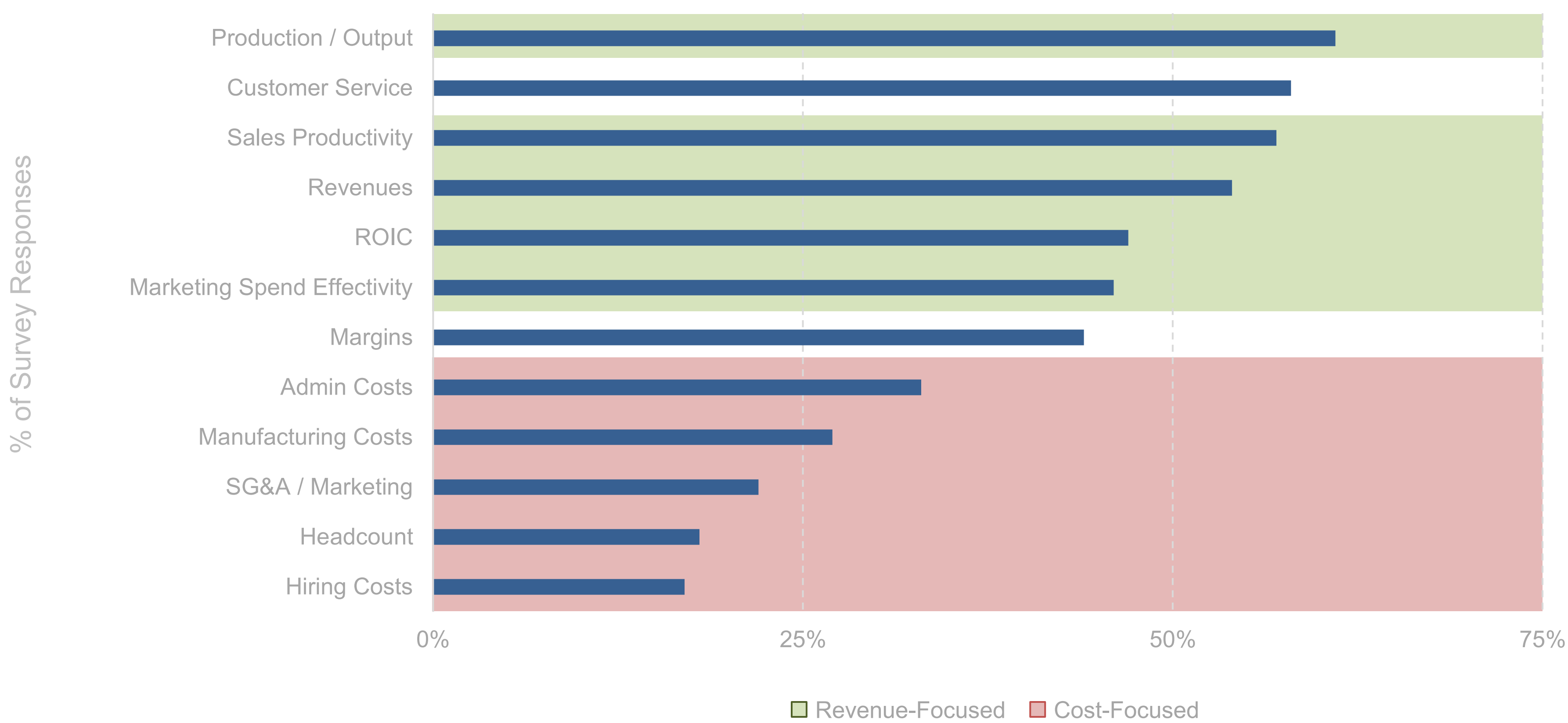
Quarterly Earnings Call Mentions of 'AI' – S&P 500 Companies (2015-2025), per Goldman Sachs Research



Source: Goldman Sachs Global Investment Research, 'S&P Beige Book: 3 themes from 4Q 2024 conference calls: Tariffs, a stronger US dollar, and AI' (2/25)

Enterprise AI Focus – Global Enterprises = Growth & Revenue…Not Cost Reduction

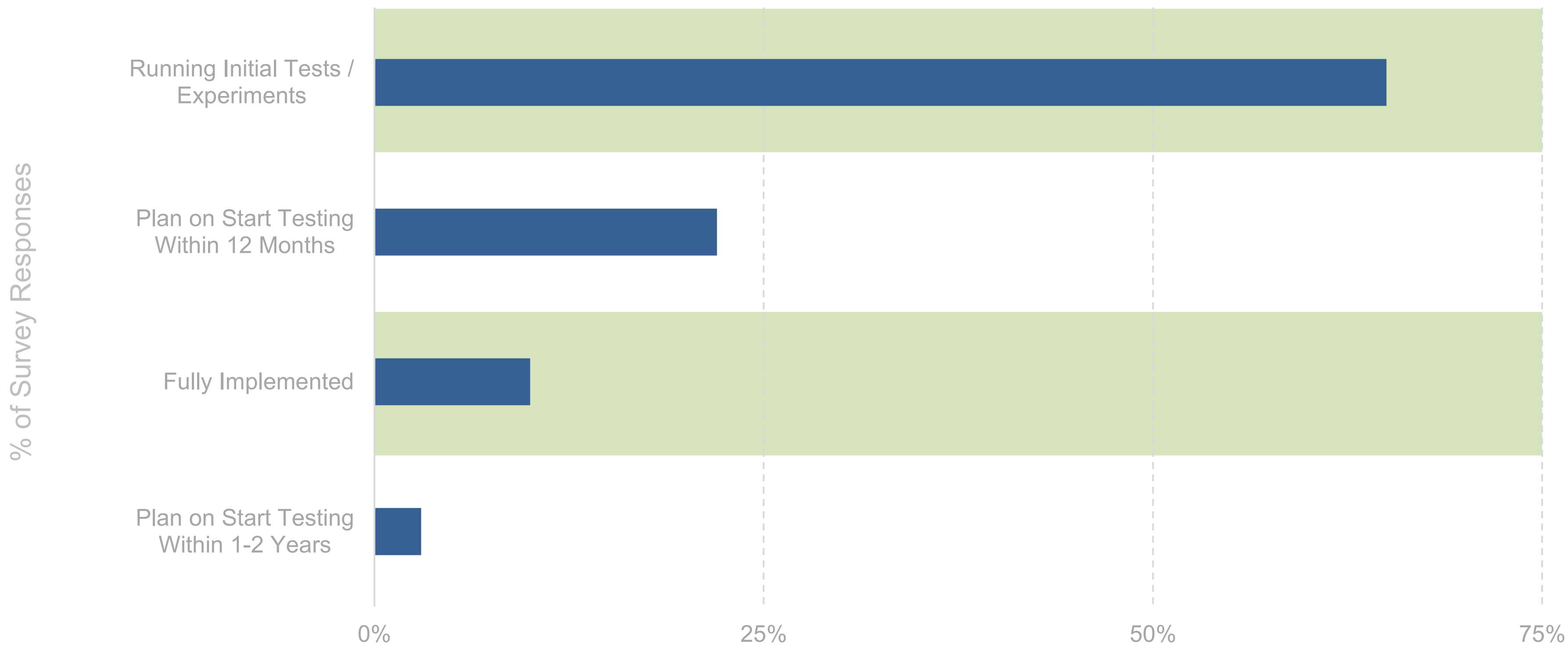
GenAI Improvements Targeted for Global Enterprises over Next 2 Years – 2024, per Morgan Stanley



Note: Survey conducted 5/24, N=427. US-based companies = 43%, Japan 15%, UK 14%, France 14%, Germany 14%. Industry mix: 18% Technology, 18% Financial Services, 17% Healthcare, 17% Manufacturing, 15% Industrials, 15% Consumer,. Revenue mix: 13% $500MM-$750MM, 25% $751MM-$1B, 36% $1B-$5B, 10% $5B-$10B, 8% $10B-$15B, 3% $15B-$20B, 5% $20B+. 'Revenue-Focused' and 'Cost-Focused' categorizations per BOND, not Morgan Stanley. Source: AlphaWise, Morgan Stanley, 'Quantifying the AI Opportunity' (12/24)

Enterprise AI Focus – Global CMOs = 75% Using / Testing AI Tools

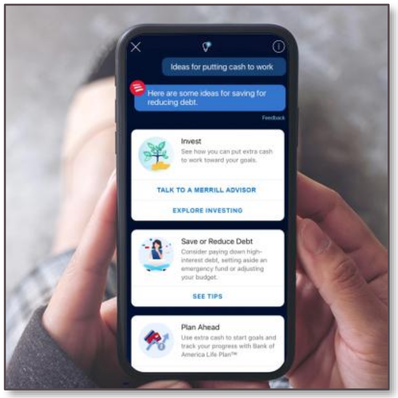
Global Chief Marketing Officer (CMO) GenAI Adoption Survey – 2024, per Morgan Stanley



Note: Survey question asked about the extent to which marketing executives worldwide are using generative AI for marketing activities. Survey conducted 7/24, N = 300 marketing executives at companies with 500+ employees worldwide. Survey geos: Australia, Belgium, Brazil, Canada, China, Denmark, Finland, France, Germany, Ireland, Italy, Japan, Luxembourg, Mexico, Netherlands, Norway, Poland, Saudi Arabia, Spain, Sweden, UAE, UK, & USA. Source: eMarketer, Morgan Stanley, 'Quantifying the AI Opportunity' (12/24)

Enterprise AI Adoption = Rising Priority…
Bank of America – Erica Virtual Assistant (6/18)

Bank of America Erica Virtual Assistant – 6/18-2/25, per Bank of America



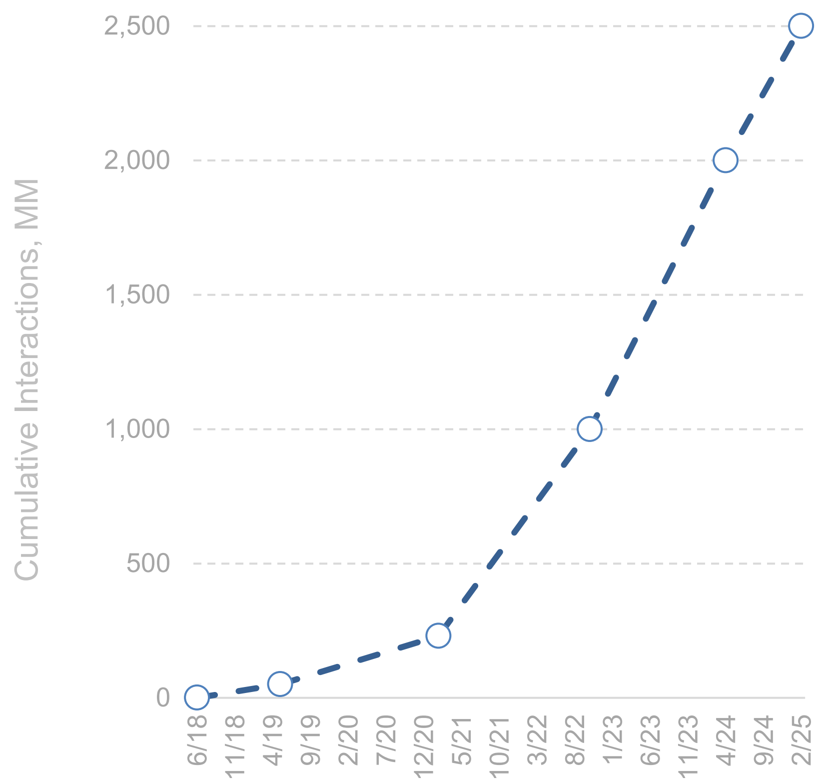
Note: Erica is a conversational AI built into Bank of America's mobile app that helps customers manage their finances by providing real-time insights, transaction search, bill reminders, and budgeting assistance. It has handled billions of interactions and serves as a 24/7 digital financial concierge for over 40 million clients.

Erica acts as both a personal concierge and mission control for our clients.

Our data science team has made more than 50,000 updates to Erica's performance since launch – adjusting, expanding and fine-tuning natural language understanding capabilities, ensuring answers and insights remain timely and relevant. 2 billion client interactions is a compelling milestone though this is only the beginning for Erica.

- Head of Digital at Bank of America Nikki Katz, 4/24

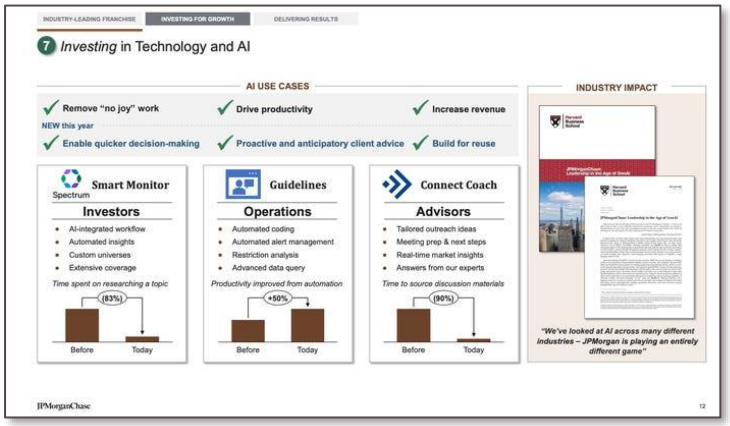
Cumulative Client Interactions with Erica Virtual Assistant (MM)



Note: We assume a start at zero users from Erica's launch in 6/18. Pilot users excluded. Source: Bank of America (2/21, 4/24, 2/25)

Enterprise AI Adoption = Rising Priority…
JP Morgan – End-to-End AI Modernization (2020)

JP Morgan End-to-End AI Modernization – 2023-2025E, per JP Morgan



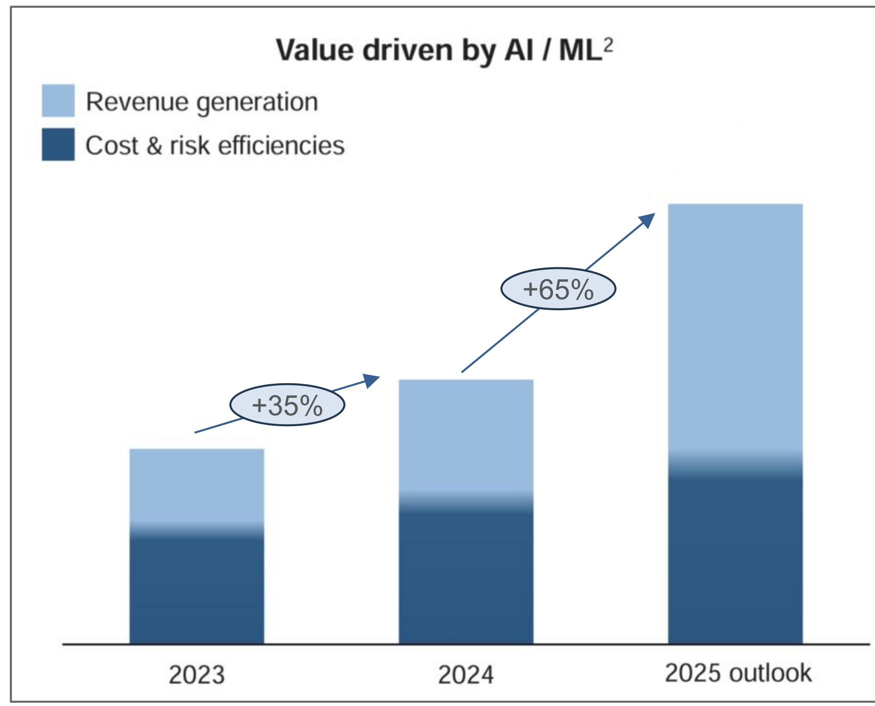
We have high hopes for the efficiency gains we might get [from AI]…

…Certain key subsets of the users tell us they are gaining several hours a week of productivity, and almost by definition, the time savings is coming from less valuable tasks…

…We were early movers in AI. But we're still in the early stages of the journey.

- JP Morgan CFO Jeremy Barnum, 5/25

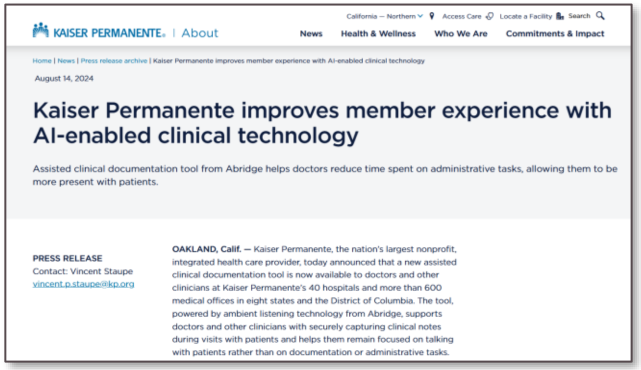
JP Morgan Estimated Value from AI / ML



Note: Superscript '2', per JP Morgan, indicates 'Value is described as benefit in revenue, lower expense, or avoidance of cost – majority is measured as the lift relative to prior analytical techniques with the remainder relative to a random baseline or holdout control.' We indicate 2020 as the start year for JP Morgan's AI Modernization (2020 Letter to Shareholders: 'We already extensively use AI, quite successfully, in fraud and risk, marketing, prospecting, idea generation, operations, trading and in other areas—to great effect, but we are still at the beginning of this journey'). Source: JP Morgan Investor Day (5/25)

Enterprise AI Adoption = Rising Priority…
Kaiser Permanente – Multimodal Ambient AI Scribe (10/23)

Kaiser Permanente Ambient AI Scribe – 10/23-12/24, per New England Journal of Medicine


KAISER PERMANENTE. | About

California – Northern | Access Care | Locate a Facility | Search

News | Health & Wellness | Who We Are | Commitments & Impact

Home | News | Press release archive | Kaiser Permanente improves member experience with AI-enabled clinical technology

August 14, 2024

Kaiser Permanente improves member experience with AI-enabled clinical technology

Assisted clinical documentation tool from Abridge helps doctors reduce time spent on administrative tasks, allowing them to be more present with patients.

PRESS RELEASE
Contact: Vincent Staupe
vincent.p.staupe@kp.org

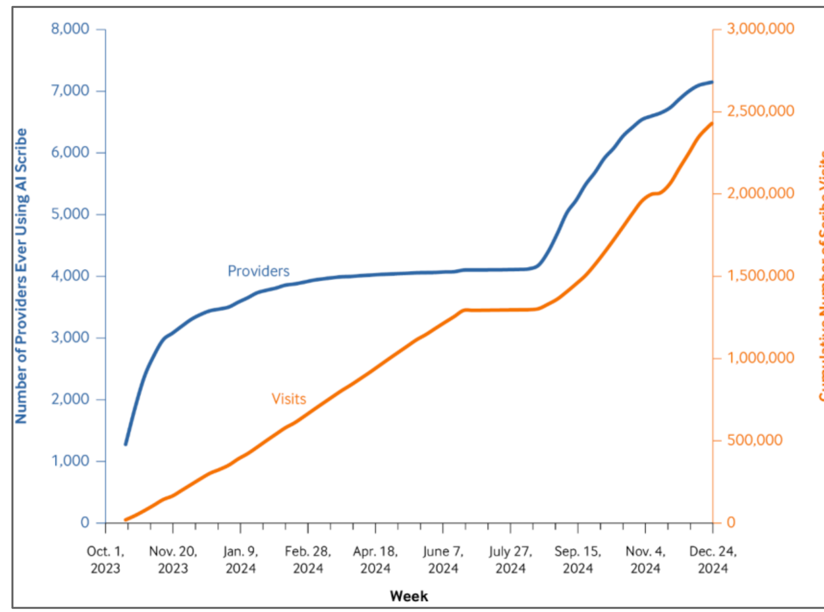
OAKLAND, Calif. — Kaiser Permanente, the nation's largest nonprofit, integrated health care provider, today announced that a new assisted clinical documentation tool is now available to doctors and other clinicians at Kaiser Permanente's 40 hospitals and more than 600 medical offices in eight states and the District of Columbia. The tool, powered by ambient listening technology from Abridge, supports doctors and other clinicians with securely capturing clinical notes during visits with patients and helps them remain focused on talking with patients rather than on documentation or administrative tasks.

Ambient artificial intelligence (AI) scribes, which use machine learning applied to conversations to facilitate scribe-like capabilities in real time, [have] great potential to reduce documentation burden, enhance physician-patient encounters, and augment clinicians' capabilities.

The technology leverages a smartphone microphone to transcribe encounters as they occur but does not retain audio recordings. To address the urgent and growing burden of data entry, in October 2023, The Permanente Medical Group (TPMG) enabled ambient AI technology for 10,000 physicians and staff to augment their clinical capabilities across diverse settings and specialties.

- New England Journal of Medicine Catalyst Research Report, 2/24

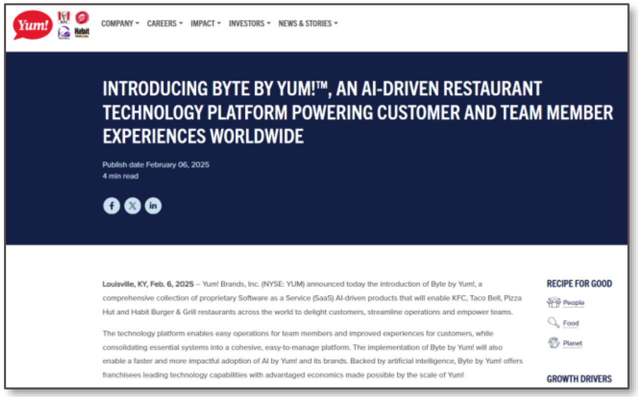
Unique Kaiser Permanente Physicians Ever Using AI Scribe & Cumulative Number of Scribe Visits



Source: Tierney, Aaron A. et al., 'Ambient Artificial Intelligence Scribes to Alleviate the Burden of Clinical Documentation' (3/24) & Tierney, Aaron A. et al., 'Ambient Artificial Intelligence Scribes: Learnings after 1 Year and over 2.5 Million Uses' (3/25) via Nestor Maslej et al., 'The AI Index 2025 Annual Report,' AI Index Steering Committee, Stanford HAI (4/25)

Enterprise AI Adoption = Rising Priority…
Yum! Brands – Byte by Yum! (2/25)

Yum! Brands Byte by Yum! – 2/24-2/25, per Yum! Brands


COMPANY CAREERS IMPACT INVESTORS NEWS & STORIES

INTRODUCING BYTE BY YUM!™, AN AI-DRIVEN RESTAURANT TECHNOLOGY PLATFORM POWERING CUSTOMER AND TEAM MEMBER EXPERIENCES WORLDWIDE

Publish date February 06, 2025
4 min read

Louisville, KY, Feb. 6, 2025 – Yum! Brands, Inc. (NYSE: YUM) announced today the introduction of Byte by Yum!, a comprehensive collection of proprietary Software as a Service (SaaS) AI-driven products that will enable KFC, Taco Bell, Pizza Hut and Habit Burger & Grill restaurants across the world to delight customers, streamline operations and empower teams.

The technology platform enables easy operations for team members and improved experiences for customers, while consolidating essential systems into a cohesive, easy-to-manage platform. The implementation of Byte by Yum! will also enable a faster and more impactful adoption of AI by Yum! and its brands. Backed by artificial intelligence, Byte by Yum! offers franchisees leading technology capabilities with advantaged economics made possible by the scale of Yum!

RECIPE FOR GOOD
People
Food
Planet

GROWTH DRIVERS

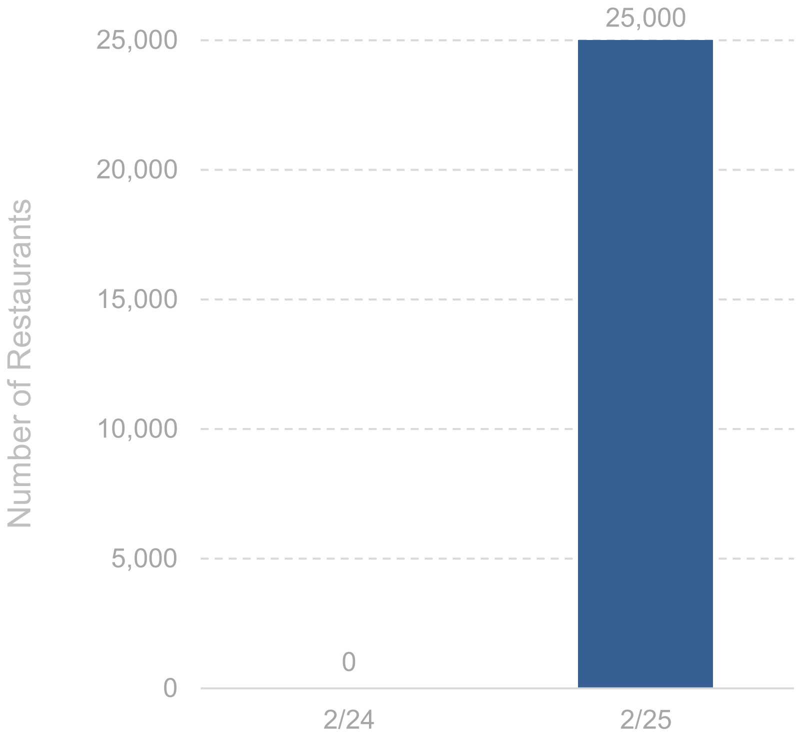
Byte is Yum! Brands' AI-powered restaurant management platform designed to optimize store operations by automating repetitive tasks like inventory tracking, scheduling, and food preparation alerts. It leverages machine learning to improve decision-making at the restaurant level, enhancing efficiency, reducing waste, and supporting staff productivity.

Backed by artificial intelligence, Byte by Yum! offers franchisees leading technology capabilities with advantaged economics made possible by the scale of Yum!.

The Byte by Yum! platform includes online and mobile app ordering, point of sale, kitchen and delivery optimization, menu management, inventory and labor management, and team member tools.

- Yum! Press Release, 2/25

Yum! Restaurants Using at Least One Byte by Yum! Product



Note: Yum! Brands names include KFC, Taco Bell, Pizza Hut, & The Habit. Byte by Yum! was officially launched in 2/25. While underlying technologies were previously in-use at restaurants in Yum!'s portfolio, the Byte by Yum! product suite had not yet officially been launched; hence, we illustratively show zero users in 2/24. Source: Yum!, 'Introducing Byte by Yum!™, an AI-driven restaurant technology platform powering customer and team member experiences worldwide'(2/25)

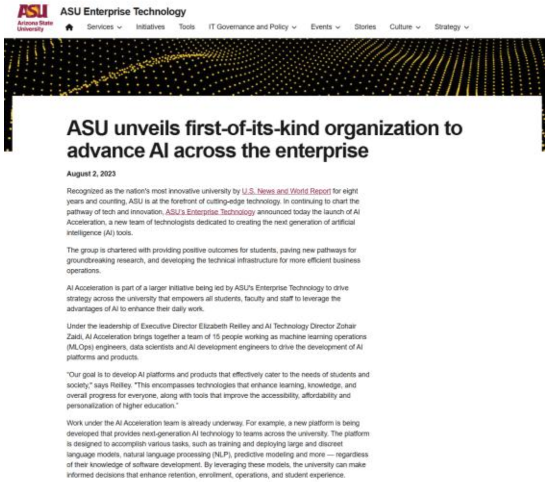
Education / Government / Research AI Adoption =

Rising Priority

Education & Government = Increasingly Announcing AI Integrations

Arizona State University's 'AI Acceleration' – 8/23

New team of technologists creating artificial intelligence (AI) tools


ASU Enterprise Technology

ASU unveils first-of-its-kind organization to advance AI across the enterprise

August 2, 2023

Recognized as the nation's most innovative university by U.S. News and World Report for eight years and counting, ASU is at the forefront of cutting-edge technology. In continuing to chart the pathway of tech and innovation, ASU's Enterprise Technology announced today the launch of AI Acceleration, a new team of technologists dedicated to creating the next generation of artificial intelligence (AI) tools.

The group is chartered with providing positive outcomes for students, paving new pathways for groundbreaking research, and developing the technical infrastructure for more efficient business operations.

AI Acceleration is part of a larger initiative being led by ASU's Enterprise Technology to drive strategy across the university that empowers all students, faculty and staff to leverage the advantages of AI to enhance their daily work.

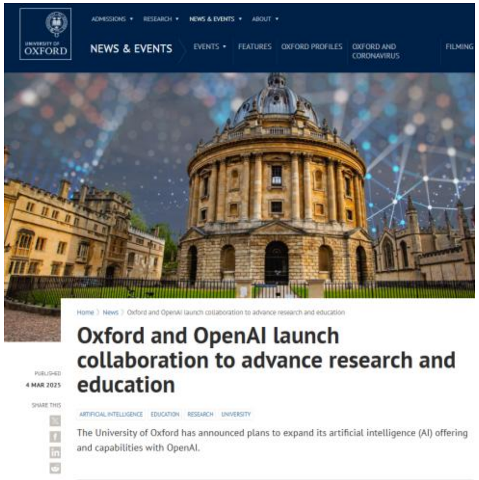
Under the leadership of Executive Director Elizabeth Reilley and AI Technology Director Zohair Zaidi, AI Acceleration brings together a team of 15 people working as machine learning operations (MLOps) engineers, data scientists and AI development engineers to drive the development of AI platforms and products.

"Our goal is to develop AI platforms and products that effectively cater to the needs of students and society," says Reilley. "This encompasses technologies that enhance learning, knowledge, and overall progress for everyone, along with tools that improve the accessibility, affordability and personalization of higher education."

Work under the AI Acceleration team is already underway. For example, a new platform is being developed that provides next-generation AI technology to teams across the university. The platform is designed to accomplish various tasks, such as training and deploying large and discreet language models, natural language processing (NLP), predictive modeling and more — regardless of their knowledge of software development. By leveraging these models, the university can make informed decisions that enhance retention, enrollment, operations, and student experience.

Oxford Partnership – 3/25

5-Year Partnership on Research & AI Literacy


NEWS & EVENTS

Oxford and OpenAI launch collaboration to advance research and education

4 MAR 2025

ARTIFICIAL INTELLIGENCE · EDUCATION · RESEARCH · UNIVERSITY

The University of Oxford has announced plans to expand its artificial intelligence (AI) offering and capabilities with OpenAI.

NextGenAI – 3/25

$50MM consortium with 15 research universities (MIT, Harvard, Caltech, etc.)


The University RECORD

News for faculty, staff and retirees

March 4, 2025

U-M, OpenAI launch partnership to expand AI research

By Don Jordan

Office of the Vice President for Research

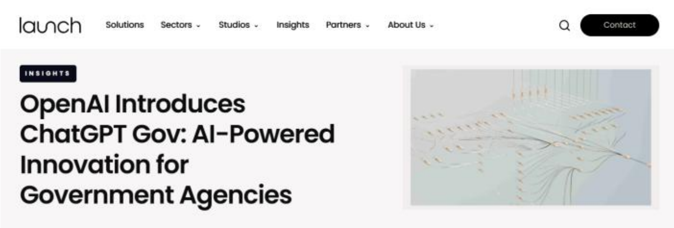
Topics: Campus News, Research

The University of Michigan has established a partnership with OpenAI that will bring additional artificial intelligence resources, research funding and computing power to campus.

The collaboration also will include joint research projects between the university and OpenAI, focusing on AI applications that broadly benefit society.

ChatGPT Gov – 1/25

ChatGPT tailored for USA federal agencies



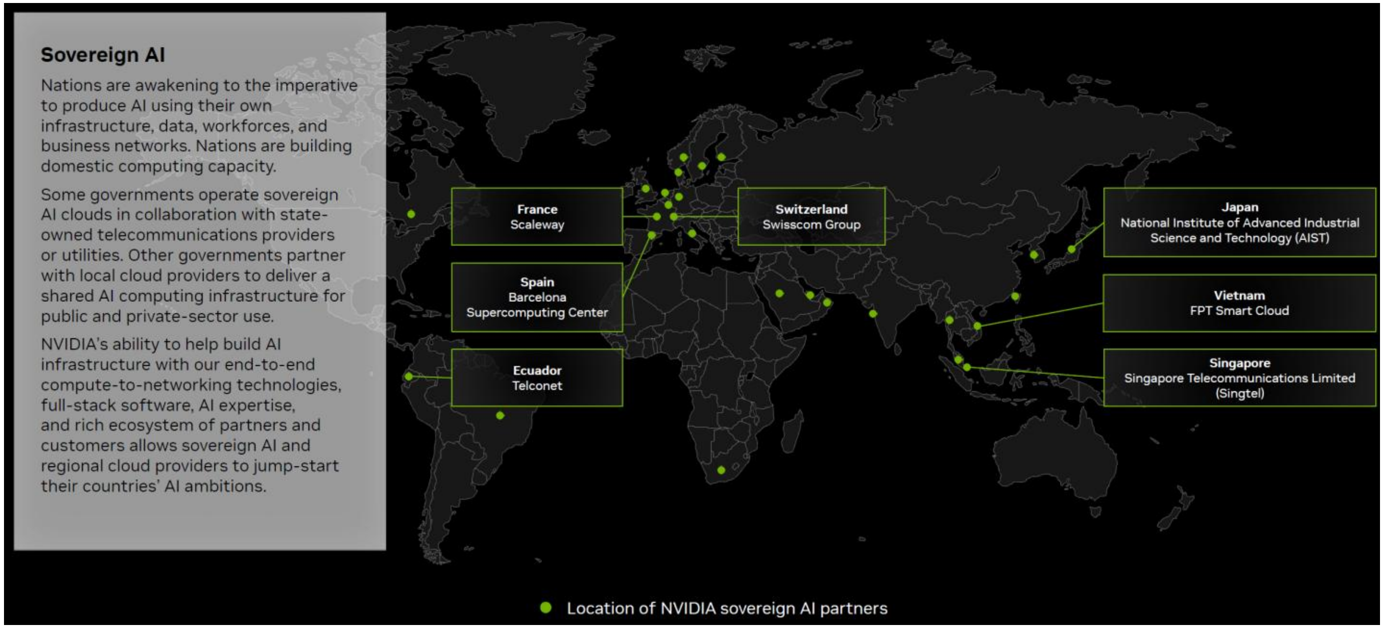
USA National Laboratories – 1/25

Partnering on Nuclear, Cybersecurity, & Scientific Breakthroughs

Source: Arizona State University (8/23), Oxford University (3/25), University of Michigan (3/25), Launch Consulting (1/25) via AI Advantage Daily News, NPR (1/25)

Government =
Increasingly Adopting Sovereign AI Policies

NVIDIA Sovereign AI Partners – 2/25, Per NVIDIA



Nations are investing in AI infrastructure like they once did for electricity and Internet.

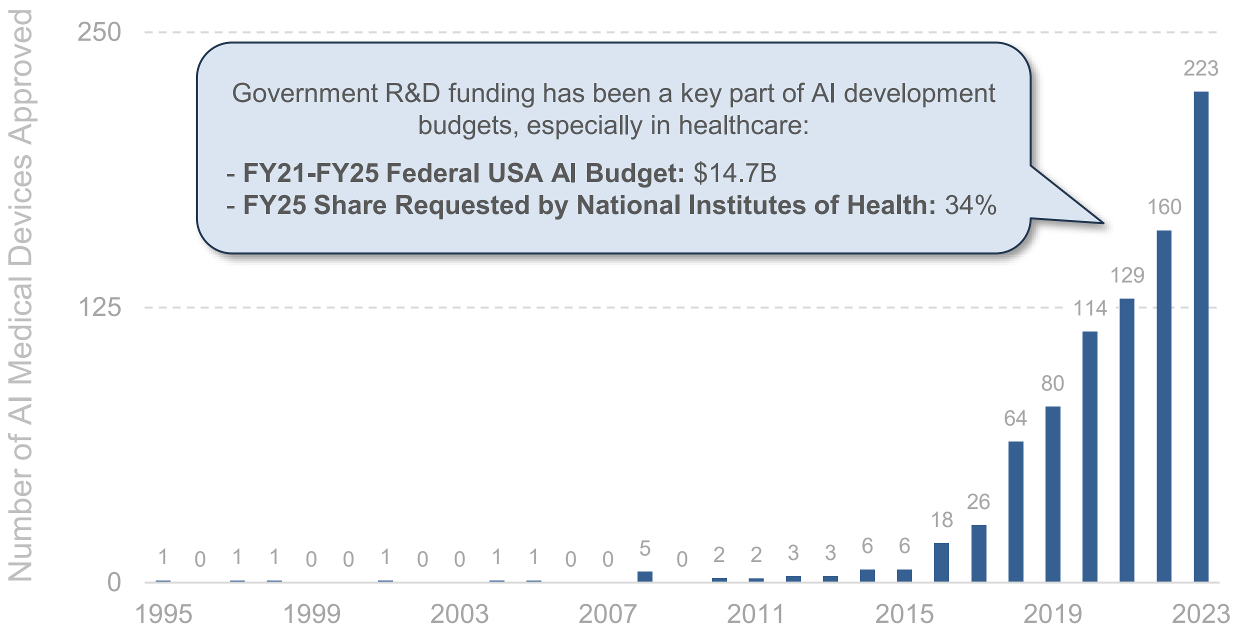
- NVIDIA Co-Founder & CEO Jensen Huang, 5/25

Source: NVIDIA (2/25 & 5/25)

Research =
Rapid Ramp in FDA-Approved AI Medical Devices, per Stanford HAI

New AI-Enabled Medical Devices Approved by USA Food & Drug Administration – 1995-2023, per Stanford HAI & USA FDA

AI-Enabled Medical Devices Approved



New USA FDA AI Policy (5/25)

In a historic first for the [USA FDA], FDA Commissioner Martin A. Makary, M.D., M.P.H., today announced an aggressive timeline to scale use of artificial intelligence (AI) internally across all FDA centers by June 30, 2025…

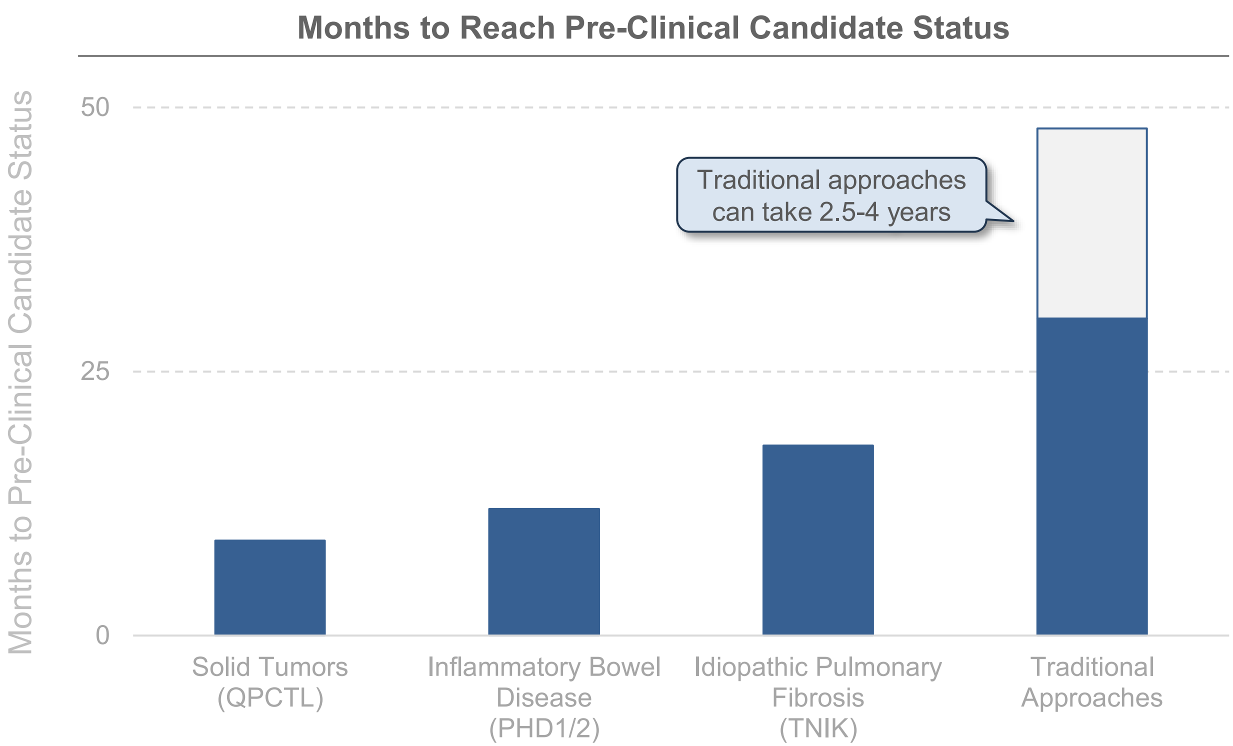
…To reflect the urgency of this effort, Dr. Makary has directed all FDA centers to begin deployment immediately, with the goal of full integration by the end of June.

- USA FDA Press Release, 5/25

Note: FY21, FY22 & FY23 USA government budget figures are actuals. FY24 data is enacted but not actual, FY25 data is requested. NIH share of total budget is requested.
Source: Nestor Maslej et al., 'The AI Index 2025 Annual Report,' AI Index Steering Committee, Stanford HAI (4/25); USA Food & Drug Administration, 'FDA Announces Completion of First AI-Assisted Scientific Review Pilot and Aggressive Agency-Wide AI Rollout Timeline' (5/25); NITRD.gov (5/25)

Research =
30%-80% Reduction in Medical R&D Timelines, per Insilico Medicine & Cradle

AI-Driven Drug Discovery – 2021-2024, Per Insilico Medicine, Cradle & BioPharmaTrend



Cradle

Pharma companies that use Cradle are seeing a 1.5x to 12x speedup in pre-clinical research and development by using our GenAI platform to engineer biologics.

- Stef van Grieken, Co-Founder & CEO of Cradle, 5/25

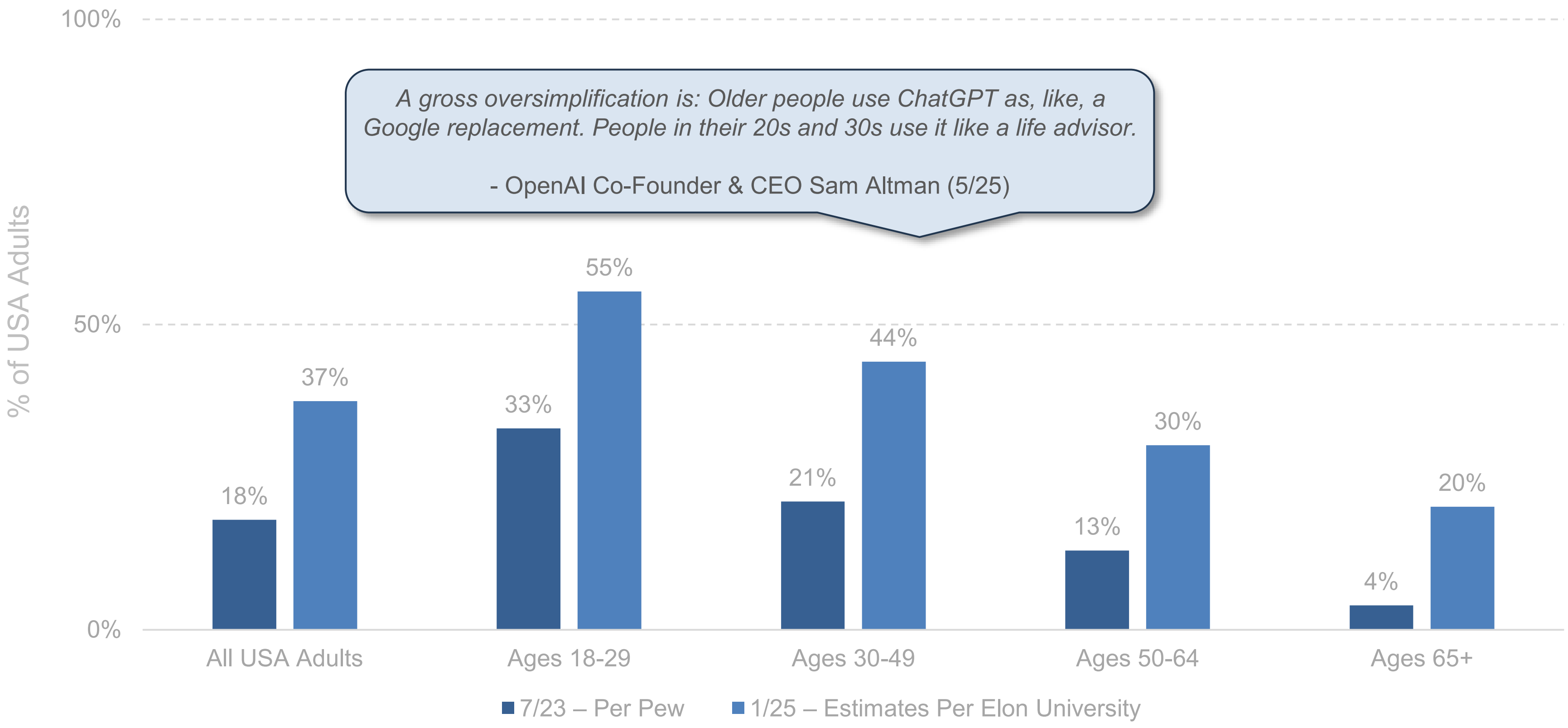
Note: Pre-Clinical Candidate Status marks the point at which a lead molecule (or biologic) has satisfied all discovery-stage gates and is officially handed off to the development organization for work related to beginning human clinical trials. Figures collected from 2021-2024. Source: Cradle, Insilico Medicine via BioPharmaTrend, 'Insilico Medicine Reports Benchmarks for its AI-Designed Therapeutics' (2/25)

*AI User + **Usage** + CapEx **Growth** =*

Unprecedented

AI Usage – ChatGPT = Rising Rapidly Across Age Groups in USA, per Pew & Elon University

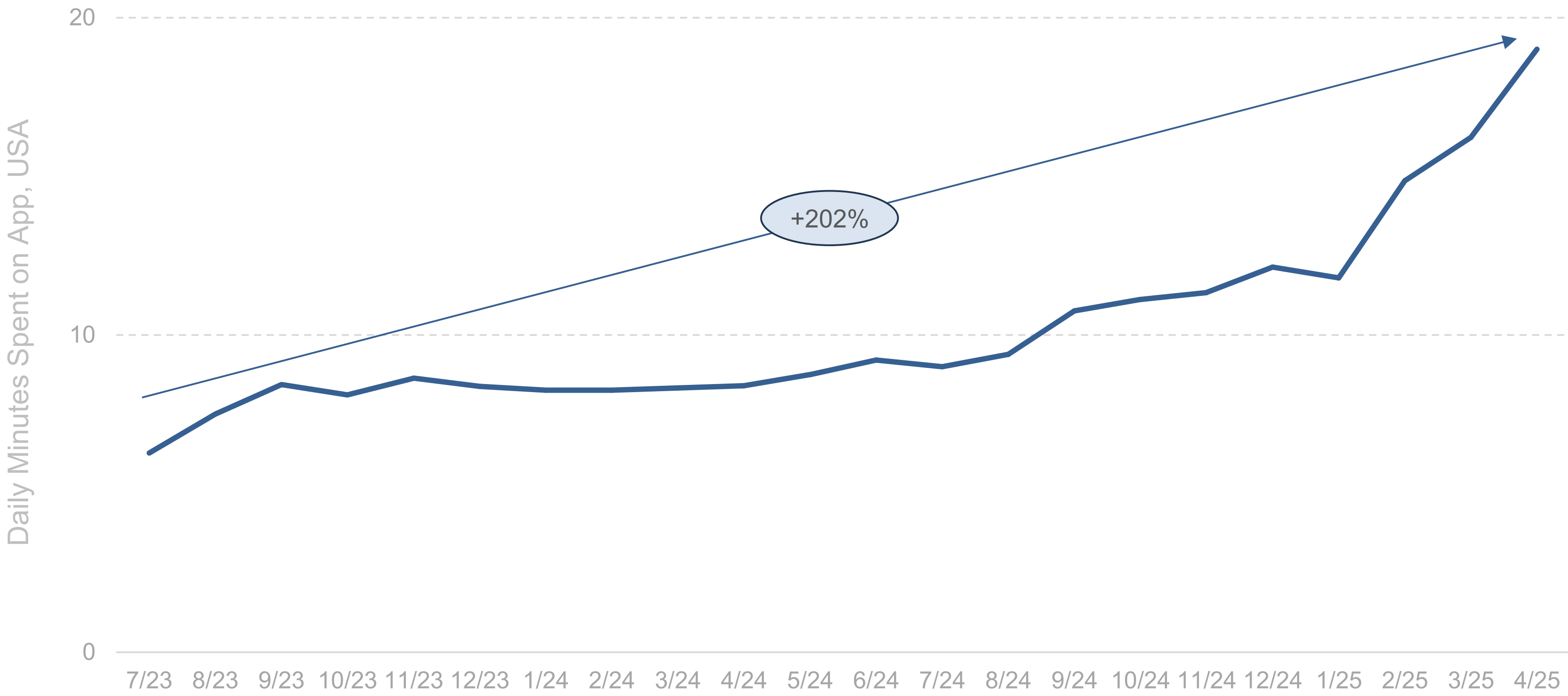
% of USA Adults Who Say They Have Ever Used ChatGPT – 7/23 per Pew & 1/25 per Elon University



Note: 7/23 data per Pew Research study on ChatGPT use, n=10,133 USA adults. Those who did not give an answer are not shown. 1/25 data per Elon University study on use of any AI models, n=500 USA adults,. Figures estimated based on overall AI tool usage adjusted for an average 72% usage rate of ChatGPT amongst respondents who use any AI tools. Actual ChatGPT penetration may vary by cohort. Note that this chart aggregates data across survey providers and as such may not be directly comparable. Source: Pew Research Center (3/26/24), Elon University (released 3/12/25), Sam Altman (5/12/25) via Fortune

AI Engagement (ChatGPT App as Proxy) = +202% Rise in Daily Time Spent Over Twenty-One Months…

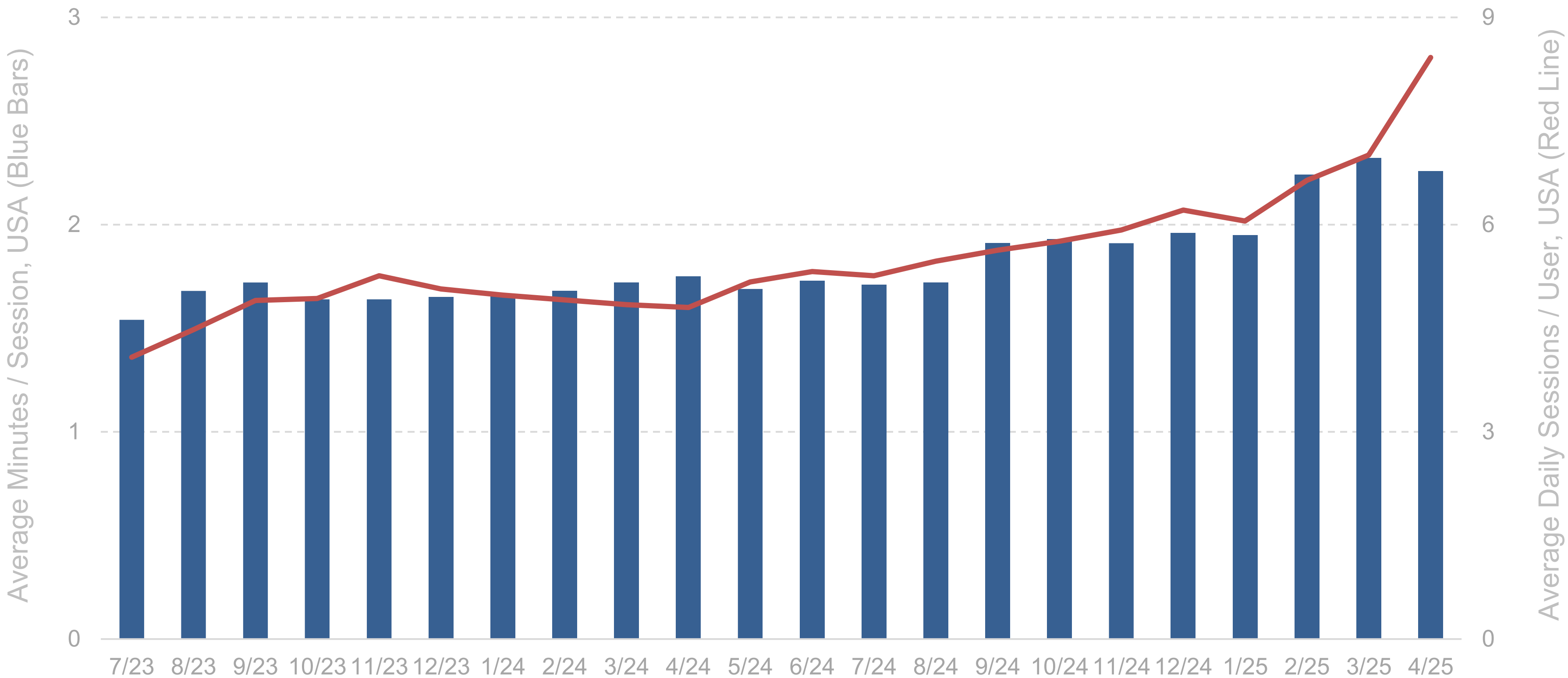
Minutes per Day that USA Active Users Spend on ChatGPT App – 7/23-4/25, per Sensor Tower



Note: Data represents USA App Store & Google Play Store monthly active users. Data for ChatGPT standalone app only. ChatGPT app not available in China, Russia and select other countries as of 5/25. Source: Sensor Tower (5/25)

…AI Engagement (ChatGPT App as Proxy) =
+106% Growth in Sessions & +47% Growth in Duration Over Twenty-One Months

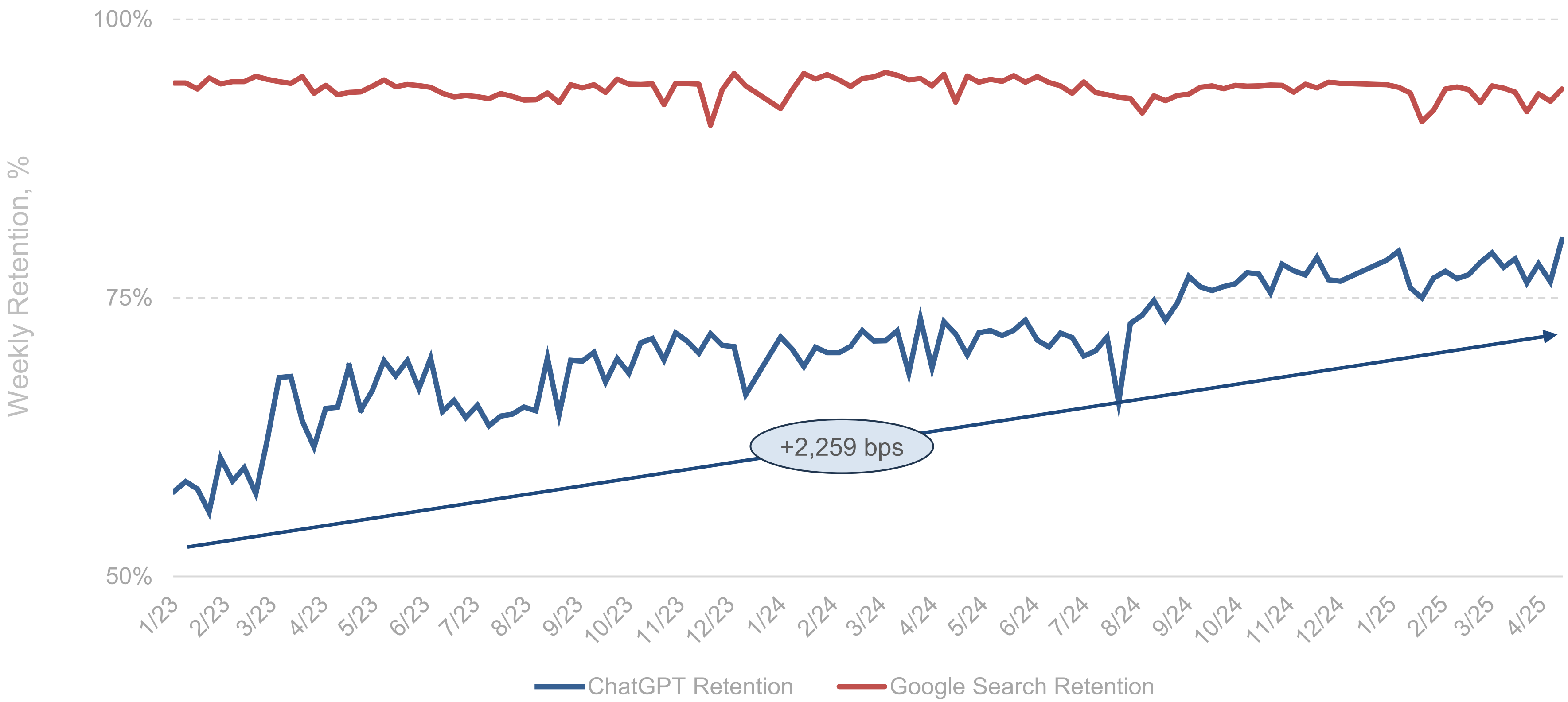
Average USA Session Duration (Minutes) & Daily Sessions per User for ChatGPT App – 7/23-4/25, per Sensor Tower



Note: Data represents USA App Store & Google Play Store monthly active users. Data for ChatGPT standalone app only. ChatGPT app not available in China, Russia and select other countries as of 5/25. Source: Sensor Tower (5/25)

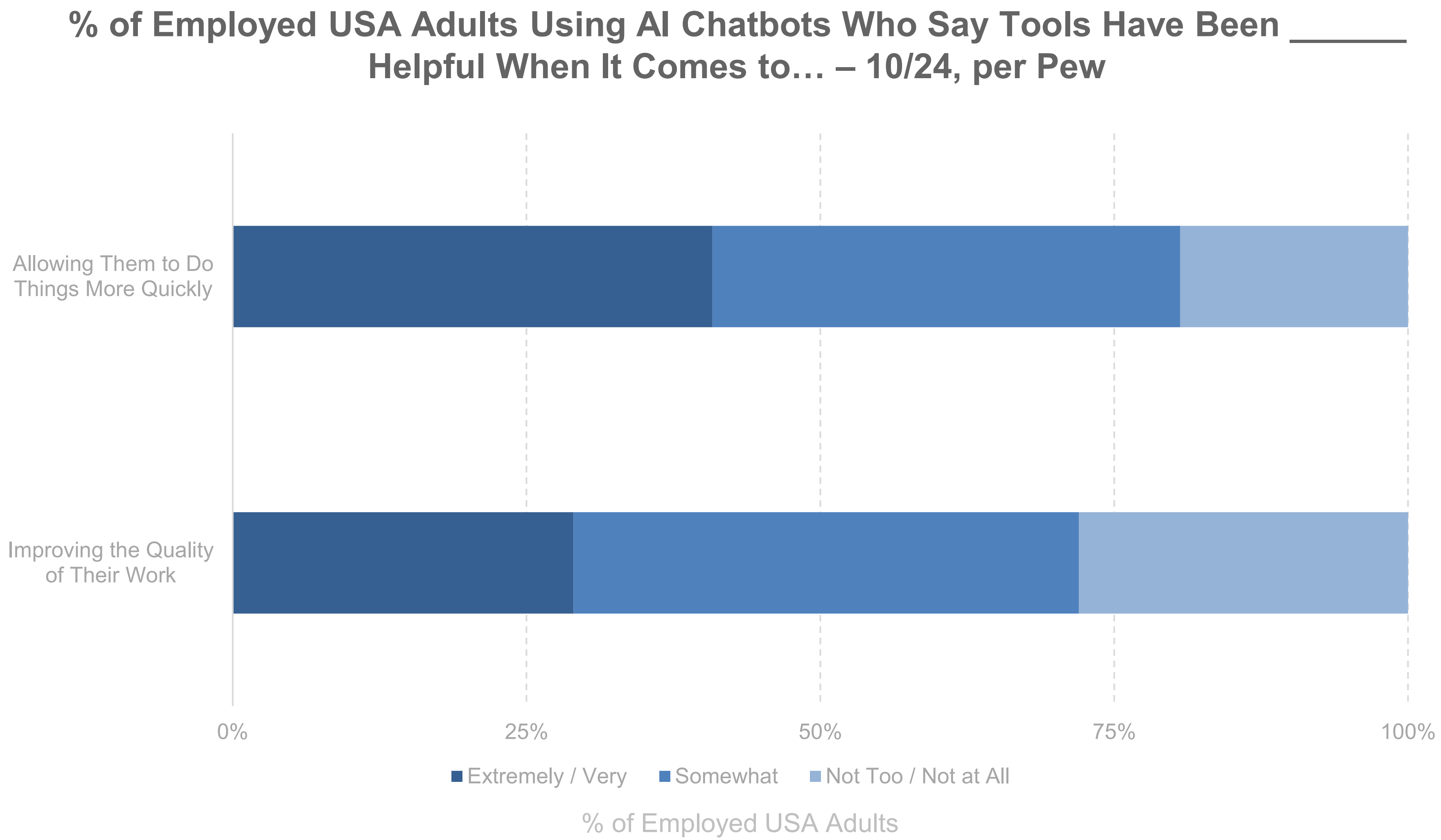
AI Retention (ChatGPT as Proxy) = 80% vs. 58% Over Twenty-Seven Months, per YipitData

Consumer ChatGPT & Google Search Global Desktop User Retention Rates (1/23-4/25), per YipitData



Note: Retention Rate = Percentage of users from the immediately preceding week that were users again in the current week. Data measures several million global active desktop users' clickstream data. Data consists of users' web requests & is collected from web services / applications, such as VPNs and browser extensions. Users must have been part of the panel for 2 consecutive months to be included. Panel is globally-representative, though China data may be subject to informational limitations due to government restrictions. Excludes anomalies in w/c 12/24/23, 12/31/23, 12/22/24, 12/29/24, 1/5/25, potentially due to holiday breaks causing less enterprise usage. Source: YipitData (5/25)

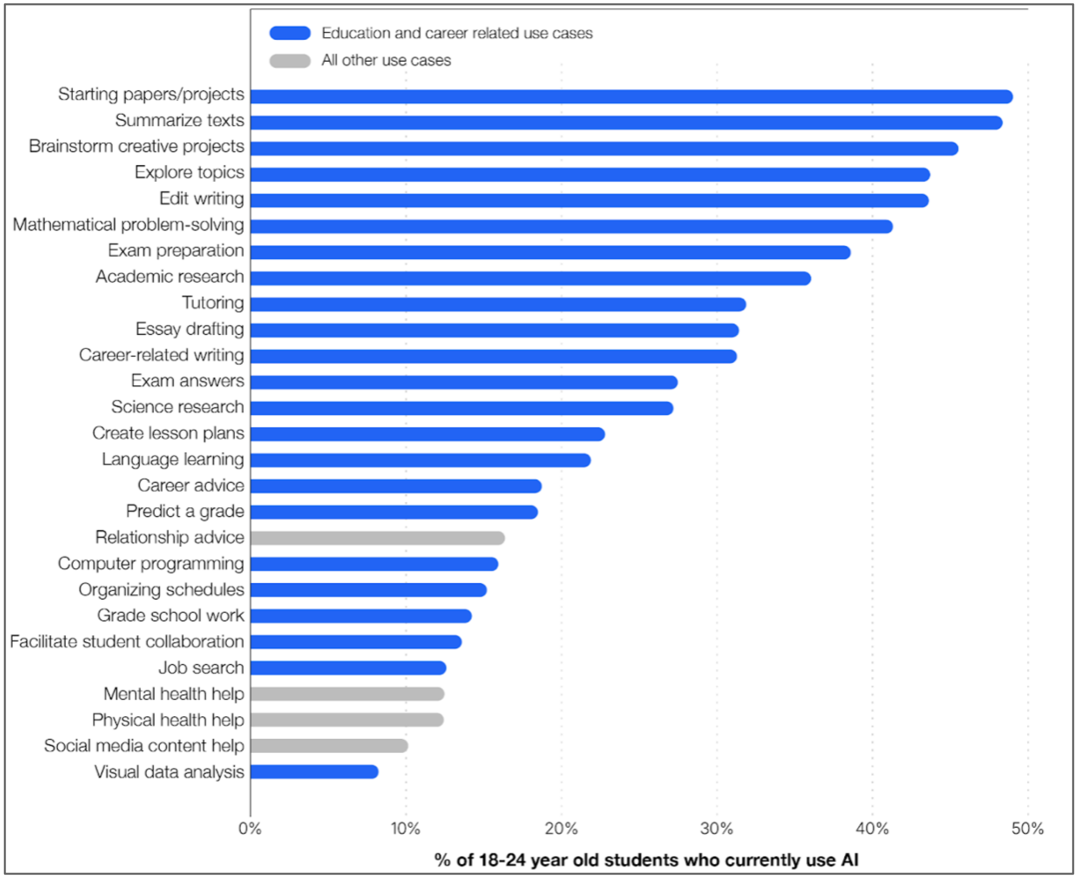
AI Chatbots @ Work Tells =
>72% Doing Things Quicker / Better



Note: N = 5,273 USA adults who are employed part time or full time and who have only one job or have more than one but consider one of them to be their primary job were surveyed.
Source: Pew Research Center (10/24)

AI Chatbots @ School Tells (ChatGPT as Proxy) = Bias to Research / Problem Solving / Learning / Advice

OpenAI ChatGPT Usage Survey, USA Students Ages 18-24 – 12/24-1/25, per OpenAI

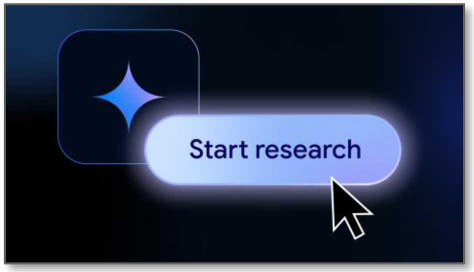


Note: Data per OpenAI survey (12/24), n = 1,299 USA college and graduate students across a mix of STEM and non-STEM disciplines; only answers from 18-24 year olds used. Sample includes both AI users and non-users but excludes "AI rejectors" – defined as non-users with little to no interest in adopting AI within the next 12 months. Source: OpenAI, 'Building an AI-Ready Workforce: A Look at College Student ChatGPT Adoption in the US' (2/25)

AI Usage Expansion – Deep Research = Automating Specialized Knowledge Work

Select AI Company Deep Research Capabilities – 12/24-2/25, per Google, OpenAI & xAI

Google Gemini Deep Research

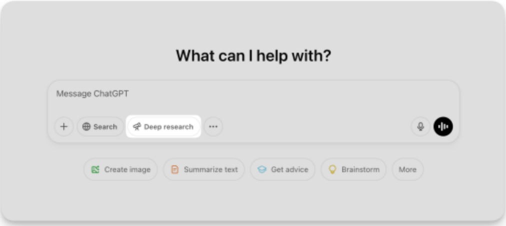


Get up to speed on just about anything with Deep Research, an agentic feature in Gemini that can automatically browse up to hundreds of websites on your behalf, think through its findings, and ***create insightful multi-page, reports that you can turn into engaging podcast-style conversations****…*

…It's a step towards more agentic AI that can move beyond simple question-answering to become a true collaborative partner.

- Google Deep Research Overview, launched 12/24

OpenAI ChatGPT Deep Research

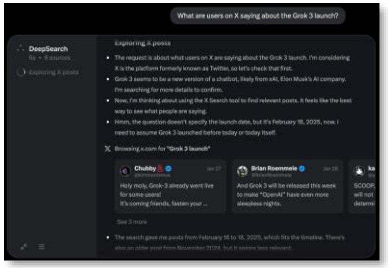


Today we're launching deep research in ChatGPT, a new agentic capability that conducts multi-step research on the internet for complex tasks. ***It accomplishes in tens of minutes what would take a human many hours****…*

…Deep research marks a significant step toward our broader goal of developing AGI, which we have long envisioned as capable of producing novel scientific research.

- OpenAI Deep Research Press Release, 2/25

xAI Grok DeepSearch



To understand the universe, we must interface Grok with the world…

…As a first step towards this vision, we are rolling out DeepSearch – our first agent. It's a ***lightning-fast AI agent built to relentlessly seek the truth across the entire corpus of human knowledge****. DeepSearch is designed to synthesize key information, reason about conflicting facts and opinions, and distill clarity from complexity.*

- xAI Grok 3 Beta Press Release, 2/25

Source: Google (5/25), OpenAI (2/25), xAI (2/25), Digital Trends (1/25)

AI Agent Evolution =

Chat Responses → Doing Work

A new class of AI is now emerging – less assistant, more service provider.
What began as basic conversational interfaces may now be evolving into something far more capable.

Traditional chatbots were designed to respond to user prompts, often within rigid scripts or narrow flows.
They could fetch answers, summarize text, or mimic conversation – but always in a reactive, limited frame.

AI agents represent a step-change forward. These are intelligent long-running processes
that can reason, act, and complete multi-step tasks on a user's behalf. They don't just answer questions –
they execute: booking meetings, submitting reports, logging into tools, or orchestrating workflows across platforms,
often using natural language as their command layer.

This shift mirrors a broader historical pattern in technology.
Just as the early 2000s saw static websites give way to dynamic web applications –
where tools like Gmail and Google Maps transformed the internet from a collection of pages into a set of utilities –
AI agents are turning conversational interfaces into functional infrastructure.

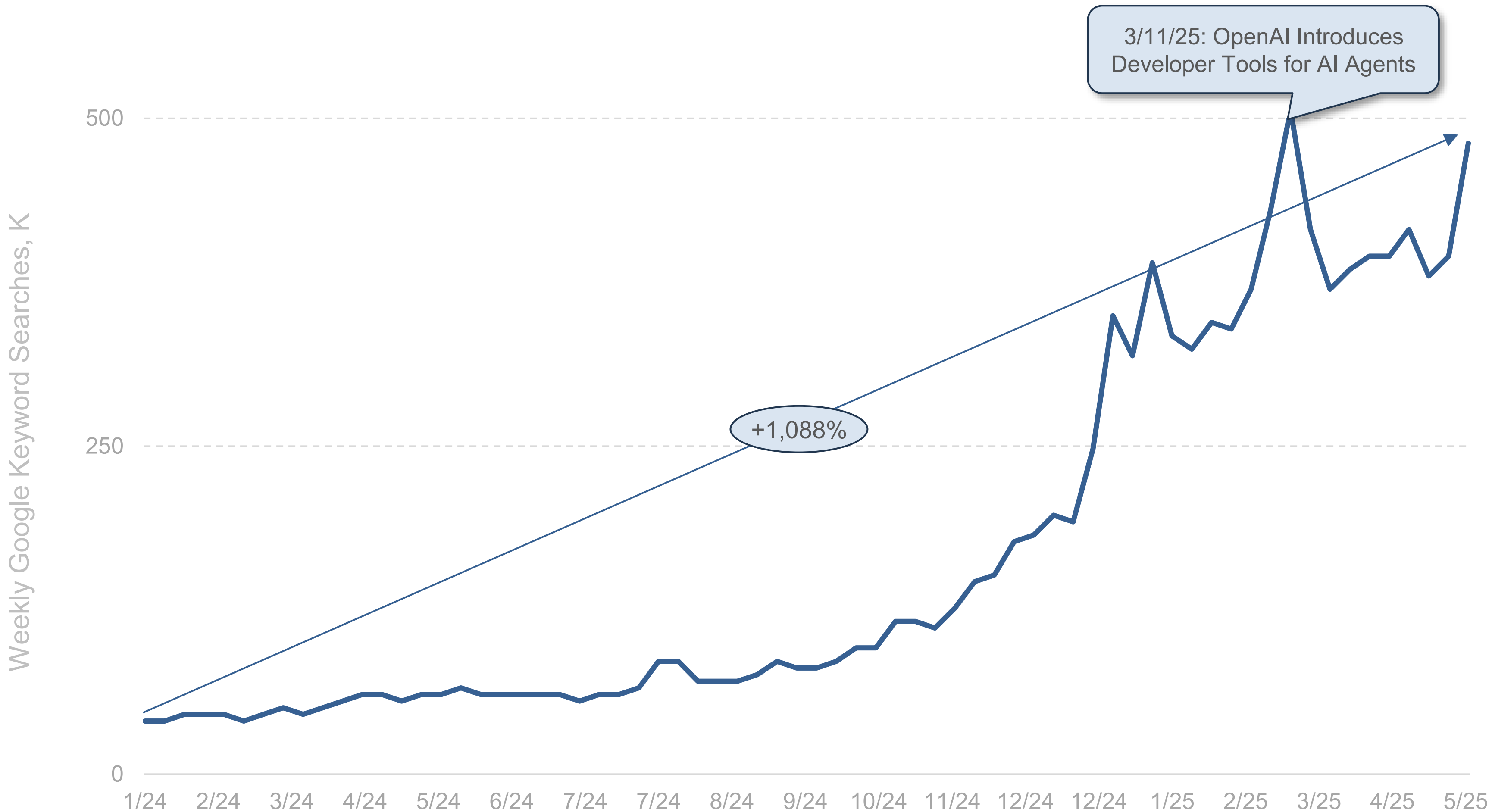
Whereas early assistants needed clear inputs and produced narrow outputs, agents promise to operate with goals,
autonomy and certain guardrails. They promise to interpret intent, manage memory, and coordinate across
apps to get real work done. It's less about responding and more about accomplishing.

While we are early in the development of these agents, the implications are just starting to emerge.
AI agents could reshape how users interact with digital systems –
from customer support and onboarding to research, scheduling, and internal operations.

Enterprises are leading the charge; they're not just experimenting with agents, but deploying them,
investing in frameworks and building ecosystems around autonomous execution.
What was once a messaging interface is becoming an action layer.

AI Agent Interest (Google Searches) = +1,088% Over Sixteen Months

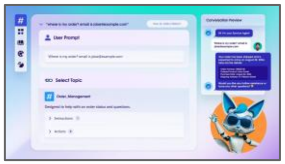
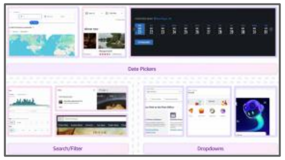
Global Google Searches for 'AI Agent' (K) – 1/24-5/25, per Google Trends



Source: Google Trends via Glimpse (5/15/24), OpenAI (3/25)

AI Agent Deployments =
AI Incumbent Product Launches Accelerating

AI Incumbent Agent Launches

Agent Released		Select Capabilities
	**Salesforce Agentforce** *(10/24 = General Release)*	• Automated customer support • Case resolution • Lead qualification • Order tracking
 	**Anthropic Claude 3.5 Computer Use** *(10/24 = Research Preview Release)*	• Control computer screen directly to perform tasks like pulling data from websites, making online purchases, etc.
 	**OpenAI Operator** *(1/25 = Research Preview Release)*	• Control computer screen directly to perform tasks like pulling data from websites, making online purchases, etc.
	**Amazon Nova Act** *(3/25 = Research Preview Release)*	• Home automation • Information collection • Purchasing • Scheduling

Source: Salesforce (10/24), Salesforce Ben, Anthropic (10/24), OpenAI (1/25), Amazon (3/25)

Next Frontier For AI =

Artificial General Intelligence

Artificial General Intelligence, or AGI, refers to systems capable of performing the full range of human intellectual tasks – reasoning, planning, learning from small data samples, and generalizing knowledge across domains. Unlike current AI models, which excel within specific (albeit broad) boundaries, AGI would be able to operate fully flexibly across disciplines and solve unfamiliar problems without retraining. It represents a major milestone in AI development – one that builds on recent exponential gains in model scale, training data, and computational efficiency.

Timelines for AGI remain uncertain, but expert expectations have shifted forward meaningfully in recent years. Sam Altman, CEO of OpenAI, remarked in January 2025, *We are now confident we know how to build AGI as we have traditionally understood it.* This is a forecast, not a dictum, but it reflects how advances in model architecture, inference* efficiency, and training scale are shortening the distance between research and frontier capability. The broader thread is clear: AI development is trending at unprecedented speed, and AGI is increasingly being viewed not as a hypothetical endpoint, but as a reachable threshold.

If / when achieved, AGI would redefine what software (and related hardware) can do. Rather than executing pre-programmed tasks, AGI systems would understand goals, generate plans, and self-correct in real time. They could drive research, engineering, education, and logistics workflows with little to no human oversight – handling ambiguity and novelty with general-purpose reasoning. These systems wouldn't require extensive retraining to handle new problem domains – they would transfer learning and operate with context, much like human experts. Additionally, humanoid robots powered by AGI would have the power to reshape our physical environment and how we operate in it.

Still, the implications warrant a measured view. AGI is not a finish line, but a phase shift in capability – and how it reshapes institutions, labor, and decision-making will depend on the safeguards and deployment frameworks that accompany it. The productivity upside may be significant, but unevenly distributed.

The geopolitical, ethical, and economic implications may evolve gradually, not abruptly. As with earlier transitions – from industrial to digital to algorithmic – the full consequences will be shaped not just by what the technology can do, but by how society chooses to adopt and govern it.

**Inference = Fully-trained model generates predictions, answers, or content in response to user inputs. This phase is much faster and more efficient than training.*

*AI User + Usage + **<u>CapEx</u> <u>Growth</u>** =*

Unprecedented

To understand where technology CapEx is heading, it helps to look at where it's been. Over the past two decades, tech CapEx has flexed upward at points through data's long arc – first toward storage / access, then toward distribution / scale, and now toward computation / intelligence.

The earliest wave saw CapEx pouring into building internet infrastructure – massive server farms, undersea cables, and early data centers that enabled Amazon, Microsoft, Google and others to lay the foundation for cloud computing. That was the first phase: store it, organize it, serve it.

The second wave – still unfolding – has been about supercharging compute for data-heavy AI workloads, a natural evolution of cloud computing. Hyperscaler* CapEx budgets now tilt increasingly toward specialized chips (GPUs, TPUs, AI accelerators…), liquid cooling, and frontier data center design.

In 2019, AI was a research feature; by 2023, it was a capital expenditure line item. Microsoft Vice Chair and President Brad Smith put it well in a 4/25 blog post:
Like electricity and other general-purpose technologies in the past, AI and cloud datacenters represent the next stage of industrialization.

The world's biggest tech companies are spending tens of billions annually – not just to gather data, but to learn from it, reason with it and monetize it in real time. It's still about data – but now, the advantage goes to those who can train on it fastest, personalize it deepest, and deploy it widest.

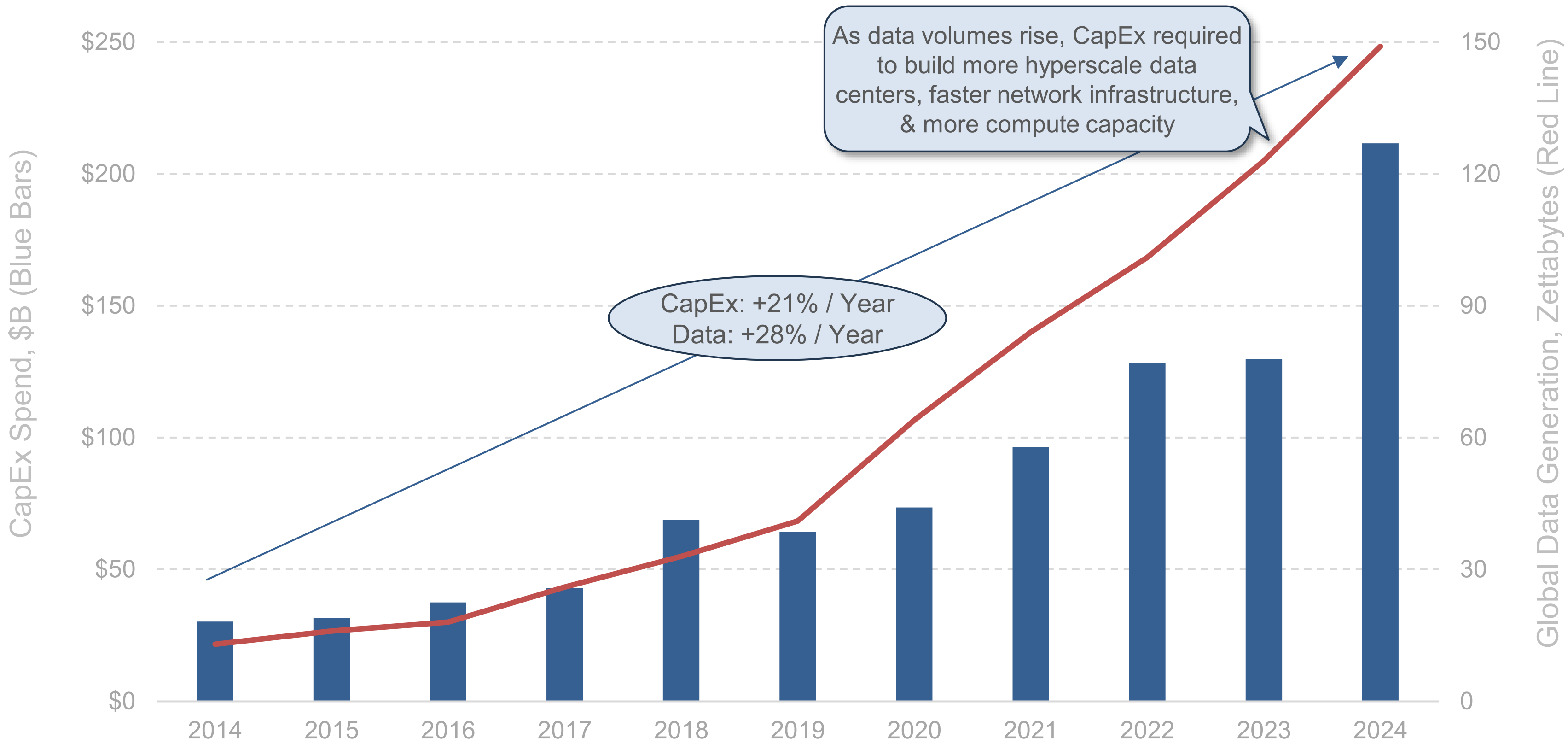
**Hyperscalers (large data center operators) are Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), Alibaba Cloud, Oracle Cloud Infrastructure (OCI), IBM Cloud & Tencent Cloud.*

CapEx Spend – Big Technology Companies =

On Rise for Years as
Data Use + Storage Exploded

CapEx Spend @ Big Six* Tech Companies (USA) = +21% Annual Growth Over Ten Years

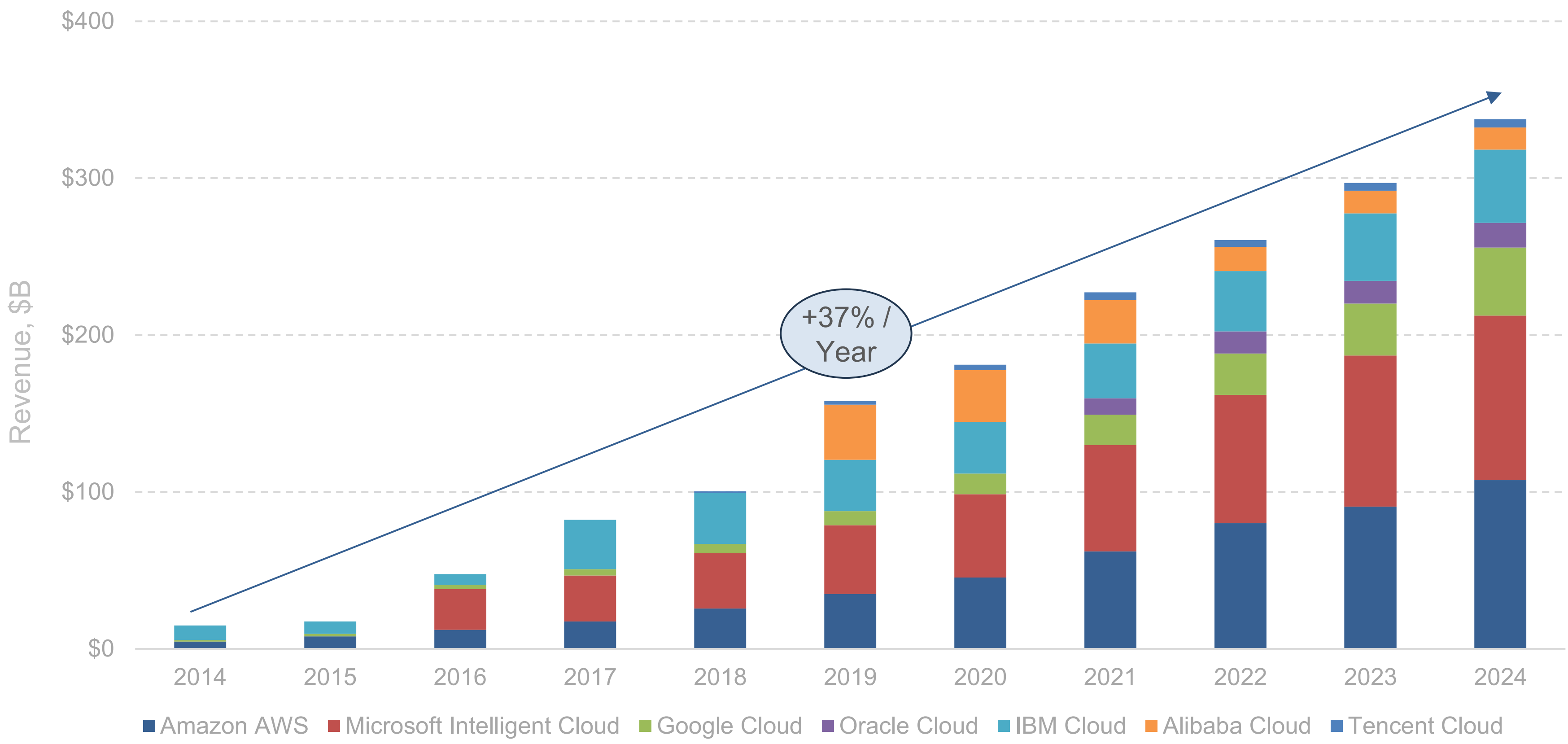
Big Six* USA Public Technology Company CapEx Spend ($B) vs. Global Data Generation (Zettabytes) – 2014-2024, per Capital IQ & Hinrich Foundation



Note: Big Six USA technology companies include Apple, Nvidia, Microsoft, Alphabet / Google, Amazon, & Meta Platforms / Facebook. Only AWS CapEx & revenue shown for Amazon (i.e. excludes Amazon retail CapEx). AWS CapEx estimated per Morgan Stanley – equals AWS net additions to property & equipment less finance leases and obligations. Global data generation figures for 2024 are estimates. Source: Capital IQ (3/25), Hinrich Foundation (3/25)

CapEx Spend for Tech Hyperscalers = Mirrored by… +37% Annual Cloud Revenue Growth Over Ten Years

Global Hyperscaler Cloud Revenue ($B) – 2014-2024, per Company Disclosures & Morgan Stanley Estimates

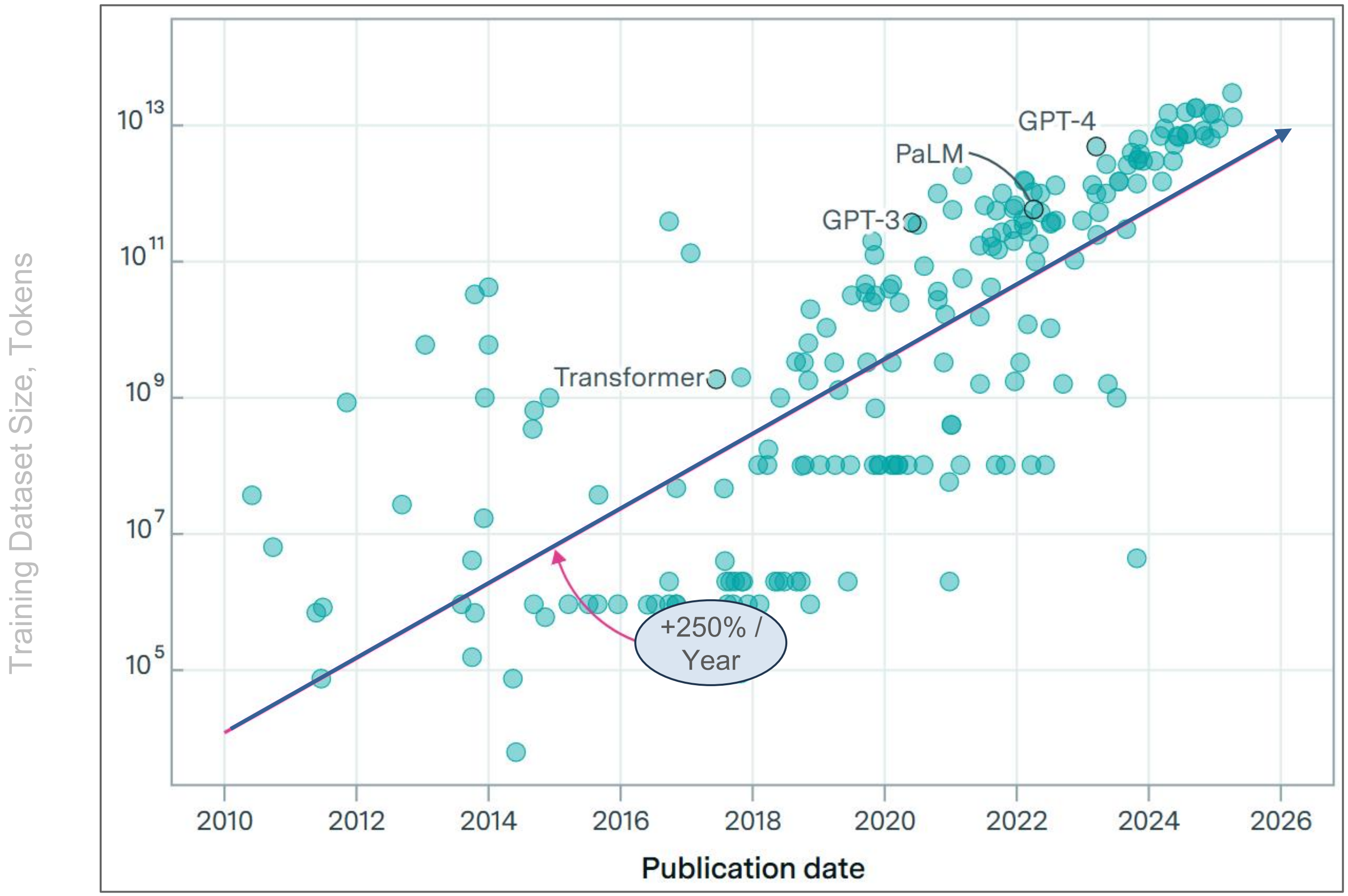


Note: Companies do not report "hyperscaler cloud revenue" on like-for-like basis so data represents best estimates and may not align between companies. Oracle Cloud revenue includes Cloud Services & License Support, as well as Cloud License & On-Premise License. IBM Cloud includes all 'Infrastructure' line items due to reporting standards. Alibaba & Tencent Cloud revenues estimated per Morgan Stanley. Source: Company disclosures, Morgan Stanley (as of 4/25)

CapEx Spend – Big Technology Companies = Inflected With AI's Rise

AI Model Training Dataset Size =
250% Annual Growth Over Fifteen Years, per Epoch AI

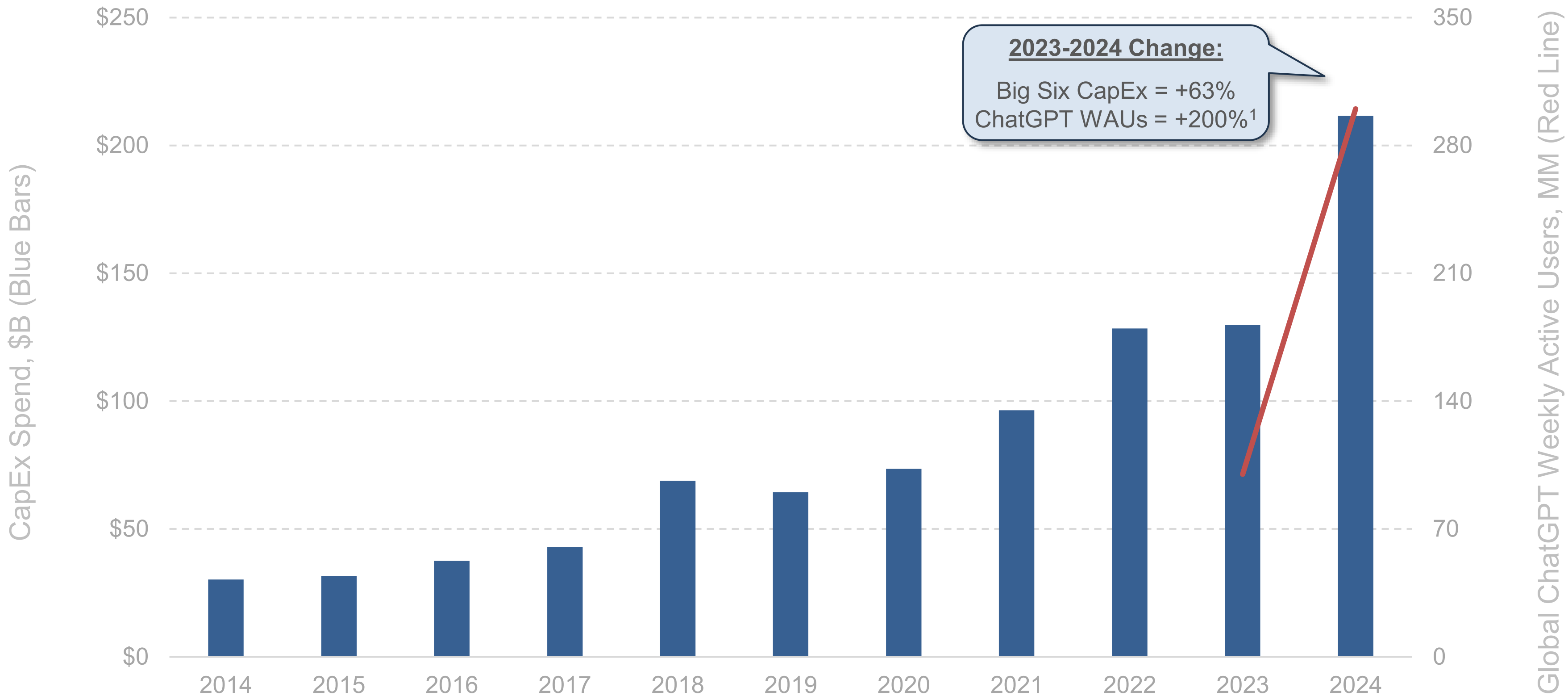
AI Model Training Dataset Size (Tokens) by Model Release Year – 6/10-5/25, per Epoch AI



Note: In AI language models, tokens represent basic units of text (e.g., words or sub-words) used during training. Training dataset sizes are often measured in total tokens processed. A larger token count typically reflects more diverse and extensive training data, which can lead to improved model performance – up to a point – before reaching diminishing returns.
Source: Epoch AI (5/25)

CapEx Spend @ Big Six* Tech Companies = +63% Y/Y & Accelerated…

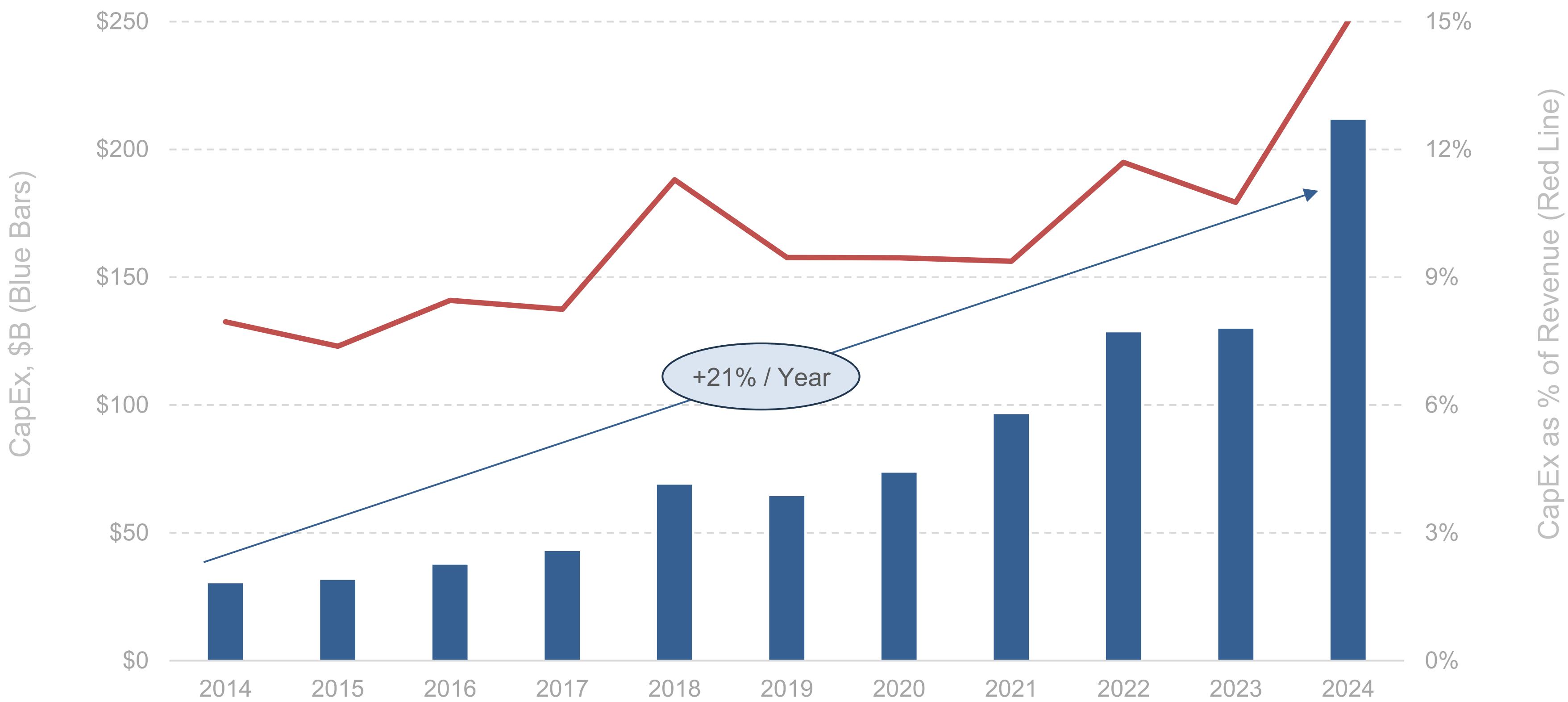
Big Six* USA Public Technology Company CapEx Spend ($B) vs. Global ChatGPT Weekly Active Users (MM) – 2014-2024, per Capital IQ & OpenAI



[1]ChatGPT WAU data as of 11/23 & 12/24 due to data availability.
*Note: Big Six USA technology companies include Apple, Nvidia, Microsoft, Alphabet / Google, Amazon, & Meta Platforms / Facebook. Only AWS CapEx & revenue shown for Amazon (i.e. excludes Amazon retail CapEx). AWS CapEx estimated per Morgan Stanley – equals AWS net additions to property & equipment less finance leases and obligations. Source: Capital IQ (3/25), OpenAI disclosures (3/25)

…CapEx Spend @ Big Six* Tech Companies = 15% of Revenue & Accelerated vs. 8% Ten Years Ago

Big Six* USA Public Technology Company – CapEx Spend ($B) vs. % of Revenue – 2014-2024, per Capital IQ & Morgan Stanley



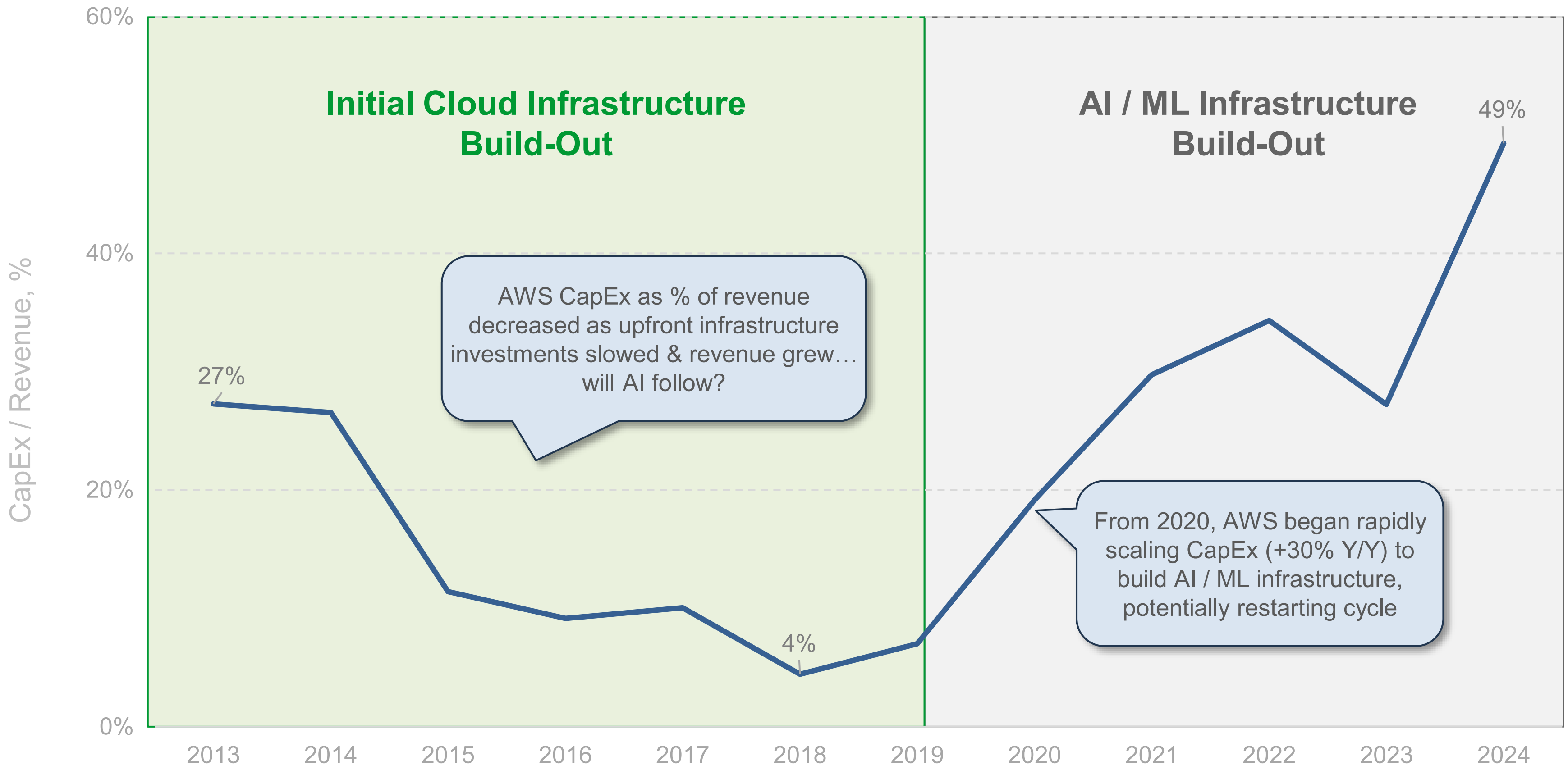
Note: Big Six USA technology companies include Apple, Nvidia, Microsoft, Alphabet / Google, Amazon, & Meta Platforms / Facebook. Only AWS CapEx & revenue shown for Amazon (i.e. excludes Amazon retail CapEx). AWS CapEx estimated per Morgan Stanley – equals AWS net additions to property & equipment less finance leases and obligations.
Source: Capital IQ (3/25), Morgan Stanley (5/25)

CapEx Spend @ Amazon AWS =

Cloud vs. AI Patterns

CapEx as % of Revenue (AWS as Proxy) – AI vs. Cloud Buildouts = 49% (2024) vs. 4% (2018) vs. 27% (2013), per Morgan Stanley

Amazon AWS CapEx as % of Revenue – 2013-2024, Estimated per Morgan Stanley



Note: Figures shown represent AWS only. AWS CapEx estimated per Morgan Stanley – equals AWS net additions to property & equipment less finance leases and obligations.
Source: Amazon, Morgan Stanley (5/25)

Tech CapEx Spend Partial Instigator =

Material Improvements in GPU Performance

NVIDIA GPU Performance = +225x Over Eight Years

Performance of NVIDIA GPU Series Over Time – 2016-2024, per NVIDIA

$1B Data Center Comparison
GPT-MoE Inference Workload[1]

	Pascal 2016	Volta 2018	Ampere 2020	Hopper 2022	Blackwell 2024	
Number of GPUs	46K	43K	28K	16K	11K	**+225x**
Factory AI FLOPS	1EF	5EF	17EF	63EF	220EF	
Annual Inference Tokens	50B	1T	5T	58T	1,375T	**+30,000x**
Annual Token Revenue	$240K	$3M	$24M	$300M	$7B	
DC Power	37MW	34MW	25MW	19MW	21MW	**+50,000x**
Token Per MW-Year	1.3B	2.9B	200B	3T	65T	

For a Theoretical $1B-Scale Data Center…

…Performance +225x over eight years while requiring 4x fewer GPUs…

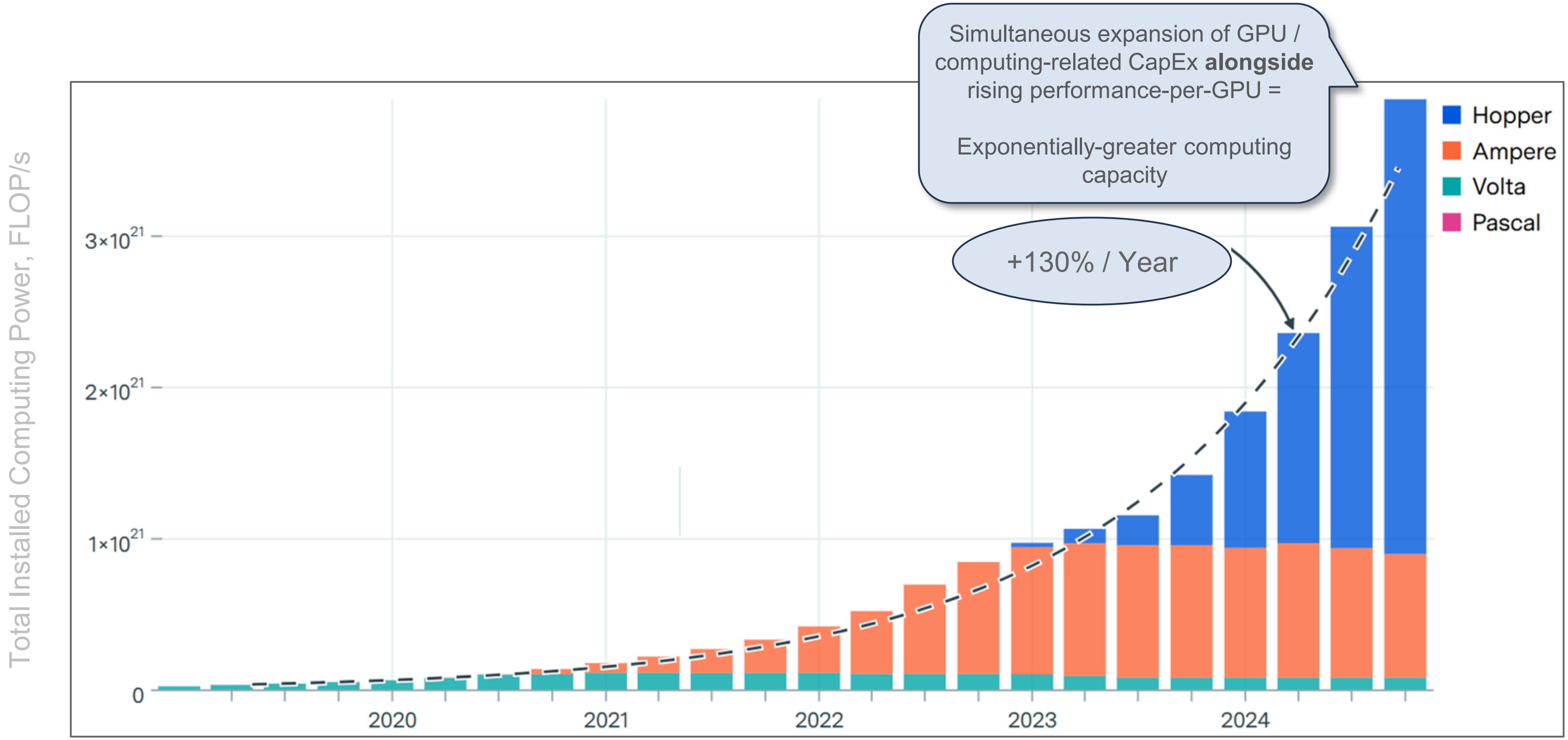
…Inference token capacity +27,500x over eight years, implying +30,000x higher theoretical token revenue…

…Data center power use down 43% over eight years, leading to +50,000x greater per-unit energy efficiency

[1] GPT-MoE Inference Workload = A type of workload where a GPT-style model with a Mixture-of-Experts (MoE) architecture is used for inference (i.e., making predictions).
Note: Annual token revenue assumes a flat per-token cost. Source: NVIDIA (5/25)

NVIDIA Installed GPU Computing Power = 100x+ Growth Over ~Six Years

Global Stock of NVIDIA GPU Computing Power (FLOP/s) – Q1:19-Q4:24, per Epoch AI



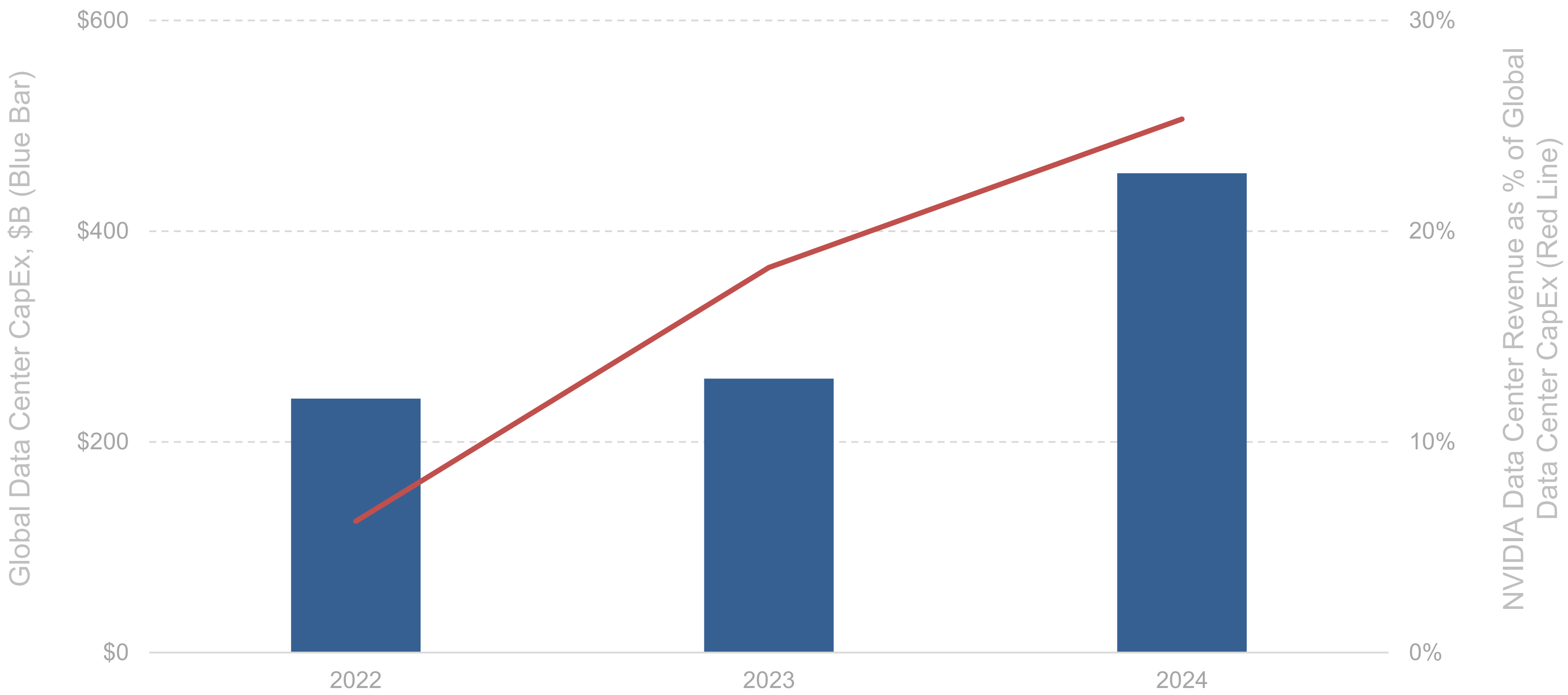
Note: Analysis does not include TPUs or other specialized AI accelerators, for which less data is available. TPUs may provide comparable total computing power to NVIDIA chips.
Source: Epoch AI (2/25)

Tech CapEx Spend Beneficiary =

NVIDIA

Key Tech CapEx Spend Beneficiary = NVIDIA…
25% & Rising of Global Data Center CapEx, per NVIDIA

Global Data Center CapEx ($B) vs. NVIDIA's Data Center Revenue as Percent of Data Center CapEx (Global) – 2022-2024, per NVIDIA @ GTC



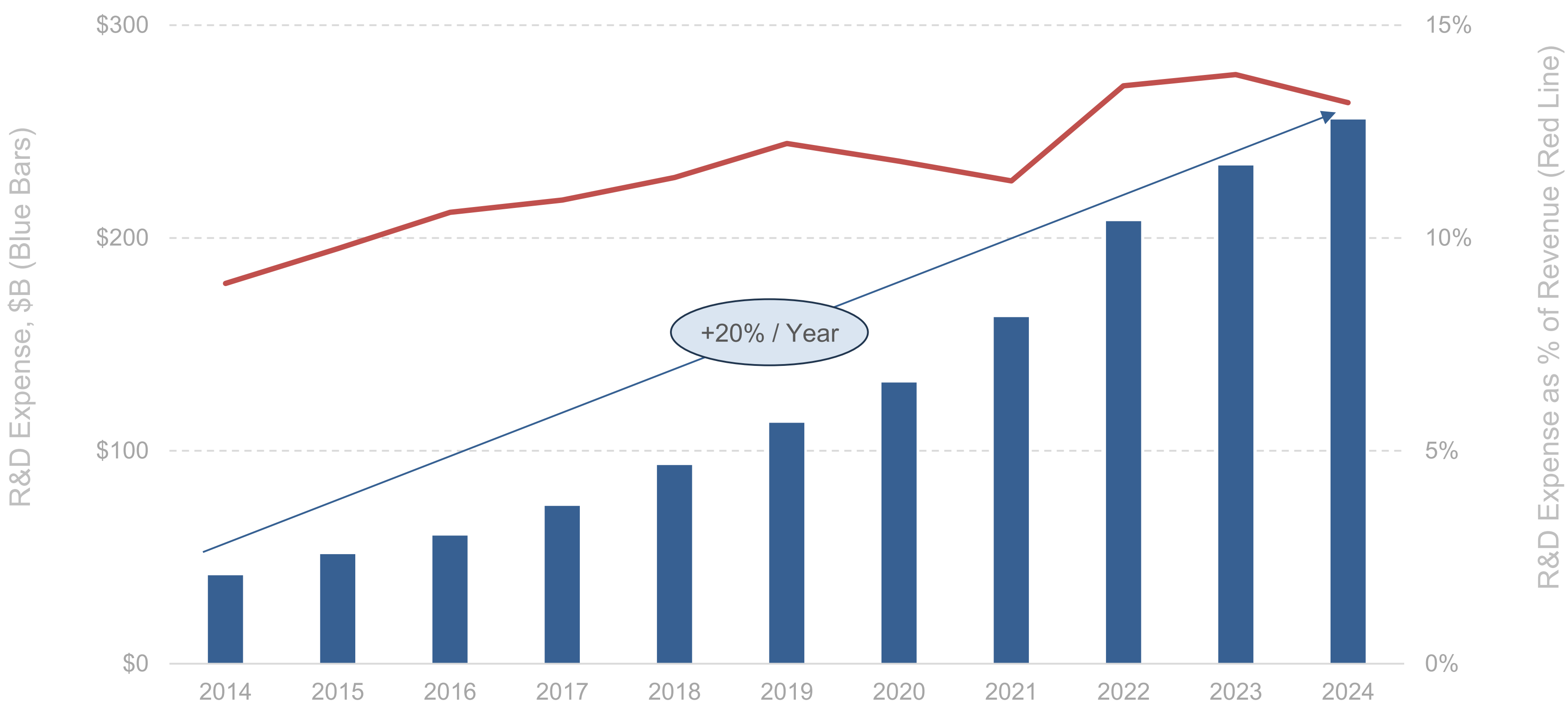
Note: NVIDIA data represents January FYE (e.g., 2024 = FY25 ending 1/25) vs calendar year for data center CapEx. Data presented by Jensen Huang at NVIDIA GTC 2025 (link).
Source: Dell'Oro Research for CapEx (3/25); NVIDIA for data center revenue (3/25)

Technology Company Spend =

R&D Rising Along with CapEx

R&D Spend @ Big Six* USA Public Tech Companies = 13% of Revenue…vs. 9% Ten Years Ago

Big Six* USA Public Technology Company – R&D Spend ($B) vs. % of Revenue – 2014-2024, per Capital IQ



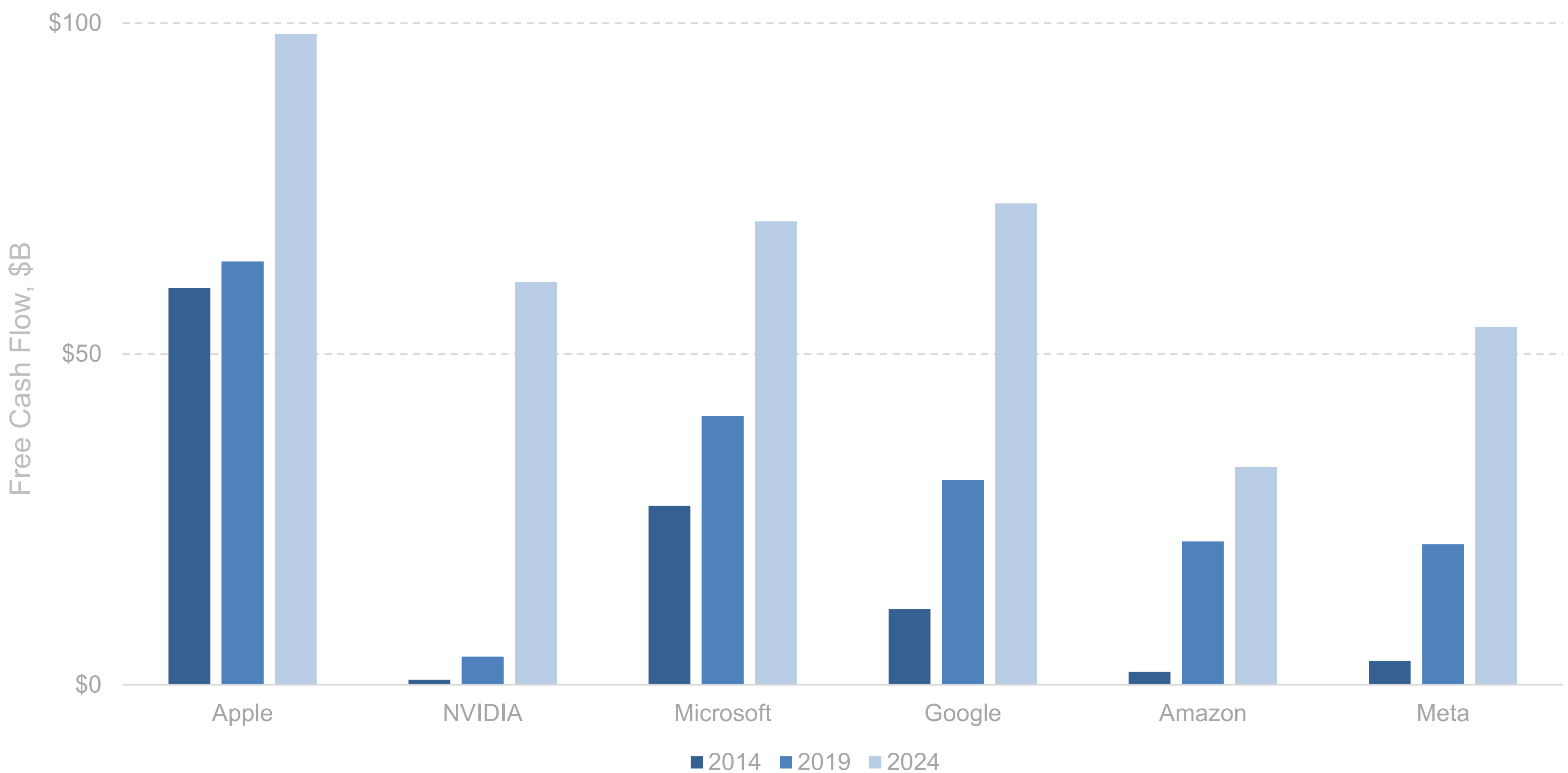
**Note: Big Six USA technology companies include Apple, Nvidia, Microsoft, Alphabet / Google, Amazon, & Meta Platforms / Facebook. R&D expense shown for Amazon, not AWS, as figures are not broken out in company financials; revenue therefore shown on like-for-like basis. Source: Capital IQ (3/25)*

Tech Big Six (USA) =

Loaded With Cash to Spend on AI & CapEx

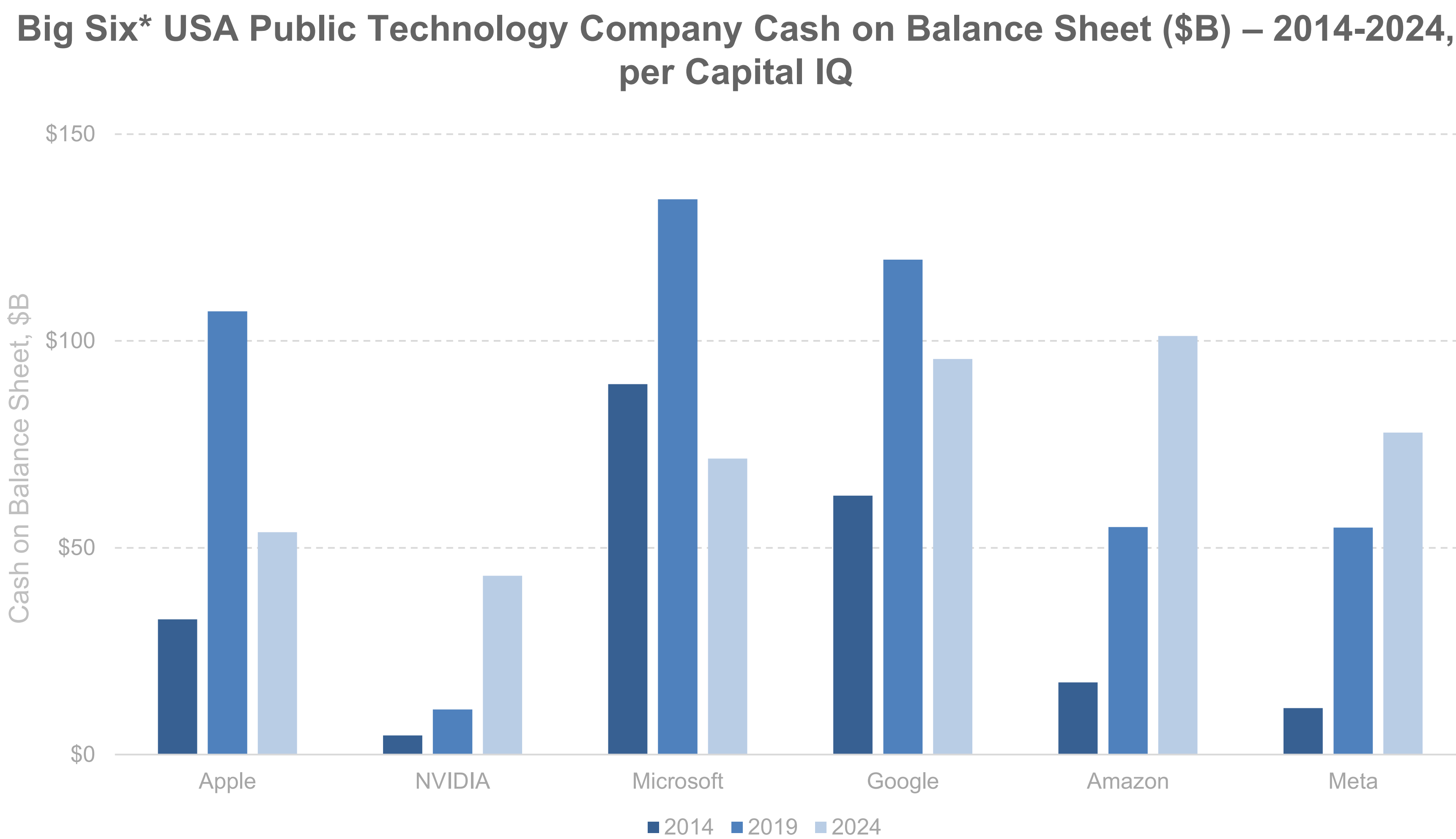
Big Six* Generating Loads of Cash =
+263% Growth in Free Cash Flow Over Ten Years to $389B…

Big Six* Public Technology Companies – Free Cash Flow ($B) – 2014-2024, per Capital IQ



Note: Big Six USA technology companies include Apple, Nvidia, Microsoft, Alphabet / Google, Amazon, & Meta Platforms / Facebook. FCF calculated as cash flow from operations less capex to standardize definitions, as only some companies subtract finance leases and Amazon adjusts FCF for gains on sale of equipment. FCF shown for Amazon, not AWS, as figures are not broken out in company financials. Source: Capital IQ (3/25)

…Big Six* Generating Loads of Cash =
+103% Growth in Cash Over Ten Years to $443B



*Note: Big Six USA technology companies include Apple, Nvidia, Microsoft, Alphabet / Google, Amazon, & Meta Platforms / Facebook. Figure measures cash and other equivalents (e.g., short-term investments and marketable securities) on companies' balance sheets. Source: Capital IQ (3/25)

Tech CapEx Spend Driver =

Compute Spend to Train & Run AI Models

To understand the evolution of AI computing economics, it's constructive to look at where costs are concentrated – And where they're headed. The bulk of spending in AI large language model (LLM) development is still dominated by compute – specifically, the compute needed to train and run models. Training costs remain extraordinarily high and are rising fast, often exceeding $100 million per model today. As Dario Amodei, CEO of Anthropic, noted in mid-2024, *Right now, [AI model training costs] $100 million. There are models in training today that are more like a billion… I think that the training of…$10 billion models, yeah, could start sometime in 2025.*

Around these core compute costs sit additional high-cost layers: research, data acquisition and hosting, and a mix of salaries, general overhead, and go-to-market operations. Even as the cost to train models climbs, a growing share of total AI spend is shifting toward inference – the cost of running models at scale in real-time. Inference happens constantly, across billions of prompts, queries, and decisions, whereas model training is episodic. As Amazon CEO Andy Jassy noted in his April 2025 letter to shareholders, *While model training still accounts for a large amount of the total AI spend, inference… will represent the overwhelming majority of future AI cost because customers train their models periodically but produce inferences constantly.* NVIDIA Co-Founder & CEO Jensen Huang noted the same in NVIDIA's FQ1:26 earnings call, saying *Inference is exploding. Reasoning AI agents require orders of magnitude more compute.*

At scale, inference becomes a persistent cost center – one that grows in parallel with usage, despite declines in unit inference costs.

The broader dynamic is clear: lower per-unit costs are fueling higher overall spend. As inference becomes cheaper, AI gets used more. And as AI gets used more, total infrastructure and compute demand rises – dragging costs up again. The result is a flywheel of growth that puts pressure on cloud providers, chipmakers, and enterprise IT budgets alike.

The economics of AI are evolving quickly – but for now, they remain driven by heavy capital intensity, large-scale infrastructure, and a race to serve exponentially expanding usage.

Data Centers =

Key Beneficiary of AI CapEx Spend

For one lens into the economics of AI infrastructure,
it's useful to look at the pace and scale of data center construction.
The current wave of AI-driven demand has pushed data center spending to historic highs.
According to Dell'Oro Research, global IT company data center CapEx
reached $455 billion in 2024 and is accelerating.

Hyperscalers and AI-first companies alike are pouring billions into building out
compute-ready capacity – not just for storage, but for real-time inference and
model training workloads that require dense, high-power hardware.
As AI moves from experimental to essential, so too do data centers.
Per NVIDIA Co-Founder and CEO Jensen Huang, *These AI data centers…are, in fact, AI factories*.

That race is moving faster than many expected.
The most striking example may be xAI's Colossus facility in Memphis, Tennessee which went
from a gutted factory to a fully operational AI data center in just 122 days.
As noted on page 122, at 750,000 square feet – roughly the size of 418 average USA homes –
it was built in half the time it typically takes to construct a single American house.

Per NVIDIA Co-Founder & CEO Jensen Huang,
What they achieved is singular, never been done before…That is, like, superhuman…

…These kinds of timelines are no longer the exception. With prefabricated modules, streamlined permitting, and vertical integration across electrical, mechanical, and software systems, new data centers are going up at speeds that resemble consumer tech cycles more than real estate development.

But beneath that velocity lies a capital model that's anything but simple. CapEx is driven by land, power provisioning, chips, and cooling infrastructure – especially as AI workloads push thermal and power limits far beyond traditional enterprise compute. OpEx, by contrast, is dominated by energy costs and systems maintenance, particularly for high-density training clusters that operate near constant load.

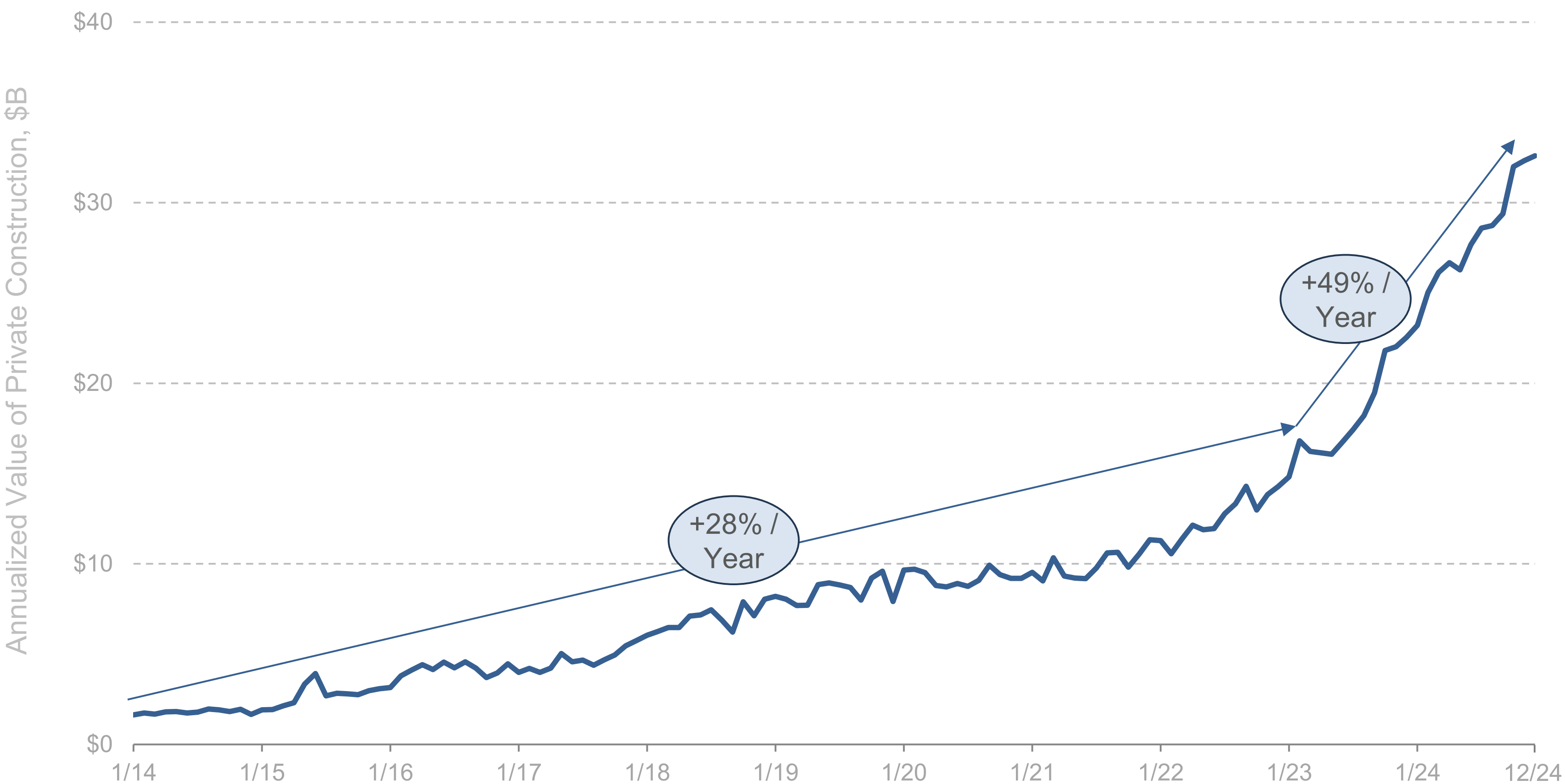
Revenue is driven by compute sales – whether in the form of AI APIs, enterprise platform fees, or internal productivity gains. But payback periods are often long, especially for vertically-integrated players building ahead of demand. For newer entrants, monetization may lag build-out by quarters or even years.

And then there's the supply chain. Power availability is becoming more of a gating factor. Transformers, substations, turbines, GPUs, cables – these aren't commodities that can be spun up overnight. In this context, data centers aren't just physical assets – they are strategic infrastructure nodes. They sit at the intersection of real estate, power, logistics, compute, and software monetization.

The companies that get this right may do more than run servers – they will shape the geography of AI economics for the next decade.

Data Center Buildout Construction Value, USA = +49% & Accelerated Annual Growth Over Two Years

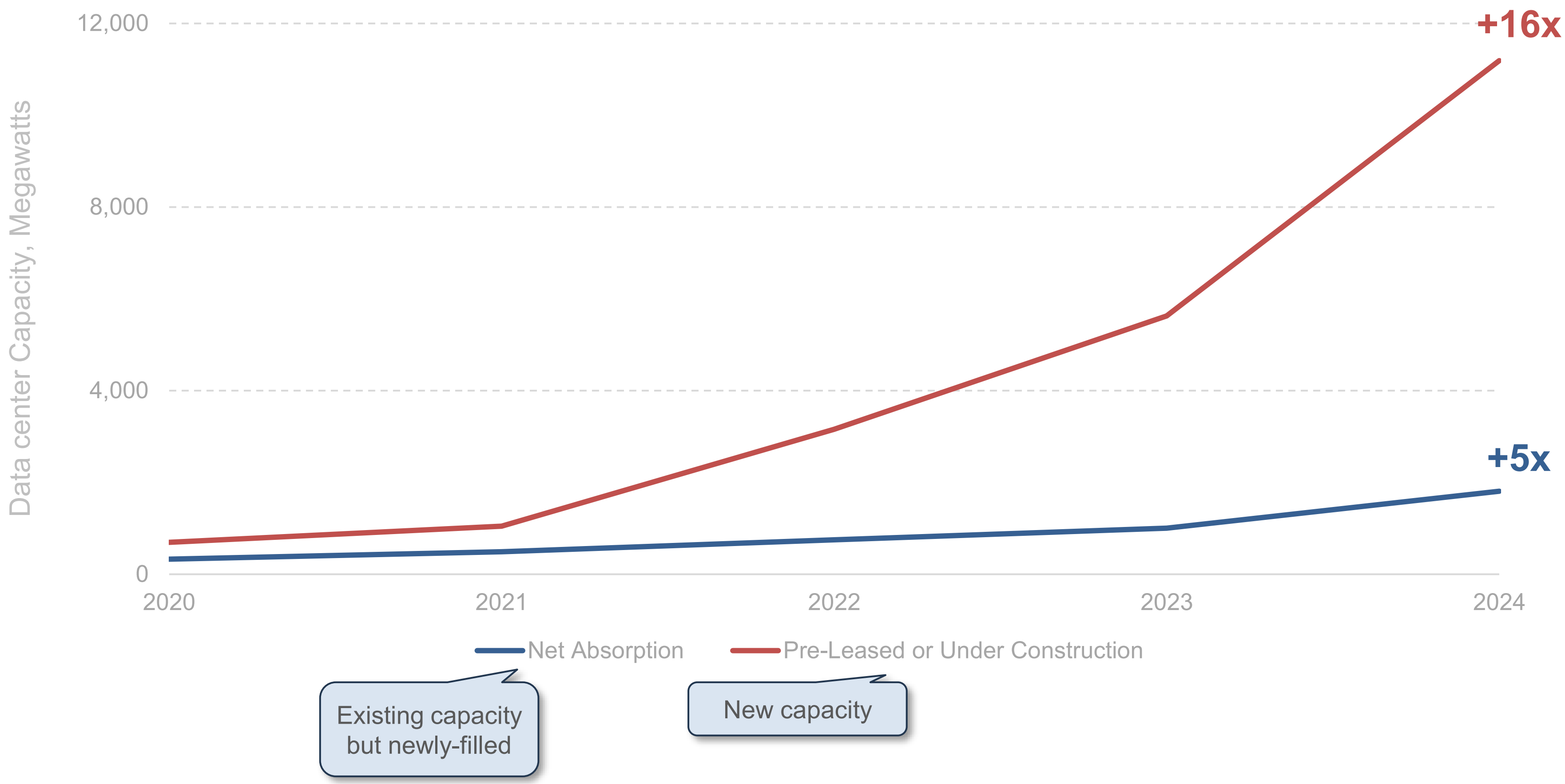
USA Data Center Annualized Private Construction Value ($B) – 1/14-12/24, per USA Census Bureau



Note: All data are seasonally adjusted. Data obtained via USA Census Bureau's Value of Construction Put in Place (VIP) Survey, which provides monthly estimates of the total dollar value of construction work done in USA. Data is annualized to avoid seasonal fluctuations. Source: USA Census Bureau

Data Center New Construction vs. Existing Capacity, USA = +16x in New vs. +5x in Existing Over Four Years

Data Center Capacity (Megawatts) by Real Estate Profile, USA Primary Markets – 2020-2024, per CBRE



Note: Primary USA markets only (Northern Virginia, Atlanta, Chicago, Phoenix, Dallas-Ft. Worth, Hillsboro, Silicon Valley, New York Tri-State.)
Source: CBRE, 'North America Data Center Trends H2 2024' (2/25)

Data Center Build Time (xAI Colossus as Proxy) = 122 Days vs. 234 for a Home

122 Days =
A Fully-Operational Data Center – 2024…
750,000 Sq. Ft = Size of 418 USA Homes

750,000 Square Feet

We were told it would take 24 months to build. So we took the project into our own hands, questioned everything, removed whatever was unnecessary, and accomplished our goal in four months.

- xAI Website

122 Days =
One Half-Built House – 2024
(Average Build Time = 234 Days)

1,792 Square Feet

Note: Median USA home size shown as of January 2025, per FRED. Colossus was built in a former Electrolux factory in Memphis, TN, USA. Average build time shown for single-unit buildings. Measures time between start of onsite work & completion. Data reported in 2024 but measures build times for homes started in 2023.
Source: xAI, USA Census Bureau, Federal Reserve Bank of St. Louis, Wikimedia Commons

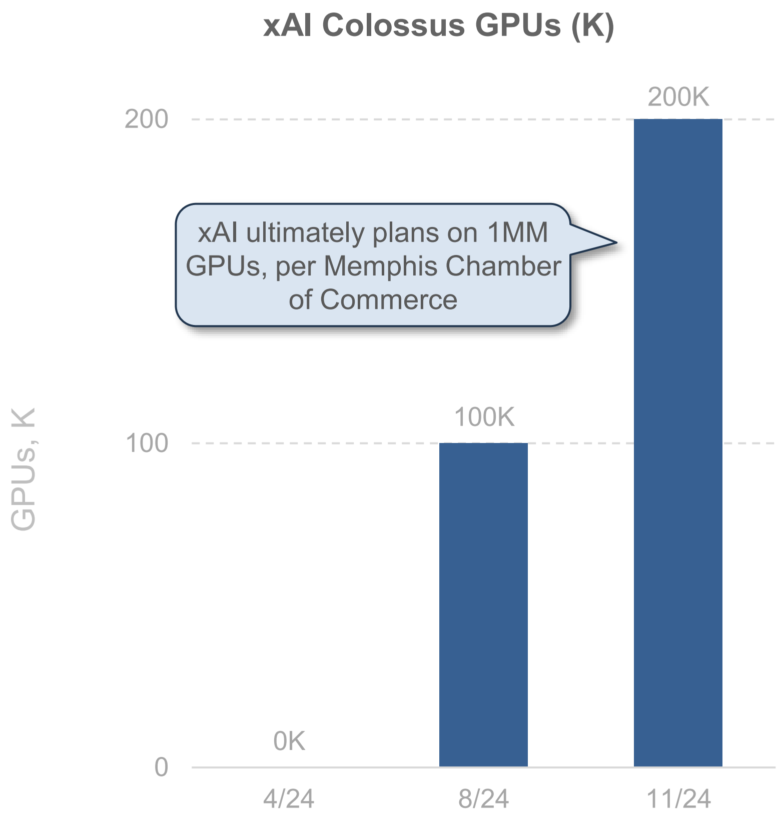
Data Center Compute (xAI Colossus as Proxy) = 0 to 200,000 GPUs in Seven Months

xAI Colossus GPUs – 4/24-11/24, per xAI

We're running the world's biggest supercomputer, Colossus. Built in 122 days – outpacing every estimate – it was the most powerful AI training system yet. Then we doubled it in 92 days to 200k GPUs. This is just the beginning…

…We doubled our compute at an unprecedented rate, with a roadmap to 1M GPUs. Progress in AI is driven by compute and no one has come close to building at this magnitude and speed.

- xAI Website, 5/25



Note: We assume 200,000 GPUs as of 11/30/24 per xAI's disclosure that 'we doubled [GPU count] in 92 days to 200K GPUs.' xAI Colossus ran its first job across 4 data halls on 8/30/24. We assume zero GPUs as of construction start date (122 days prior to assumed opening date of 8/30/24).
Source: xAI (5/25), Memphis Chamber of Commerce (12/24)

Data Centers =

Electricity Guzzlers

AI and energy observations / quotes (in italics) here and the two pages that follow are from 'World Energy Outlook Special Report – Energy and AI' (link) from IEA (International Energy Agency)* – 4/10/25

To understand where energy infrastructure is heading, it helps to examine the rising tension between AI capability and electrical supply. The growing scale and sophistication of artificial intelligence is demanding an extraordinary amount of computational horsepower, primarily from AI-focused data centers. These facilities – purpose-built to train and serve models – are starting to rival traditional heavy industry in their electricity consumption.

There is no AI without energy – specifically electricity (p. 3). Data centers *accounted for around 1.5% of the world's electricity consumption in 2024 (p. 14).* Energy demand growth has been rapid: *Globally, data centre electricity consumption has grown by around 12% per year since 2017, more than four times faster than the rate of total electricity consumption (p. 14).* As power demand rises, so too does its concentration: *The United States accounted for…[45% of global data centre electricity consumption], followed by China (25%) and Europe (15%)… nearly half of data centre capacity in the United States is in five regional clusters (p. 14).*

The flipside is true as well: *Emerging and developing economies other than China account for 50% of the world's internet users but less than 10% of global data centre capacity (p. 18)…*

…AI's power demands are increasing – and its progress is increasingly bottlenecked not by data or algorithms, but by the grid and strains related to demand.

While AI presently places considerable demands on the energy sector,
it is also already unlocking major energy efficiency and operational gains…
AI is already being deployed by energy companies to transform and optimize energy and mineral supply, electricity generation and transmission, and energy consumption (p. 16).

Current AI-driven demand is extremely high.
This is forecast to continue, especially as capital gushes into model providers that, in turn, spend on more compute. At some point, these model builders will need to turn a profit to be able to spend more.

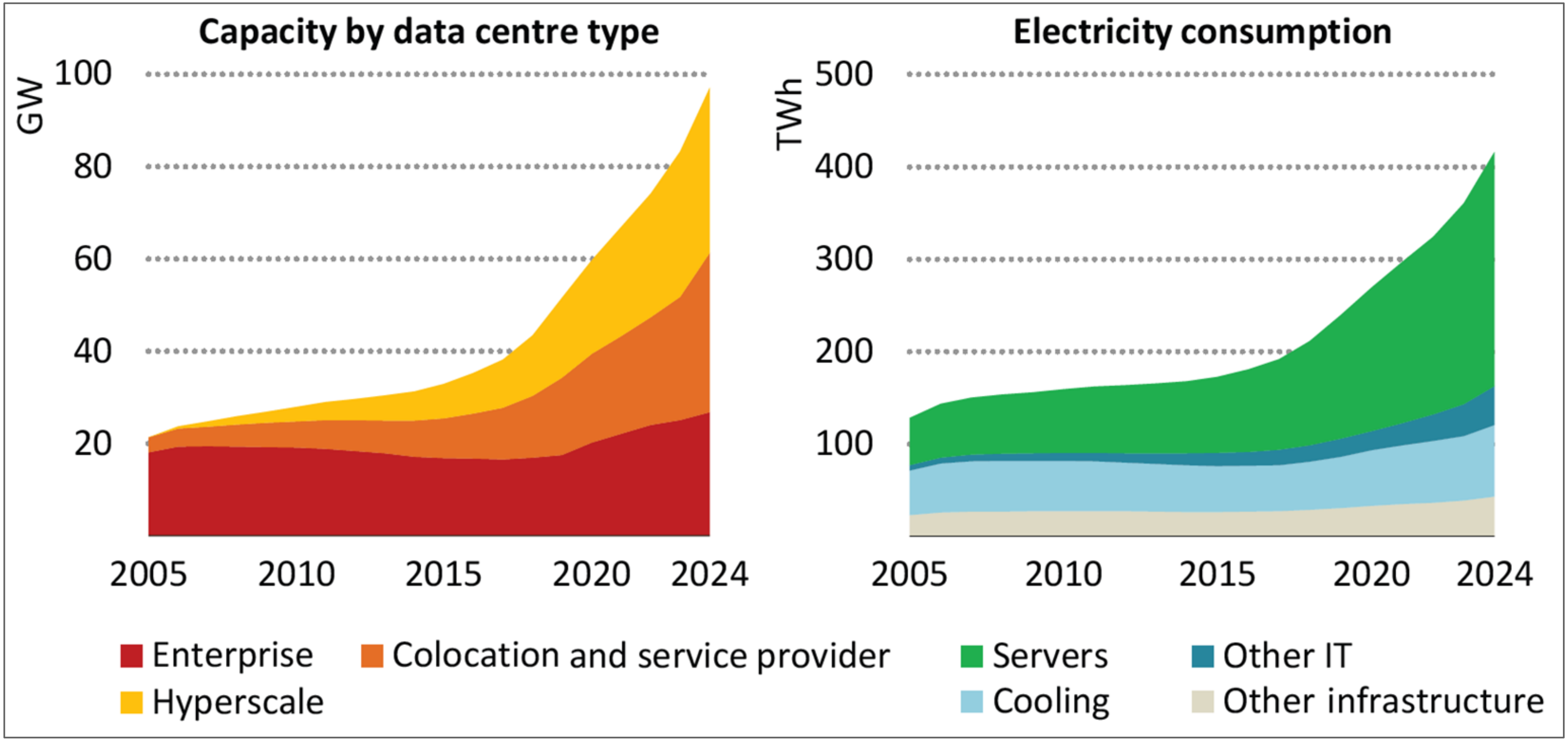
While demand – for both compute and energy – will inevitably continue to rise as consumer and business usage does the same, data centers will ultimately only serve those who pay their bills.

***IEA member countries include Australia, Austria, Belgium, Canada, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Japan, S. Korea, Latvia, Lithuania, Luxembourg, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Slovak Republic, Spain, Sweden, Switzerland, Republic of Türkiye, United Kingdom, and United States. IEA Association countries include Argentina, Brazil, China, Egypt, India, Indonesia, Kenya, Morocco, Senegal, Singapore, S. Africa, Thailand, and Ukraine.**

All data shown, unless otherwise specified, is global. Italicized text is directly quoted from the report.

Data Center Electricity Consumption, Global = +3x Over Nineteen Years, per IEA

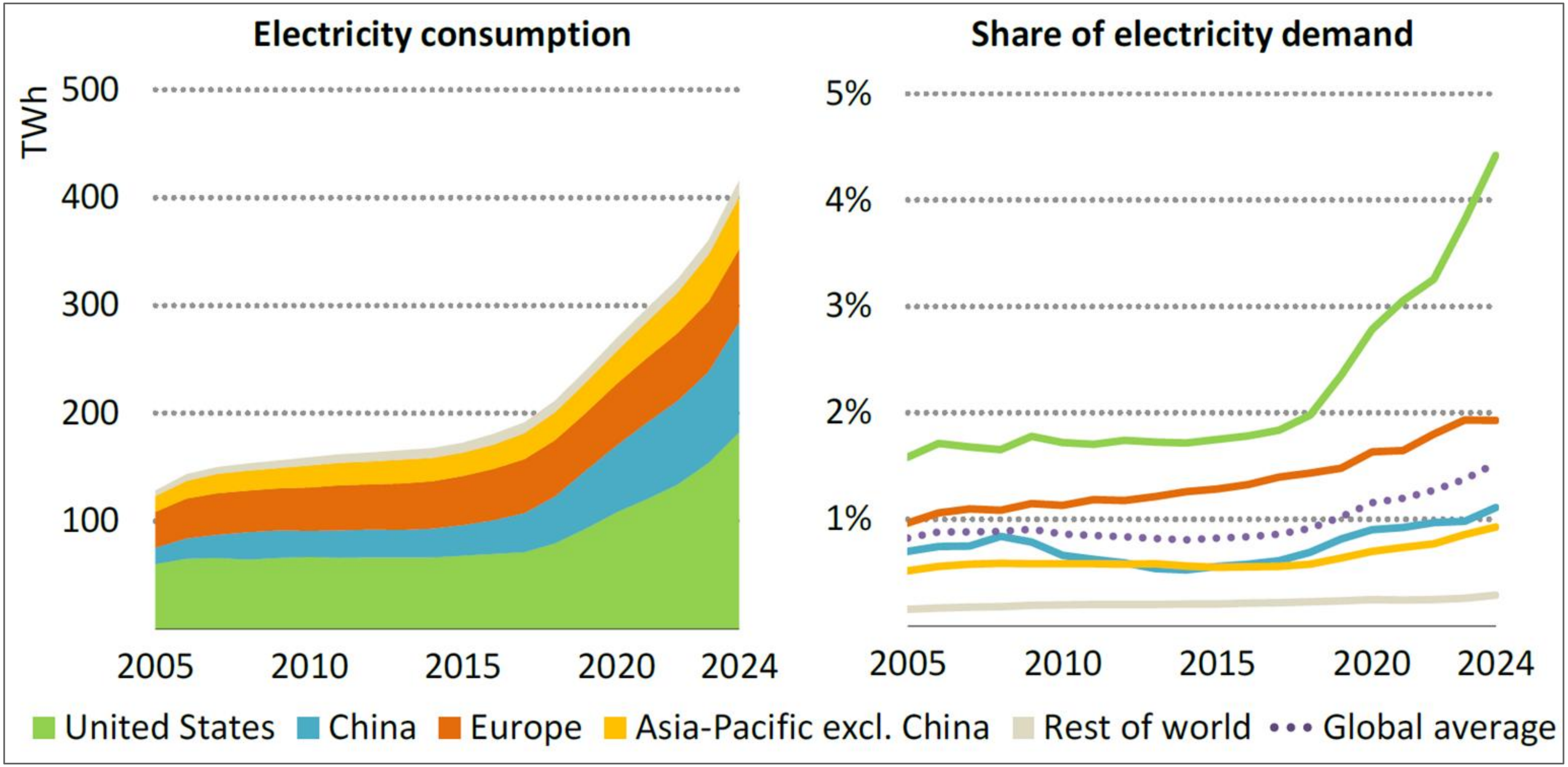
Data Center Energy Consumption by Data Center Type & Equipment, Global – 2005-2024, per IEA



Source: International Energy Agency (IEA), 'Energy and AI' (4/25)

Data Center Electricity Consumption by Region = USA Leads, per IEA

Data Center Electricity Consumption by Region – 2005-2024, per IEA



Source: International Energy Agency (IEA), 'Energy and AI' (4/25)

Outline

1. **Seem Like Change Happening Faster Than Ever?**
   Yes, It Is
2. **AI User + Usage + CapEx Growth =**
   Unprecedented
3. **AI Model Compute Costs High / Rising + Inference Costs Per Token Falling =**
   Performance Converging + Developer Usage Rising
4. **AI Usage + Cost + Loss Growth =**
   Unprecedented
5. **AI Monetization Threats =**
   Rising Competition + Open-Source Momentum + China's Rise
6. **AI & Physical World Ramps =**
   Fast + Data-Driven
7. **Global Internet User Ramps Powered by AI from Get-Go =**
   Growth We Have Not Seen Likes of Before
8. **AI & Work Evolution =**
   Real + Rapid

To understand where AI model economics may be heading, one can look at the mounting tension between capabilities and costs.

Training the most powerful large language models (LLMs) has become one of the most expensive / capital-intensive efforts in human history. As the frontier of performance pushes toward ever-larger parameter counts and more complex architectures, model training costs are rising into the billions of dollars.

Ironically, this race to build the most capable general-purpose models may be accelerating commoditization and driving diminishing returns, as output quality converges across players and differentiation becomes harder to sustain. At the same time, the cost of applying/using these models – known as inference – is falling quickly. Hardware is improving – for example, NVIDIA's 2024 Blackwell GPU consumes 105,000x less energy per token than its 2014 Kepler GPU predecessor. Couple that with breakthroughs in models' algorithmic efficiency, and the cost of inference is plummeting.

Inference represents a new cost curve, and – unlike training costs – it's arcing down, not up. As inference becomes cheaper and more efficient, the competitive pressure amongst LLM providers increases – not on accuracy alone, but also on latency, uptime, and cost-per-token*. What used to cost dollars can now cost pennies. And what cost pennies may soon cost fractions of a cent.

The implications are still unfolding. For users (and developers), this shift is a gift: dramatically lower unit costs to access powerful AI. And as end-user costs decline, creation of new products and services is flourishing, and user and usage adoption is rising.

For model providers, however, this raises real questions about monetization and profits. Training is expensive, serving is getting cheap, and pricing power is slipping. The business model is in flux. And there are new questions about the one-size-fits-all LLM approach, with smaller, cheaper models trained for custom use cases** now emerging.

Will providers try to build horizontal platforms? Will they dive into specialized applications? Only time will tell. In the short term, it's hard to ignore that the economics of general-purpose LLMs look like commodity businesses with venture-scale burn.

**Cost-per-token = The expense incurred for processing or generating a single token (a word, sub-word, or character) during the operation of a language model. It is a key metric used to evaluate the computational efficiency and cost-effectiveness of deploying AI models, particularly in applications like natural language processing.*
***E.g., OpenEvidence*

<u>AI Model Compute Costs High / Rising</u>

+

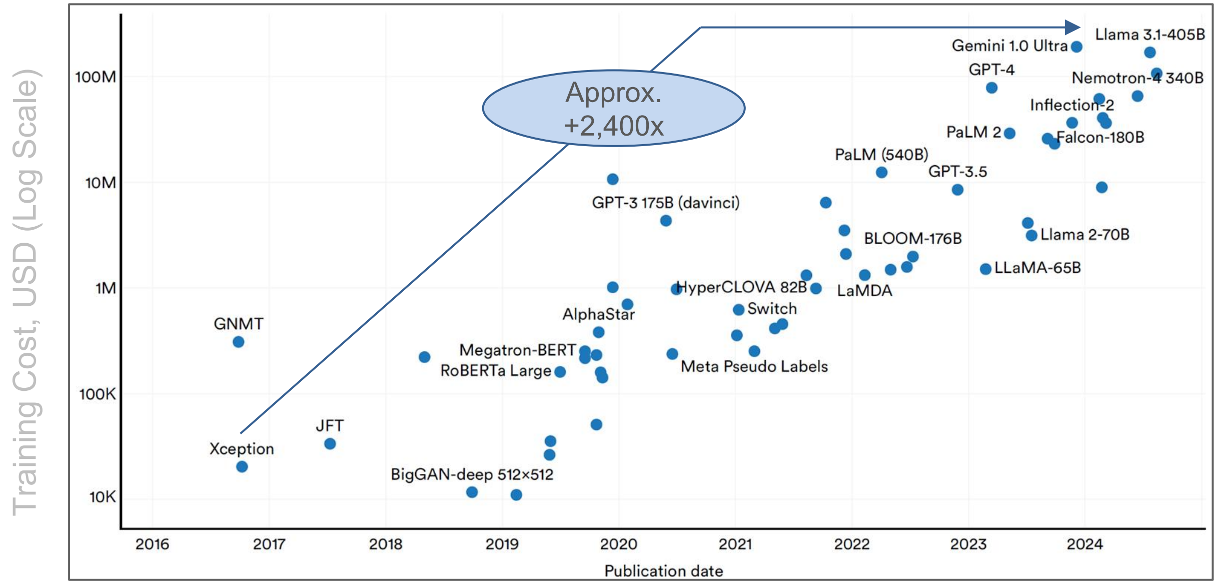
Inference Costs Per Token Falling

=

Performance Converging + Developer Usage Rising

AI Model Training Compute Costs = ~2,400x Growth Over Eight Years, per Epoch AI & Stanford

Estimated Training Cost of Frontier AI Models – 2016-2024, per Epoch AI & Stanford



Right now, [AI model training costs] $100 million. There are models in training today that are more like a billion. Right. I think if we go to $10 or $100 billion, and I think that will happen in 2025, 2026, maybe 2027…

…I think that the training of…$10 billion models, yeah, could start sometime in 2025.

- Anthropic Co-Founder & CEO Dario Amodei (6/24)

Note: Costs are estimates. Excludes most Chinese models due to lack of reliable cost data. Source: Epoch AI via Nestor Maslej et al., 'The AI Index 2025 Annual Report,' AI Index Steering Committee, Stanford HAI (4/25); In Good Company podcast (6/24)

AI Model Compute Costs High / Rising

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Inference Costs Per Token Falling

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Performance Converging + Developer Usage Rising

To understand the trajectory of AI compute, it helps to revisit an idea from the early days of PC software. 'Software is a gas…it expands to fill its container,' said Nathan Myhrvold, then CTO of Microsoft in 1997. AI is proving no different. As models get better, usage increases – and as usage increases, so does demand for compute. We're seeing it across every layer: more queries, more models, more tokens per task. The appetite for AI isn't slowing down. It's growing into every available resource – just like software did in the age of desktop and cloud.

But infrastructure is not just standing still. In fact, it's advancing faster than almost any other layer in the stack, and at unprecedented rates. As noted on page 136, NVIDIA's 2024 Blackwell GPU uses 105,000 times less energy to generate tokens than its 2014 Kepler predecessor. It's a staggering leap, and it tells a deeper story – not just of cost reduction, but of architectural and materials innovation that is reshaping what's possible at the hardware level.

These improvements in hardware efficiency are critical to offset the strain of increasing AI and internet usage on our grid. So far, though, they have not been enough. This trend aligns with Jevons Paradox, first proposed back in 1865* – that technological advancements that improve resource efficiency actually lead to increased overall usage of those resources. This is driving new focus on expanding energy production capacity – and new questions about the grid's ability to manage.

Yet again, we see this as one of the perpetual 'a-ha's' of technology: costs fall, performance rises, and usage grows, all in tandem. This trend is repeating itself with AI.

**British economist William Stanley Jevons first observed this phenomenon in 19th-century Britain, where he noticed that improvements in the efficiency of coal-powered steam engines were not reducing coal consumption but rather increasing it. In his book The Coal Question, he noted 'It is wholly a confusion of ideas to suppose that the economical use of fuel is equivalent to diminished consumption. The very contrary is the truth.'*

AI Inference ‘Currency’ = Tokens

What are tokens and how to count them?

Updated over 3 months ago

What are tokens?

Tokens can be thought of as pieces of words. Before the API processes the request, the input is broken down into tokens. These tokens are not cut up exactly where the words start or end - tokens can include trailing spaces and even sub-words. Here are some helpful rules of thumb for understanding tokens in terms of lengths:

- 1 token ~= 4 chars in English
- 1 token ~= ¾ words
- 100 tokens ~= 75 words

Or

- 1-2 sentence ~= 30 tokens
- 1 paragraph ~= 100 tokens
- 1,500 words ~= 2048 tokens

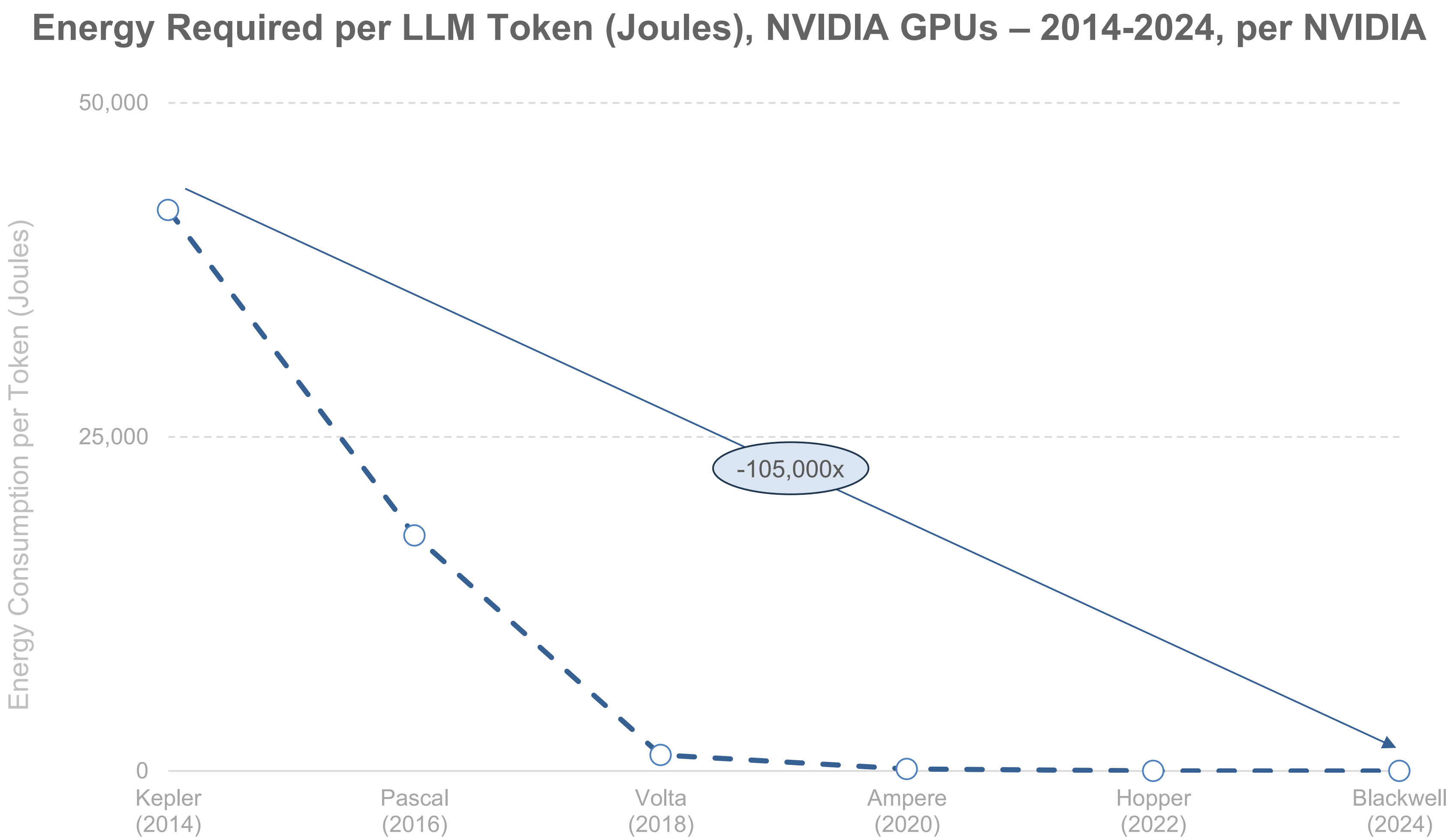
To get additional context on how tokens stack up, consider this:

- Wayne Gretzky’s quote "*You miss 100% of the shots you don't take*" contains 11 tokens.
- OpenAI’s charter contains 476 tokens.
- The transcript of the US *Declaration of Independence* contains 1,695 tokens.

Additional context: 1MM tokens = ~750,000 words…roughly
- 3,500 pages of a standard book (12-point font, double-spaced)
- 5,000 ChatGPT responses*

*Assumes that the average ChatGPT interaction consumes 200 total tokens (input + output), or 150 words. Thus, 1MM tokens equates to roughly 5,000 ChatGPT responses.
Source: OpenAI (1/25)

AI Inference Costs – NVIDIA GPUs =
-105,000x Decline in Energy Required to Generate Token Over Ten Years

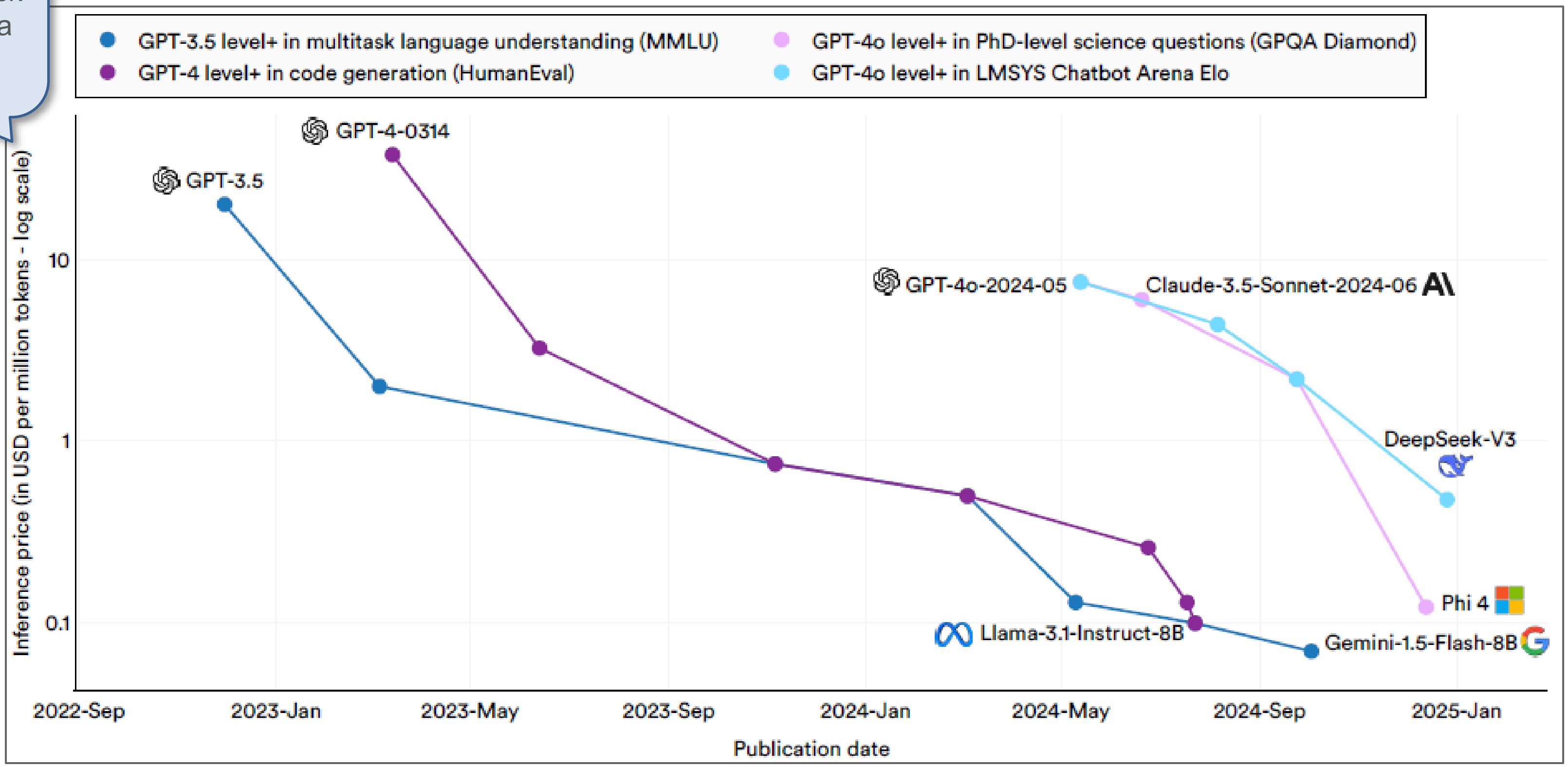


Note: Kepler released in 2012. NVIDIA materials mark performance threshold shown above for Kepler as of 2014. Source: NVIDIA Company Overview (2/25)

AI Inference Costs – Serving Models = 99.7% Lower Over Two Years, per Stanford HAI

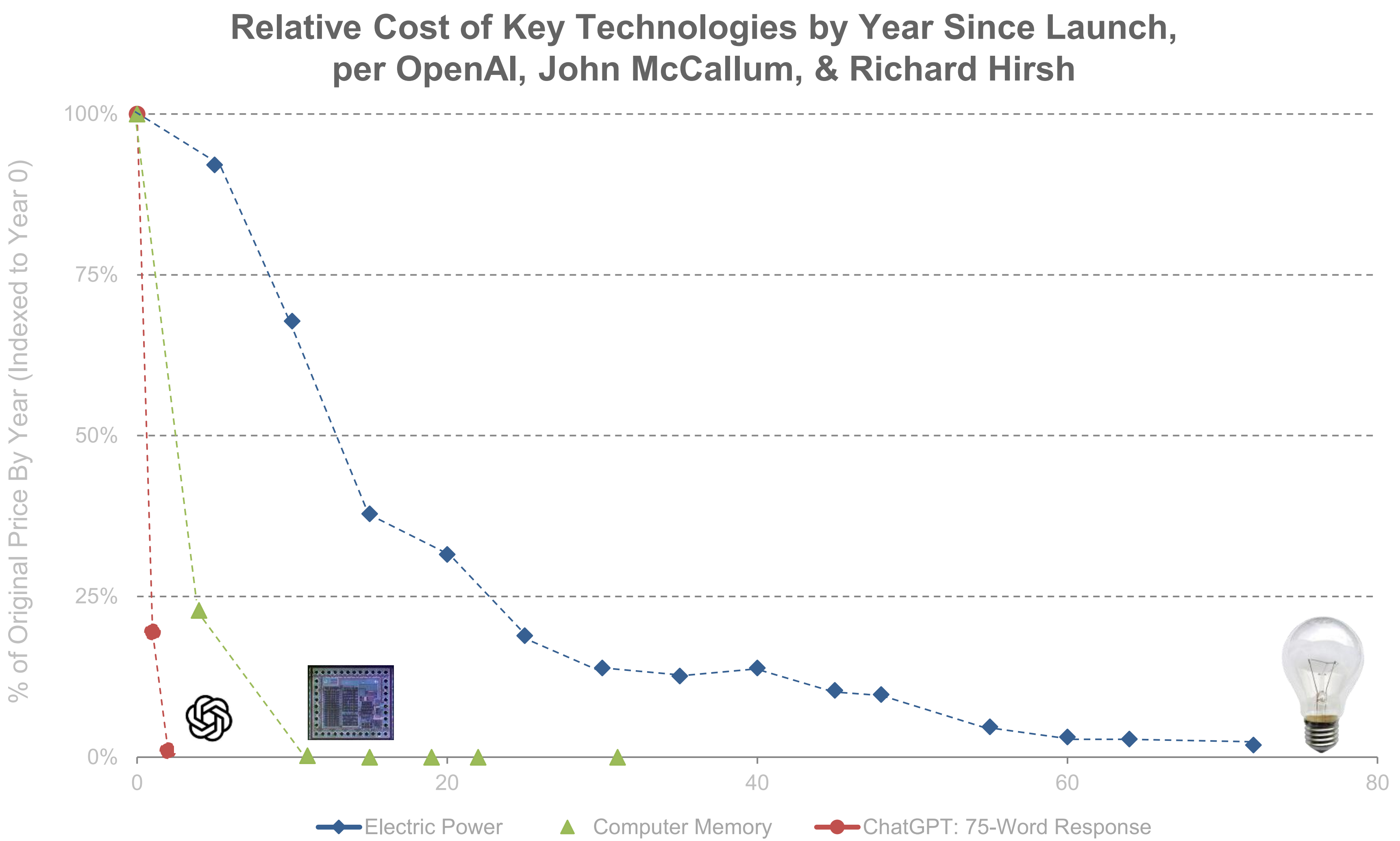
AI Inference Price for Customers (per 1 Million Tokens) – 11/22-12/24, per Stanford HAI

Note: Axis is logarithmic; every axis tick represents a 10x price change



Source: Nestor Maslej et al., 'The AI Index 2025 Annual Report,' AI Index Steering Committee, Stanford HAI (4/25)

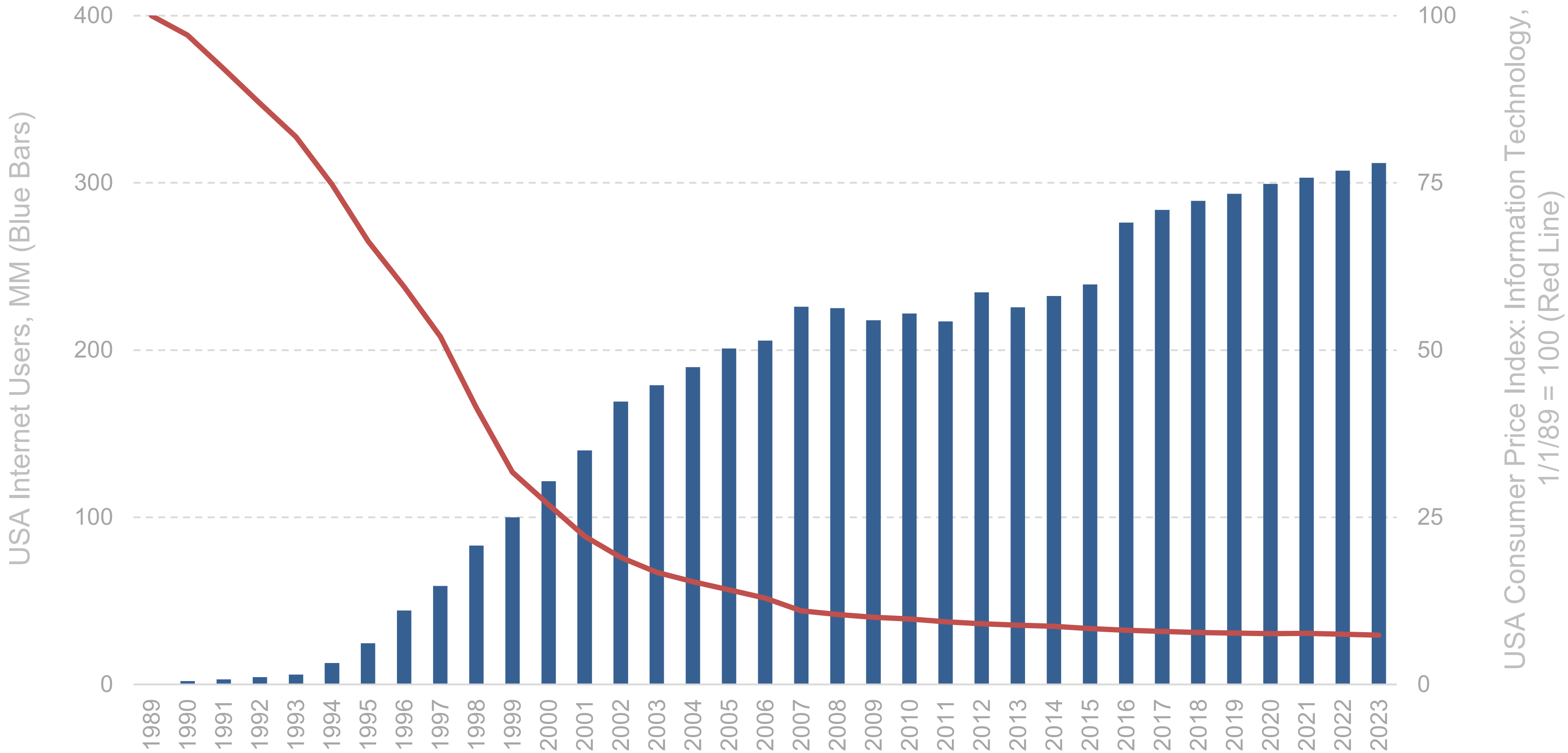
AI Cost Efficiency Gains = Happening Faster vs. Prior Technologies



Note: Price change in consumer goods and services in the United States is measured as the percentage change since 1997. Data is based on the consumer price index (CPI) for national average urban consumer prices. Per OpenAI, 100 AI 'tokens' generates approximately 75 words in a large language model response; data shown indexes to this number of tokens. 'Year 0' is not necessarily the year that the technology was introduced, but rather the first year of available data.
Source: Electricity Costs – Technology and Transformation in the American Electric Utility Industry, Richard Hirsh (1989); Computer Memory Storage Costs – John C. McCallum, with data aggregated from 72 primary sources and historical company sales documents; OpenAI, Wikimedia Commons

Tech's Perpetual A-Ha =
Declining Costs + Improving Performance → Rising Adoption…

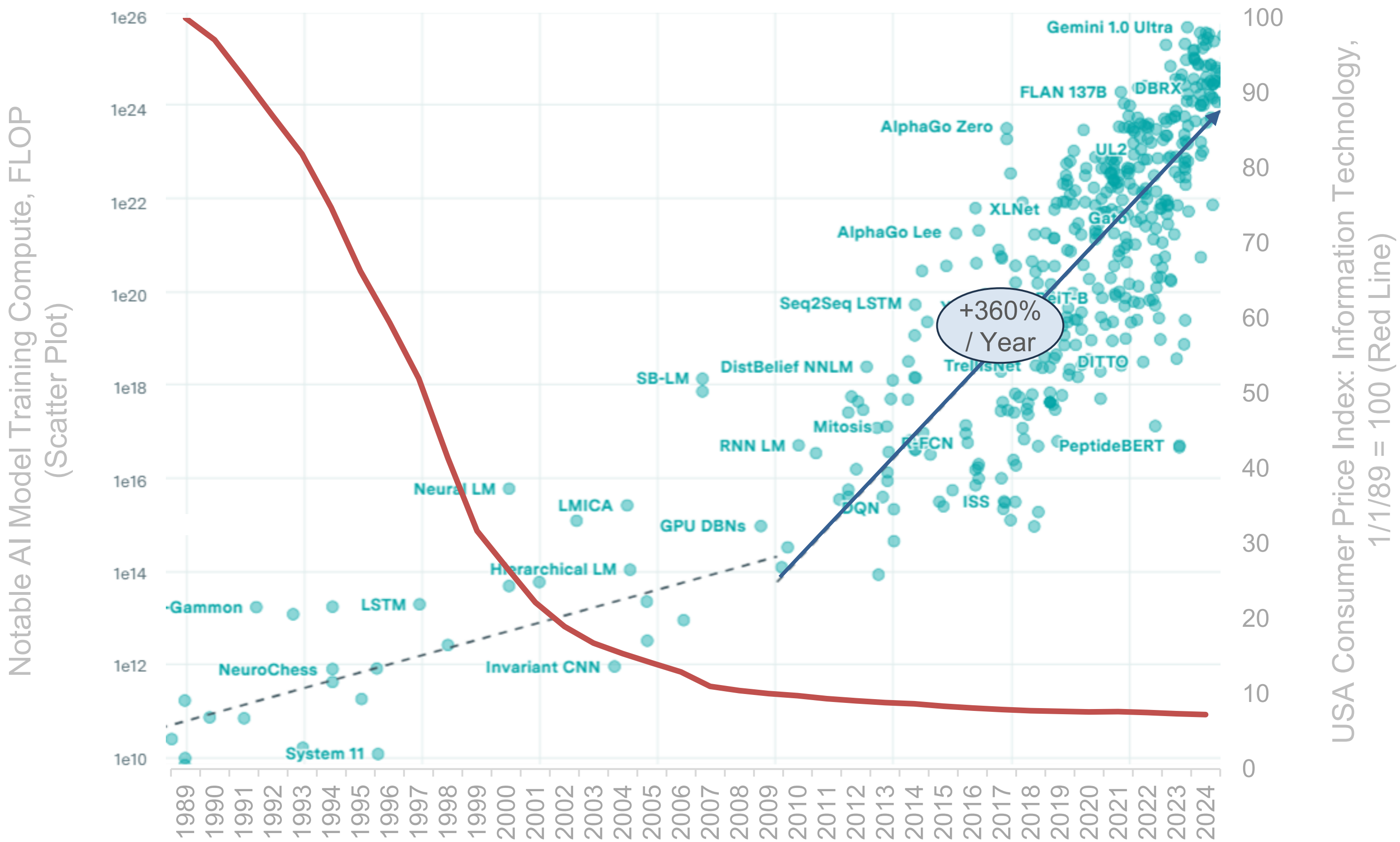
USA Internet Users (MM) vs. Relative IT Cost – 1989-2023, per FRED & ITU



Note: FRED data shows 'Consumer Price Index for All Urban Consumers: Information Technology, Hardware and Services in U.S. City Average.' Source: USA Federal Reserve Bank of St. Louis (FRED), International Telecommunications Union (via World Bank) (4/25)

…Tech's Perpetual A-Ha = Prices Fall + Performance Rises

AI Model Training Compute (FLOP) vs. Relative IT Cost – 1989-2024, per Epoch AI & FRED



Note: A petaFLOP/s-day represents the total computational work performed by a system operating at 1 petaFLOP/s (10^{15} floating-point operations per second) for 24 hours, equivalent to approximately 8.64 × 10^{19} operations. This metric is commonly used to quantify the compute required for large-scale tasks like training machine learning models. FRED data shows 'Consumer Price Index for All Urban Consumers: Information Technology, Hardware and Services in U.S. City Average.' Note that, while training compute is not a direct measurement of model performance, it is typically closely correlated with performance. Source: USA Federal Reserve Bank of St. Louis (FRED); Epoch AI (5/25)

AI Model Compute Costs High / Rising

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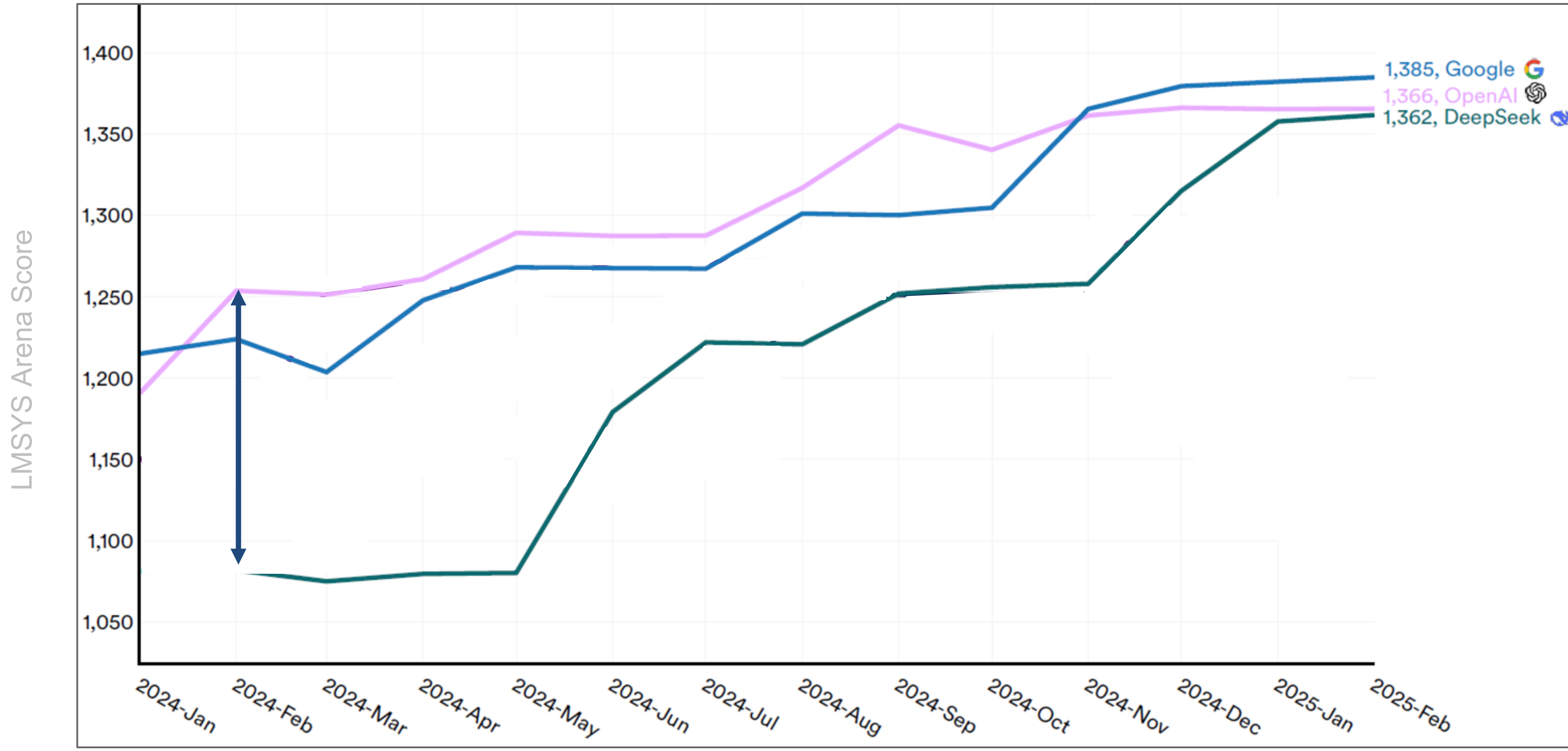
Inference Costs Per Token Falling

=

<u>Performance Converging</u> *+ Developer Usage Rising*

AI Model Performance = Converging Rapidly, per Stanford HAI

Performance of Top AI Models on LMSYS Chatbot Arena – 1/24-2/25, per Stanford HAI



Note: The LMSYS Chatbot Arena is a public website where people compare two AI chatbots by asking them the same question and voting on which answer is better. The results help rank how well different language models perform based on human judgment. Only the highest-scoring model in any given month is shown in this comparison.
Source: Nestor Maslej et al., 'The AI Index 2025 Annual Report,' AI Index Steering Committee, Stanford HAI (4/25)

AI Model Compute Costs High / Rising

+

Inference Costs Per Token Falling

=

Performance Converging + ***<u>Developer Usage Rising</u>***

To understand the surge in AI developer activity, it's instructive to look at the extraordinary drop in inference costs and the growing accessibility of capable models.

Between 2022 and 2024, the cost-per-token to run language models fell by an estimated 99.7% – a decline driven by massive improvements in both hardware and algorithmic efficiency. What was once prohibitively expensive for all but the largest companies is now within reach of solo developers, independent app builders, researchers on a laptop, and mom-and-pop shop employees.

The cost collapse has made experimentation cheap, iteration fast, and productization feasible for virtually anyone with an idea.

At the same time, performance convergence is shifting the calculus on model selection. The gap between the top-performing frontier models and smaller, more efficient alternatives is narrowing.

For many use cases – summarization, classification, extraction, or routing – the difference in real-world performance is negligible. Developers are discovering they no longer need to pay a premium for a top-tier model to get reliable outputs. Instead, they can run cheaper models locally or via lower-cost API providers and achieve functionally similar results, especially when fine-tuned on task-specific data.

This shift is weakening the pricing leverage of model incumbents and leveling the playing field for AI development…

…At the platform level, a proliferation of foundation models has created a new kind of flexibility. Developers can now choose between dozens of models – OpenAI's ChatGPT, Meta's Llama, Mistral's Mixtral, Anthropic's Claude, Google's Gemini, Microsoft's Phi, and others – each of which excels in different domains. Some are optimized for reasoning, others for speed or code generation. The result is a move away from vendor lock-in.

Instead of consolidating under a single provider who can gate access or raise prices, developers are distributing their efforts across multiple ecosystems. This plurality of options is empowering a new wave of builders to choose the best-fit model for their technical or financial needs.

What's emerging is a flywheel of developer-led infrastructure growth. As more developers build AI-native apps, they also create tools, wrappers and libraries that make it easier for others to follow. New front-end frameworks, embedding pipelines, model routers, vector databases, and serving layers are multiplying at an accelerating rate.

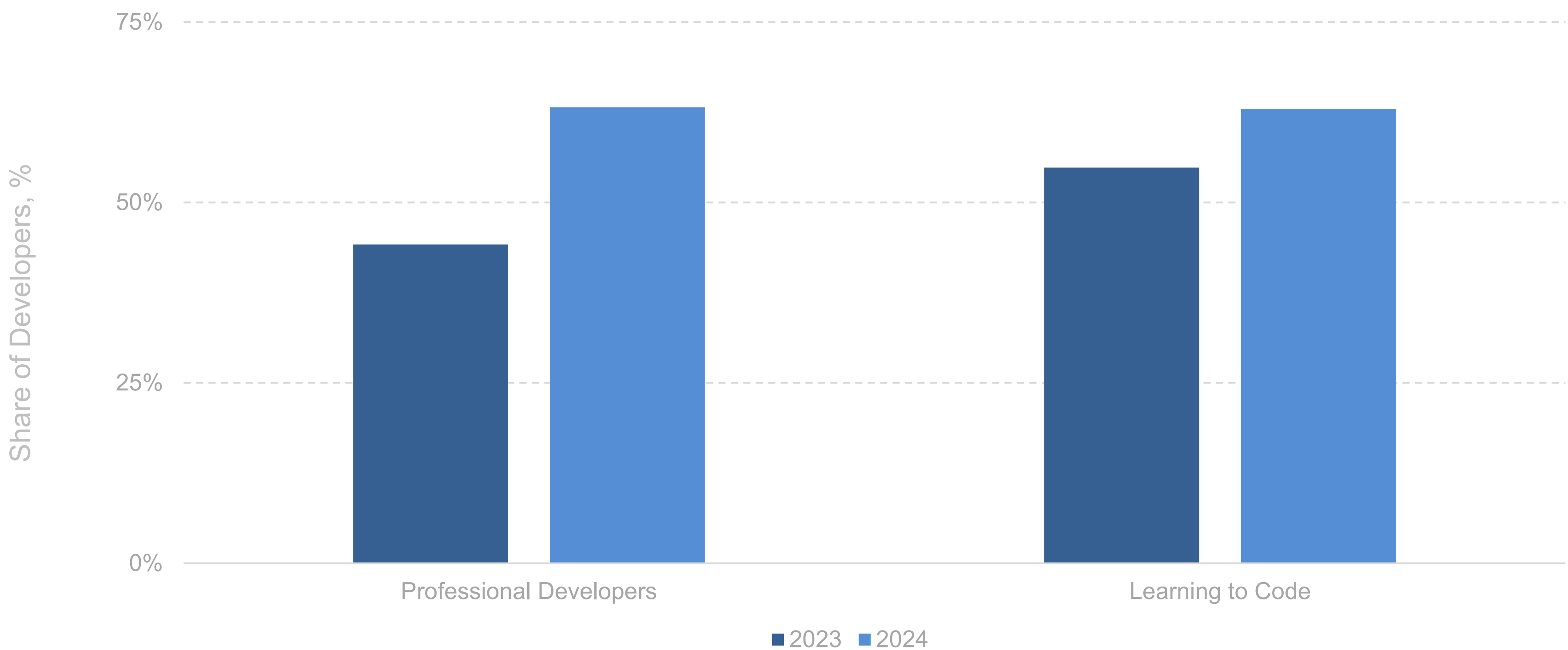
Each wave of developer activity reduces the friction for the next, compressing the time from idea to prototype and from prototype to product. In the process, the barrier to building with AI is collapsing – not just in cost, but in complexity. This is no longer just a platform shift. It's an explosion of creativity.

Technology history has shown – as memorialized by then-Microsoft President Steve Ballmer's repeat *Developers! Developers! Developers…* at a 2000 Microsoft Developers Conference (link) – the platform that gets the most consistent developer user and usage momentum – and can scale and steadily improve – wins.

The AI Developer Next Door

AI Tool Adoption by Developers = 63% vs. 44% Y/Y

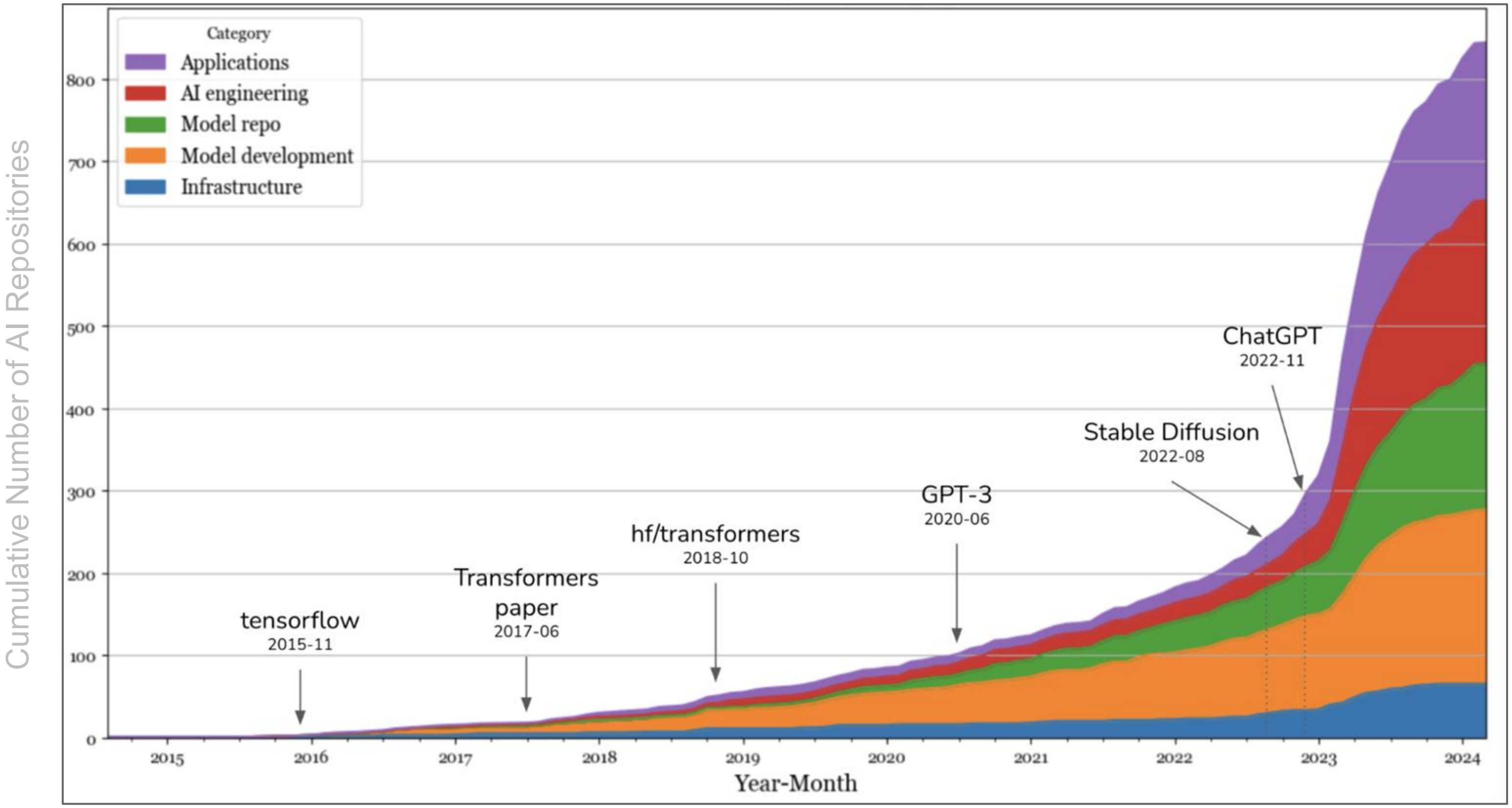
Share of Developers Currently Using AI in Development Processes – 2023-2024, per Stack Overflow



Note: 2023 N=89,184; 2024 N=65,437. Respondents are global. Source: Stack Overflow Developer Surveys (5/23 & 5/24-6/24)

AI Developer Repositories – GitHub = ~175% Increase Over Sixteen Months

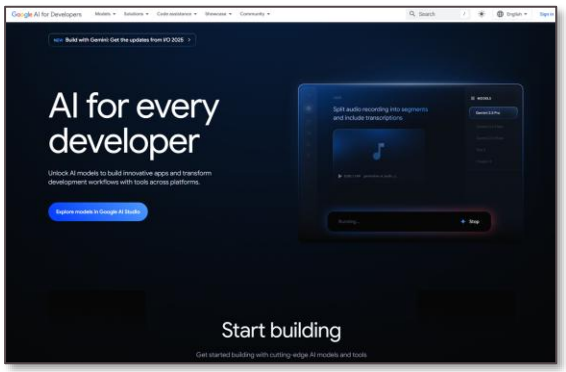
Number of AI Developer Repositories* on GitHub – 11/22-3/24, per Chip Huyen



*A repository is an online storage space where developers share and manage code, models, data, and documentation related to artificial intelligence projects. These enable collaboration, reuse, and distribution of AI tools and assets. Analysis shown includes GitHub repositories with 500+ stars. Infrastructure = tools for model serving, compute management, vector search & databases. Model development = frameworks for modeling & training, inference optimization, dataset engineering, & model evaluation. Application development = custom AI-powered applications (varied use cases). Source: Chip Huyen via GitHub (3/24)

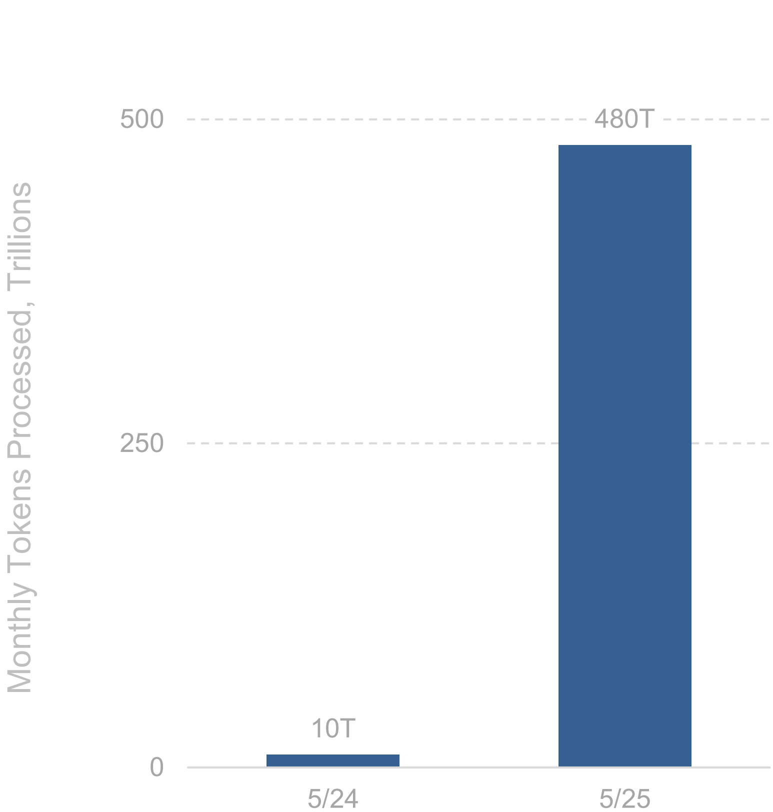
AI Developer Ecosystem – Google = +50x Monthly Tokens Processed Y/Y

Google Monthly Tokens Processed (T) – 5/24-5/25, per Google



This time last year, we were processing 9.7 trillion tokens a month across our products and APIs. Now, we're processing over 480 trillion – that's 50 times more.

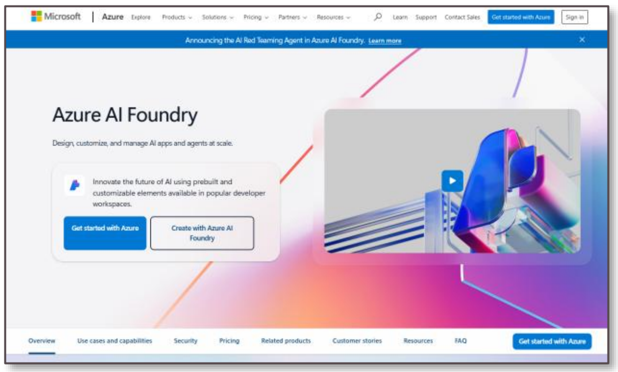
- Google I/O 2025 Press Release, 5/25



Note: Token usage shown across Google products & APIs. Per Google in 5/25, 'This time last year, we were processing 9.7 trillion tokens a month across our products and APIs. Now, we're processing over 480 trillion — that's 50 times more…Over 7 million developers are building with Gemini, five times more than this time last year.' Source: Google, 'Google I/O 2025: From research to reality' (5/25)

AI Developer Ecosystem – Microsoft Azure AI Foundry = +5x Quarterly Tokens Processed Y/Y

Microsoft Azure AI Foundry Quarterly Tokens Processed (T) – Q1:24-Q1:25, per Microsoft

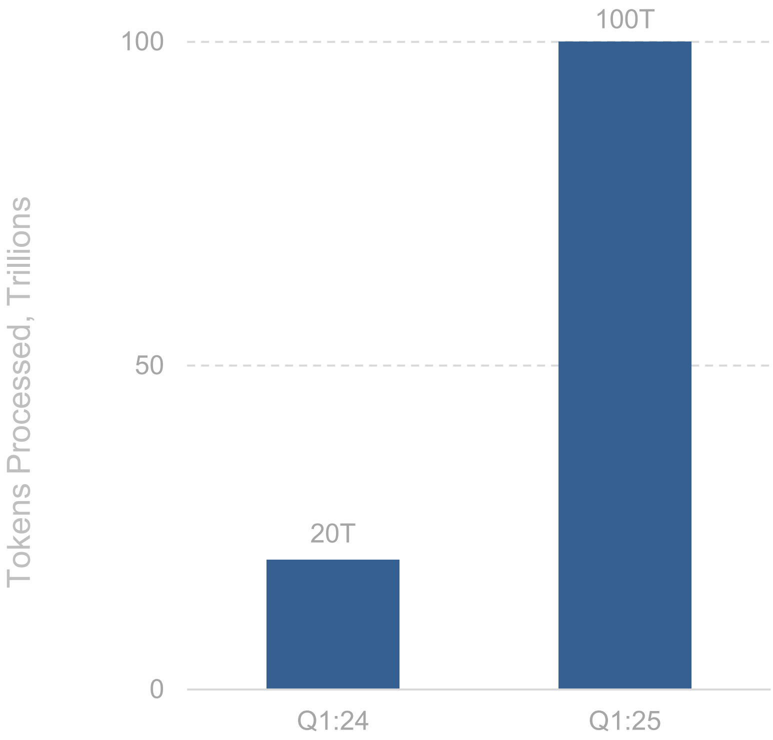


[Microsoft Azure AI] Foundry is the agent and AI app factory.

It is now used by developers at over 70,000 enterprises and digital natives – from Atomicwork, to Epic, Fujitsu, and Gainsight, to H&R Block and LG Electronics – to design, customize, and manage their AI apps and agents.

We processed over 100 trillion tokens this quarter, up 5x year-over-year – including a record 50 trillion tokens last month alone.

- Microsoft FQ3:25 Earnings Call, 4/25



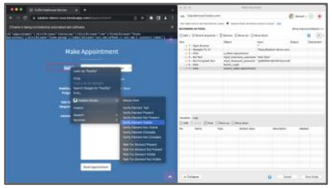
Note: Source: Microsoft FQ3:25 earnings call (4/25)

AI Developer Use Cases = Broad & Varied

AI Developer Use Cases – 2024, per IBM

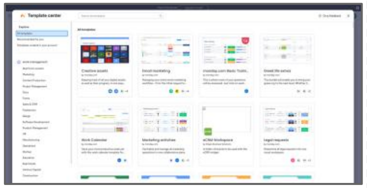
Code Generation

Bug Detection & Fixing

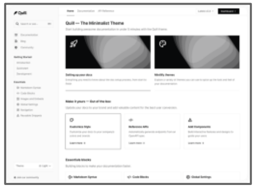


Testing Automation

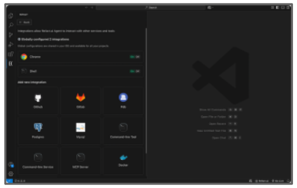
Project / Workflow Management



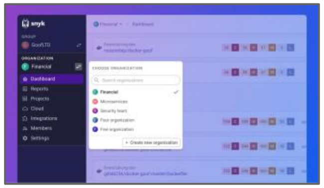
Documentation



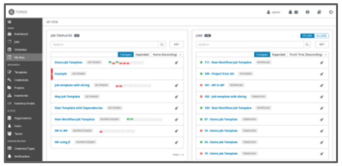
Refactoring & Optimization



Security Enhancement

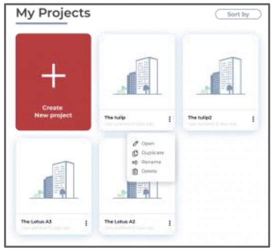


DevOps & CI / CD Pipelines



User Experience Design

Architecture Design



Note: CI / CD pipelines are continuous integration / continuous deployment pipelines.
Source: IBM, 'AI in Software Development' (2024); Anthropic; Katalon; AccelQ; Monday; Quill; Mintlify; Snyk; Ansible; UX Pilot; Ark Design AI